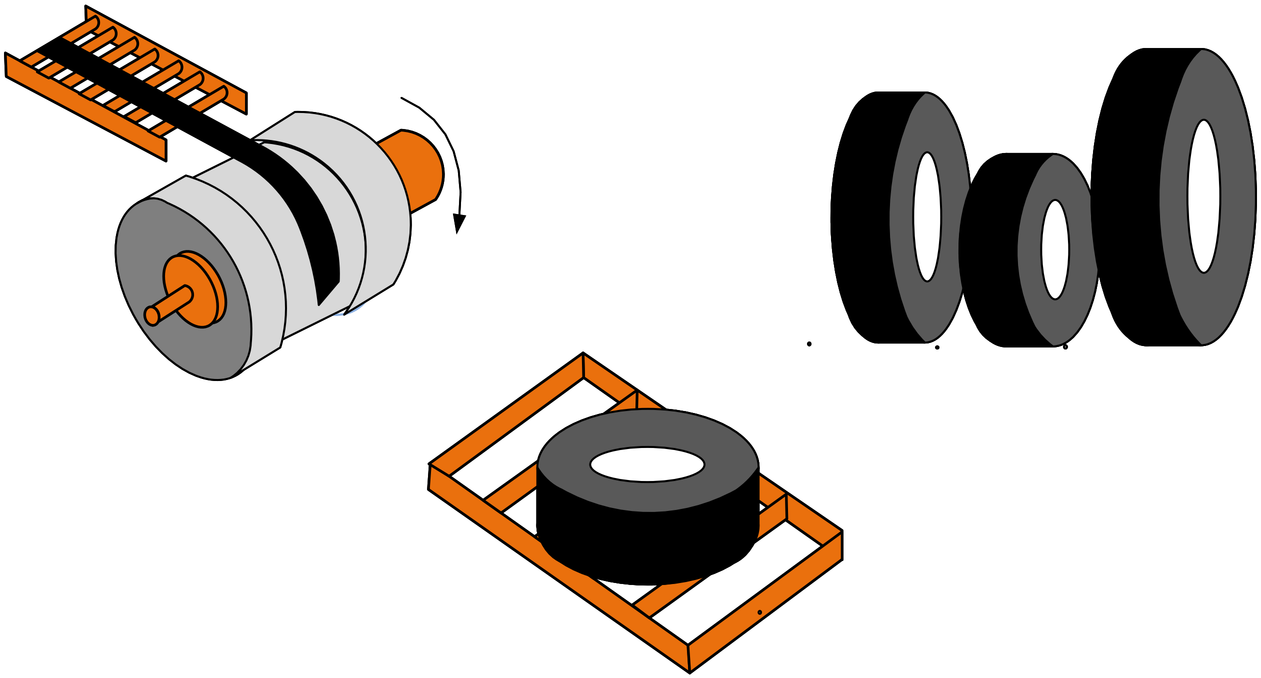
eVSM Mix Tire

Quick Mix Tire allows analytical value stream mapping of rubber tire production with variants on tire sizes and the makeup of each tire.



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How to Use this File

This file contains the reading materials and the exercise pages from the course (title on previous page). While the course can only be taken on a computer, this booklet can be useful for note taking and later for refresher training.

This booklet is designed for on screen and print use. For on screen use, we recommend Acrobat Reader with the page display set to "Single Page View". If you are using this booklet on-screen while going through the exercises in eVSM, a second monitor is very helpful.

For hardcopy use, print the file on 8.5x11 or A4, and bind along the long edge.

Table of Contents

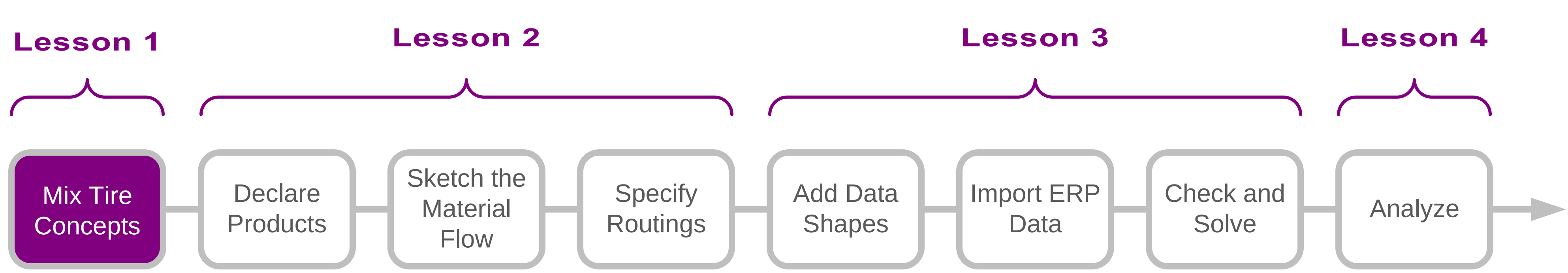
Mix Tire Value Stream Mapping Concepts

eVSM Quick Mix Tire allows analytical value stream mapping of tire production with visualization of capacity and lead times.

In this lesson, we will take a look at the concepts supported in this industry specific VSM application.

Before starting this course, you must complete the Fast Draw and the Quick Mix Time courses. These teach basic eVSM mechanics.

Quick Mix Tire Learning Path

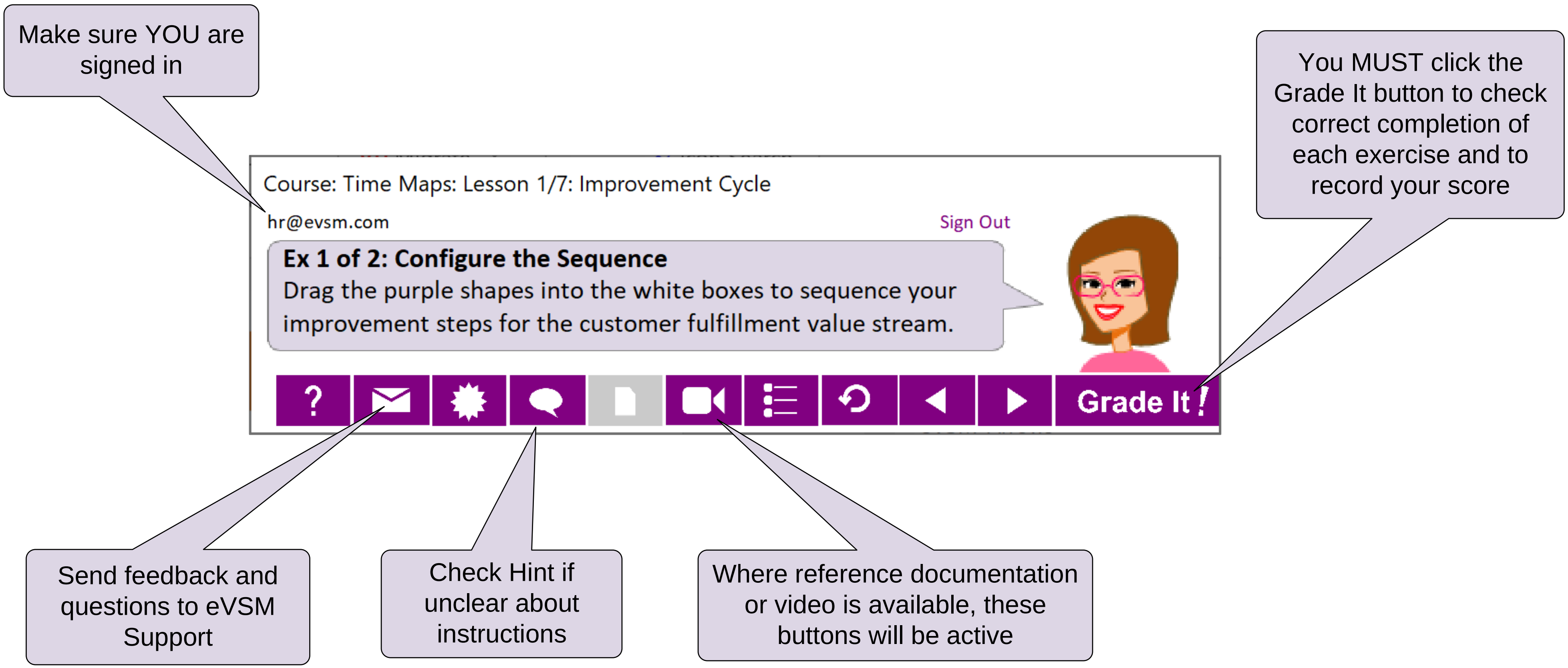


You must have eVSM v12.39 or later to work on the exercises in this course.

Mix Tire Value Stream Mapping Concepts

Working with eLeanor

The eLeanor learning system includes a range of useful functions:

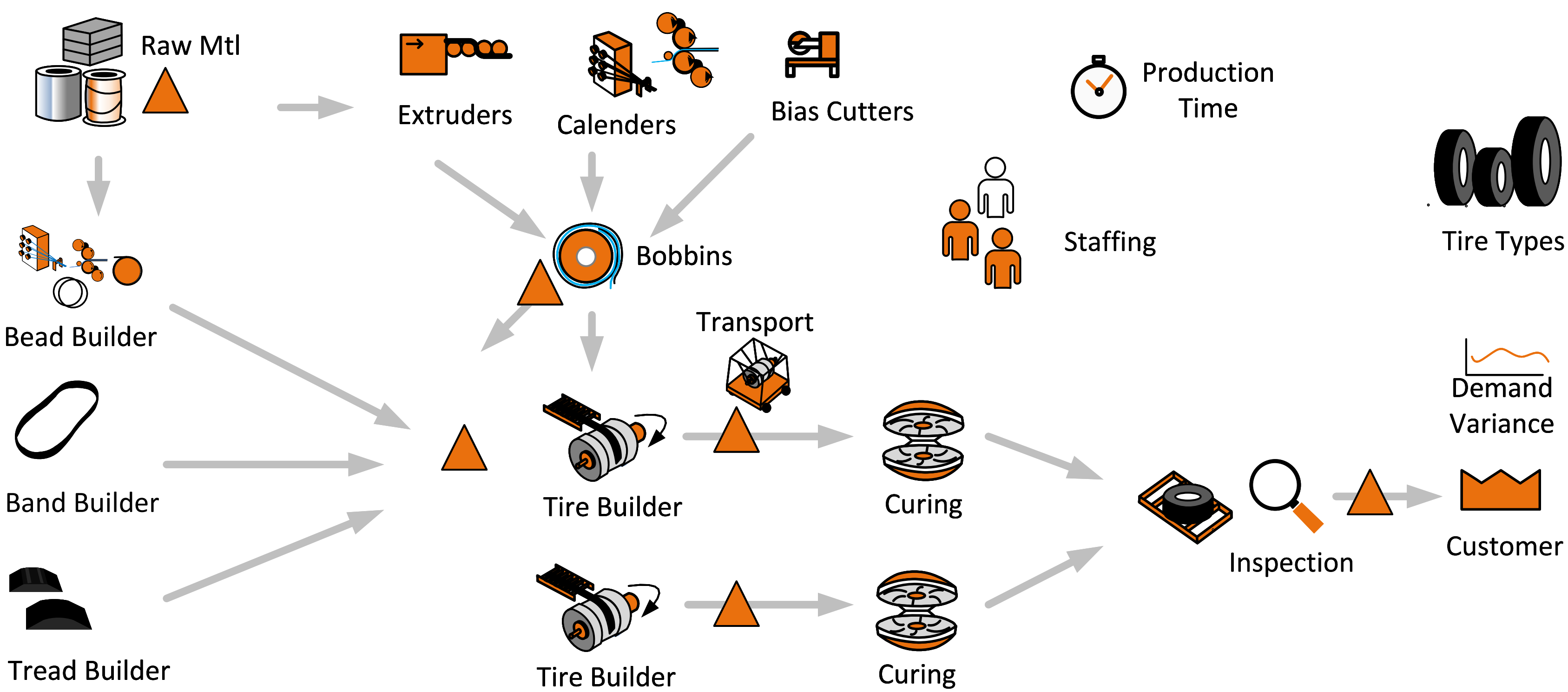


Important Notes

1. When you complete an exercise, you **MUST** click the “Grade It” button.
2. Points are deducted for incorrect attempts.
3. If you are stuck on an exercise, check the Hint. If that does not help, go back and review the preceding Readme pages. If you are still unsure, click the Feedback button and ask your question.

Tire Production Process

The graphic below depicts some elements in a tire production process that we will walk through in this concepts section to assess impact on flow, capacity, and lead time for our analytical value stream map.

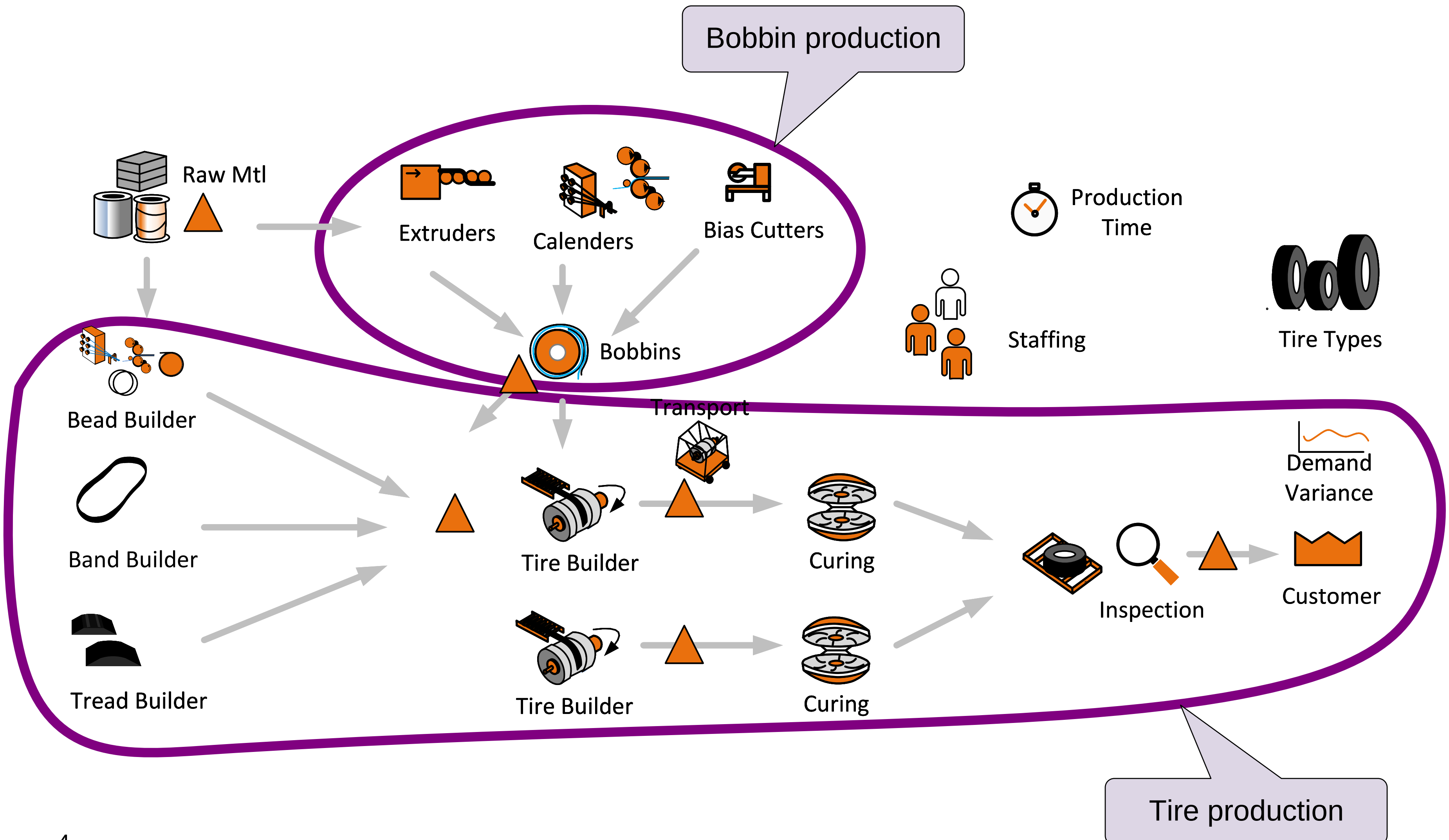


Bobbin & Tire Builder Factories

In this sketch, you can see two sub factories within the plant. The top one makes bobbins from raw materials and sometimes from other bobbins.

The bottom one makes the tire specific components (beads, bands, treads) and then together with the bobbins makes the individual tires.

For some plants there is a third factory retreading tires (The retread process is not shown here).



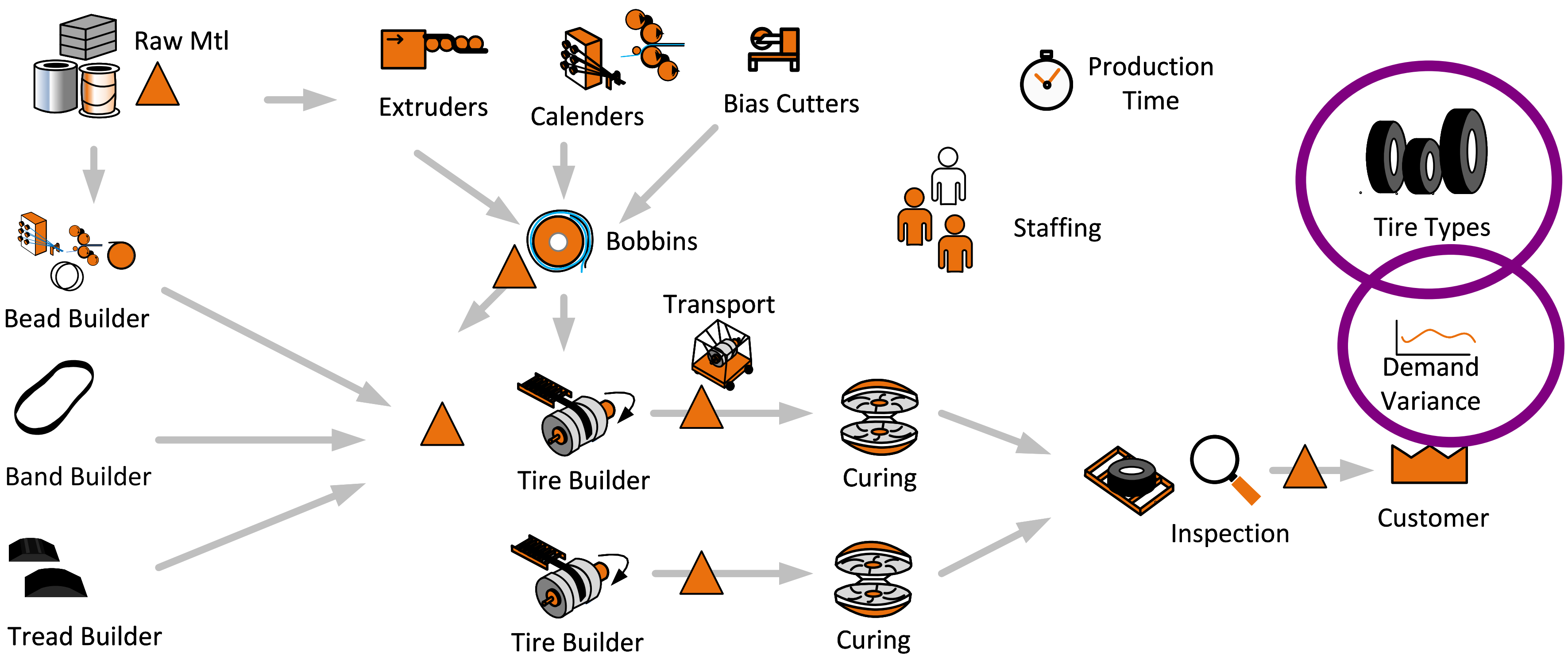
Tire Types & Demand

Most plants make multiple tire types using shared equipment and human resources. For each tire type we require the following input data

- Tire Demand: Demand per period
- Tire Weight : Weight Per Tire (useful for calculating high level plant key metrics like total weight/day)
- Demand Surge: Potential demand surge for this tire (The impact of surge will be reflected on capacity charts)

Example:

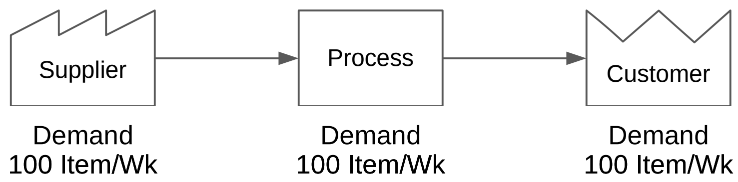
- Plant demand mix is 1000 of Tire 1, 2000 of Tire 2, 5000 or Tire 3
- Tire Weights are 50Kg, 75Kg and 100Kg respectively
- Demand Surge values are 0%, 5% and 7%



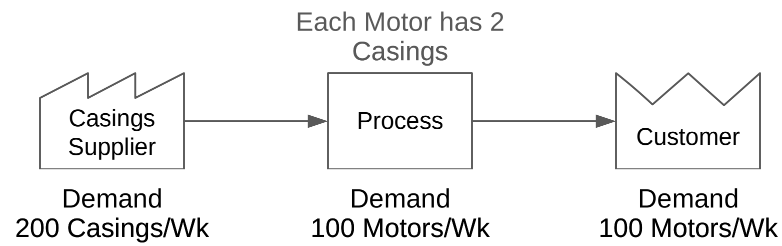
Value Stream Demand

Demand for the value stream is set by the customer and flows upstream through all the activities towards the suppliers. Generally, the demand at an activity is equal to the next downstream operation. Exceptions occur when activities are shared resources, have scrap, have parallel processing, etc. Here are some examples:

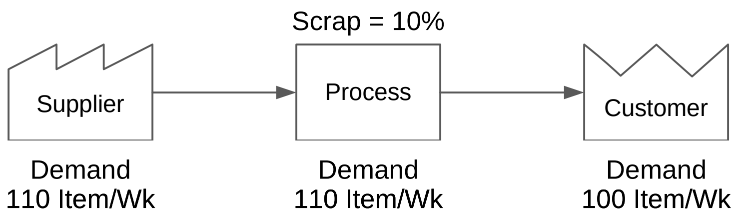
Simple straight flow



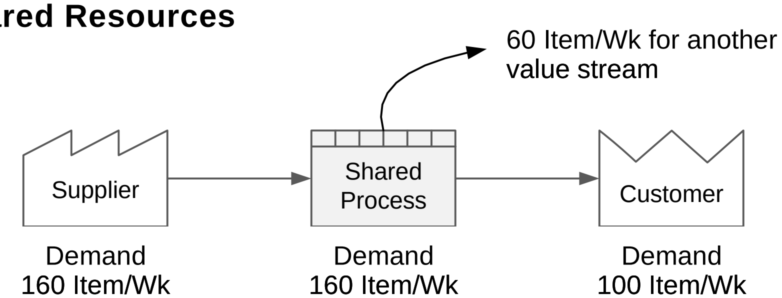
Components Demand



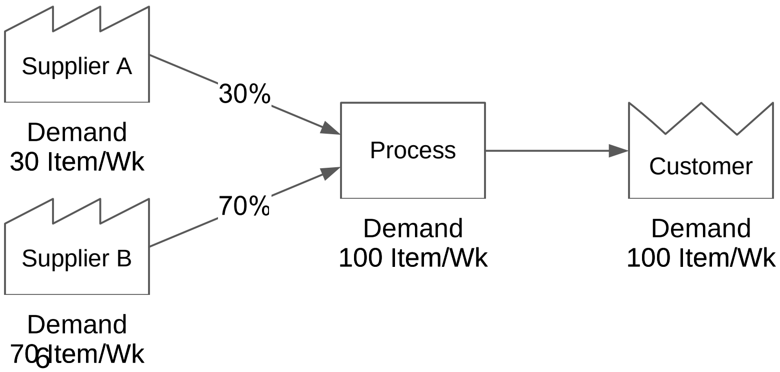
Scrap



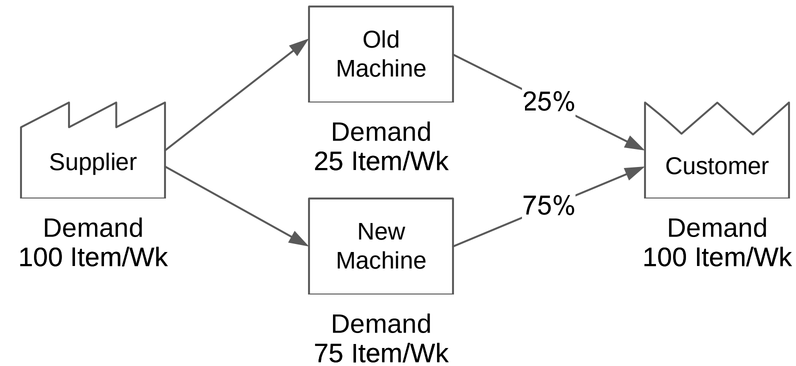
Shared Resources



Multiple Suppliers

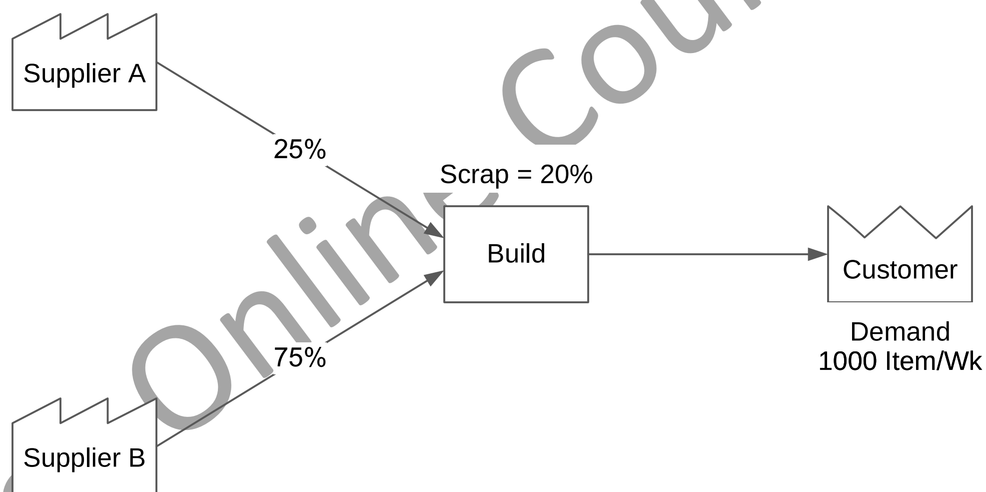


Parallel Processes



Q. What is the weekly demand at Supplier B?

- ◎ 800 Item/Wk
- ◎ 1000 Item/Wk
- ◎ 900 Item/Wk
- ◎ 750 Item/Wk



Bobbin Input Data

For each component that is stored on a bobbin, we need to connect it to tire demand via the following variables

For Component :

Bobbin Length (m/Bbn)

Bobbin Unused (%)

For each Tire Type using Component :

Length Needed (m/Tire)

Example :

Tire demands are 100/week for Tire A and 200/week for Tire B.

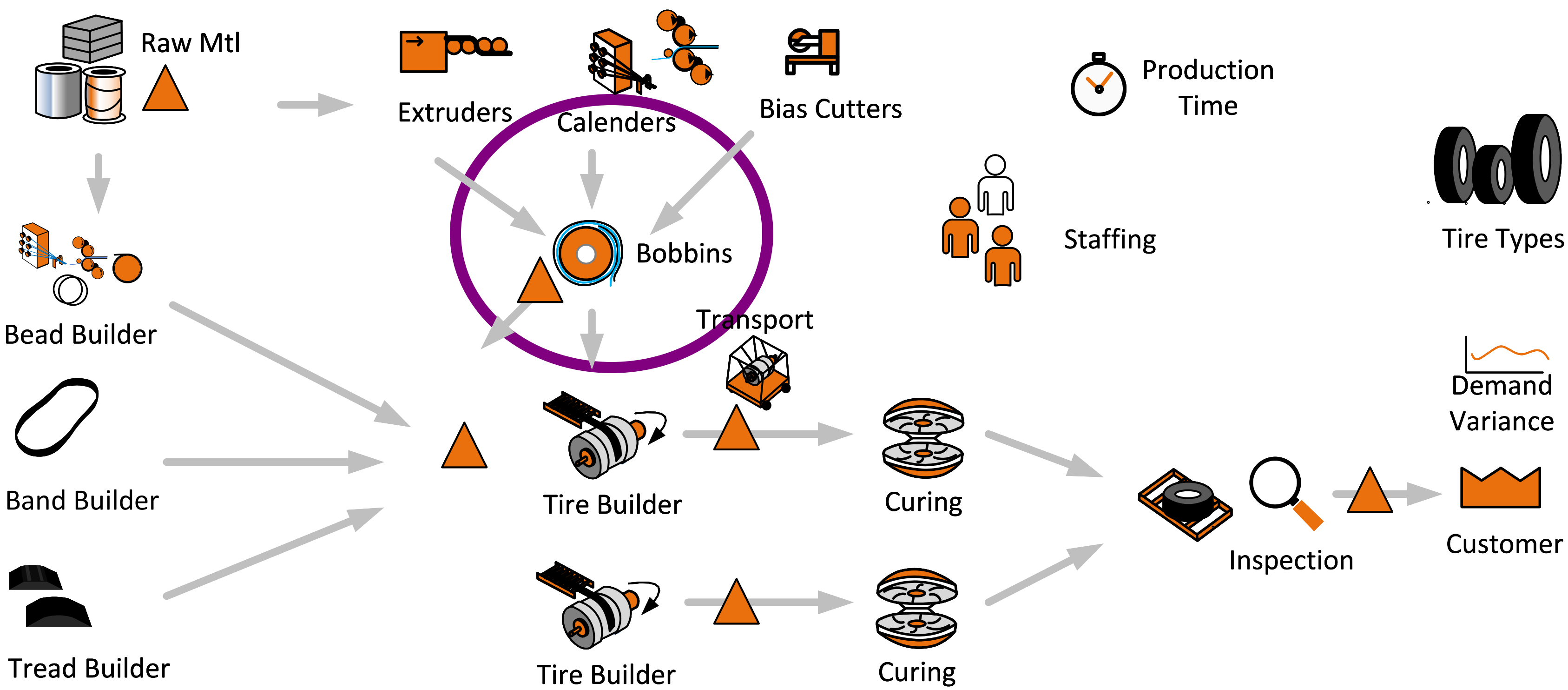
“Length Needed” of component is 15m for Tire A and 10m for Tire B

‘Bobbin Length’ for Component is 25m

‘Bobbin Unused’ is 5% for Component

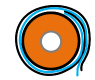
Bobbin Demand Calculation (Bbn/Wk)

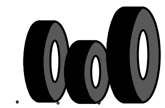
(100*15 + 200*10) / (25*.95)



Q. What is the total demand for Bobbin-A per Week?

Customer Demand	Length needed per Tire	Bobbin-A Size
Model X = 300 Tires/Wk	Model X = 5 m	Bobbin Length = 25 m
Model Y = 500 Tires/Wk	Model Y = 3 m	Bobbin Unused = 6%
Model Z = 250 Tires/Wk	Model Z = 3.8 m	


Bobbins


Tire Types

- ◎ 180 Bbn/Wk
- ◎ 172 Bbn/Wk
- ◎ 164 Bbn/Wk
- ◎ 168 Bbn/Wk

Bead Production

Multiple beads are needed per tire. The input values we use at a bead production center are

Cycle Time: This is the time per bead assuming the station is making one bead at a time

Qty Per Item: This is the number of beads per tire

OEE Input Percent

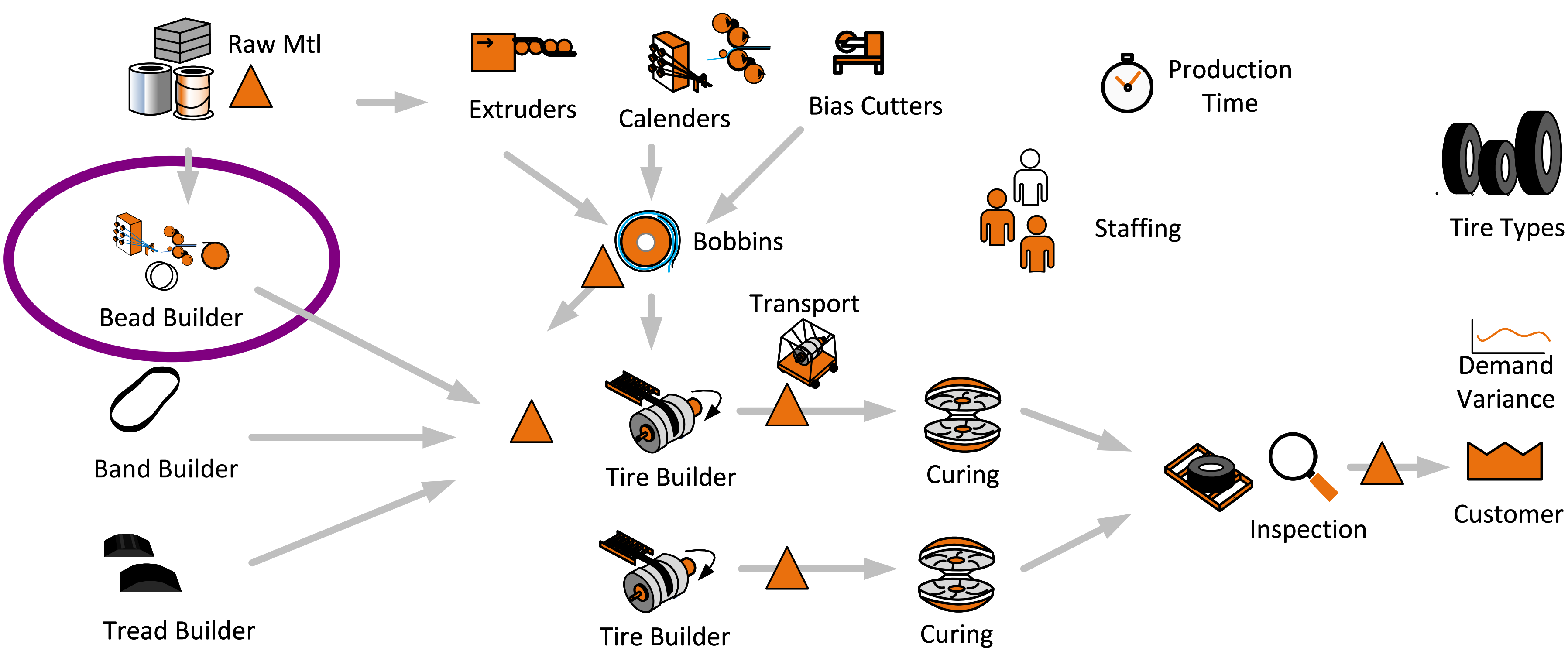
Example

A tire needs 7 beads per tire. Cycle time is 10 mins per bead. OEE Input Percent is 75%. Station operating 24 hours per day

Then

Capacity (Tires/Day) = [Total time in Minutes] * [OEE] / ([Beads per Tire] * [Cycle Time per Bead])

Capacity (Tires/Day) = (24*60*0.75)/(7*10)



Q. What is the Beads Production Capacity per Day (Tires/Day)?

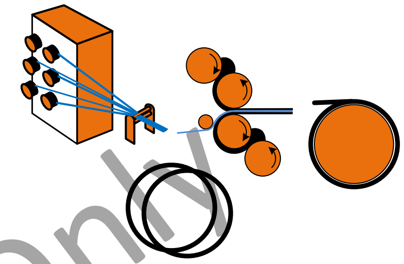
<u>Data</u>

Cycle Time = 5 Min

Beads per Tire = 5

OEE = 80%

Operation Time = 24 Hr/Day



- ◎ 46 Tires/Day
- ◎ 48 Tires/Day
- ◎ 50 Tires/Day
- ◎ 52 Tires/Day

Band & Tread Production

Bands are made from components, stored and then slipped over a tire in succession as part of the tire build process. Each band can take a different cycle time to make and each tire can have a different number of bands associated with them

The data input at a band production center:

Cycle Time 1: Cycle time for band type 1

Cycle Time 2: Cycle time for band type 2

...

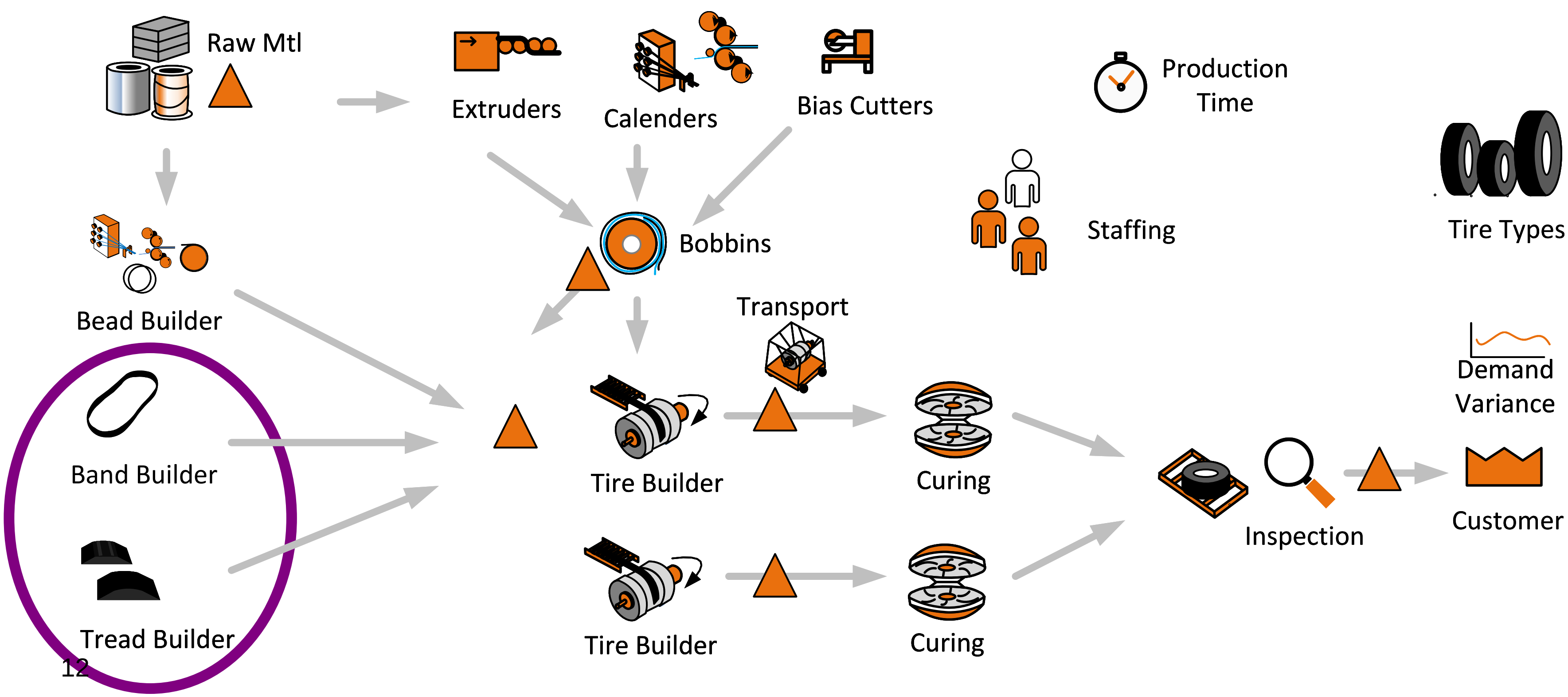
Cycle time n: Cycle time for the nth type band

Example

A tire needs 2 bands per tire. Cycle time 1 is 10 mins and Cycle Time 2 is 5 mins. OEE Input Percent is 75%. Station operating 24 hours per day

Capacity (Tires/Day) = .(24*60*0.75)/(10+5)

Similar input data and calculations are done for the treads on a tire



Q. What is the Band Production Capacity per Day (Tires/Day)?

<u>Data</u>

This tire needs 4 bands. Here are the cycle times for each:

Band 1 = 6 mins

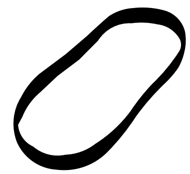
Band 2 = 3 mins

Band 3 = 4 mins

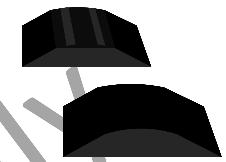
Band 4 = 4 mins

Station OEE is 75%

Operating Time = 24 Hrs/Day



Band Builder



Tread Builder

- ◎ 60 Tires/Day
- ◎ 72 Tires/Day
- ◎ 70 Tires/Day
- ◎ 64 Tires/Day

Inventory Accessories

There are inventories throughout the plant for components and tires. Some of these inventories have associated reusable accessories that support the inventory that's held (ex: rack space, bobbin liners, tire chariots).

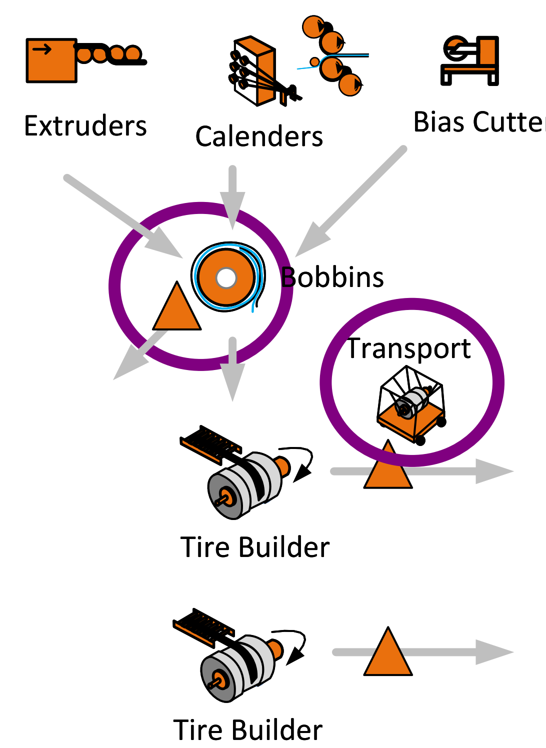
For items in an inventory location, there may be multiple size-based accessories. For this reason, we allow all tires and components to have a Group ID and then specify an accessory as applicable to items in the inventory that have a specific Group ID

For input we specify

1. The inventory location to which the accessory applies
2. The Group ID of items in the inventory that this accessory is needed for
3. Number of accessories need per unit inventory
4. Accessory Safety Factor (Percent value) to ensure adequate accessory availability

Example

Bobbin Inventory at a location each need a blue plastic liner sheet between the rubber for Bobbins with Group ID "PL" . The goal is to have 10% extra liners so they do not become a limiting factor. The peak design inventory (peak cycle stock + safety stock) of Bobbins with that Group ID is 200 at that location. So liner accessories needed would be 220.



Lead Time

The Lead Time contribution at a station is the Effective Cycle Time (Cycle Time / OEE Input Percent)

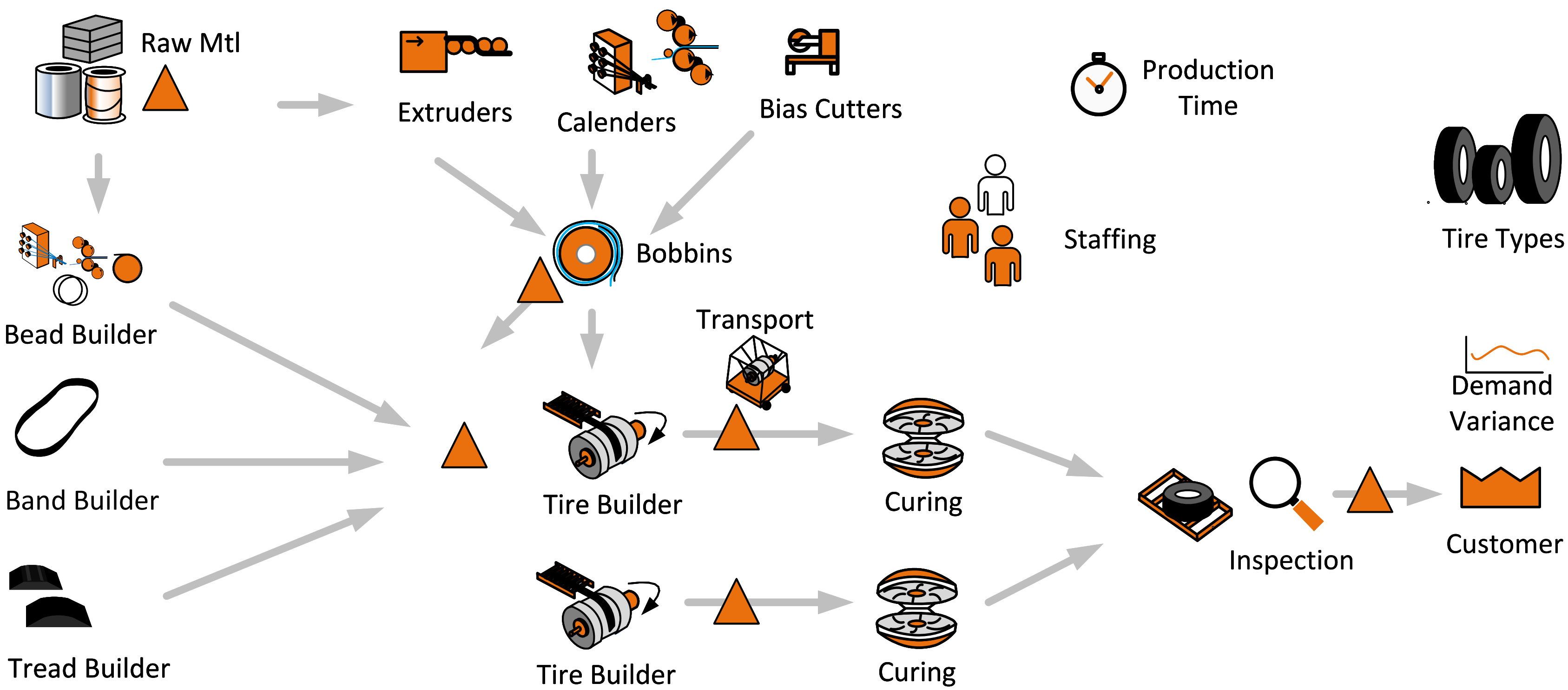
The Lead Time contribution for an inventory location is calculated as Inventory Qty / Demand

The Lead Time contribution for a Transport center is the transport time

Example

A station has a cycle time of 2 hour and an OEE Input percent of 60 %. Its lead time contribution = (2/0.6) Hr

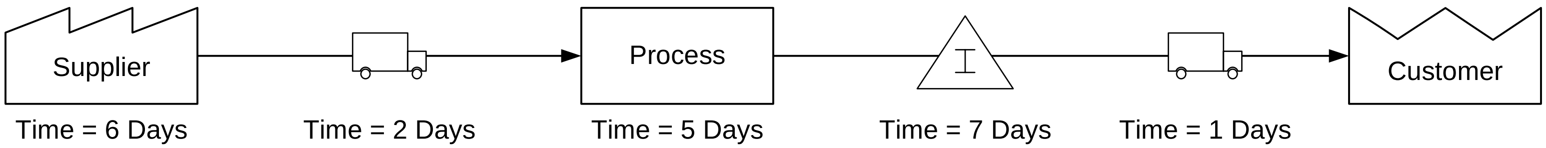
An inventory has 100 items and the daily demand is 10 items/day. Its lead time contribution = (100/10) day



Lead Time

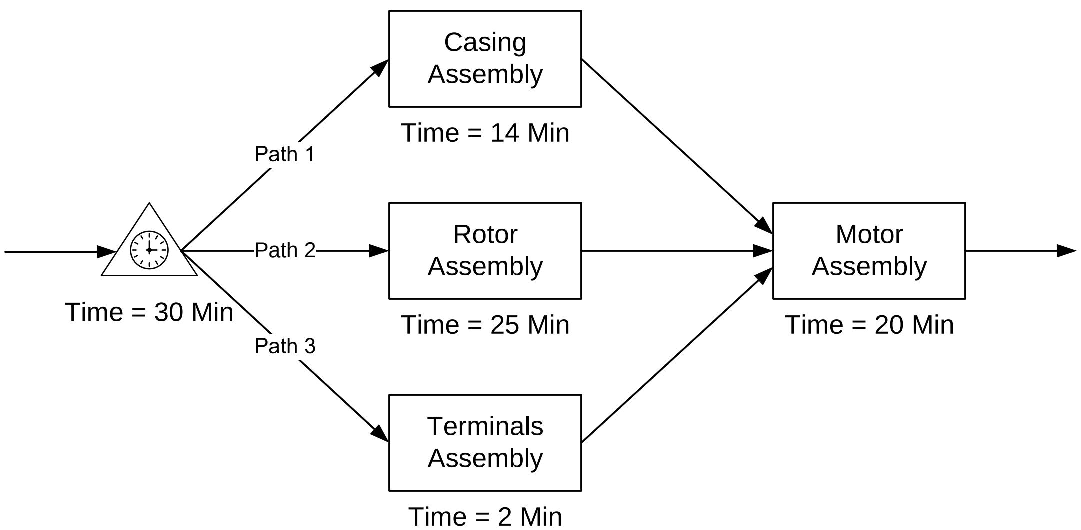
Lead time is the time it takes for an item of work to go through a process. It includes all of the time components such as work time, wait time, transport time, etc. The process may be an individual activity, a segment of a value stream, or the whole value stream. So it is important to clarify what part of the value stream the lead time is for.

Example 1 – Lead Time is the Sum of all times



Lead Time = 6 + 2 + 5 + 7 + 1 = 20 Days

Example 2 – Critical Path is the Lead Time of the longest path

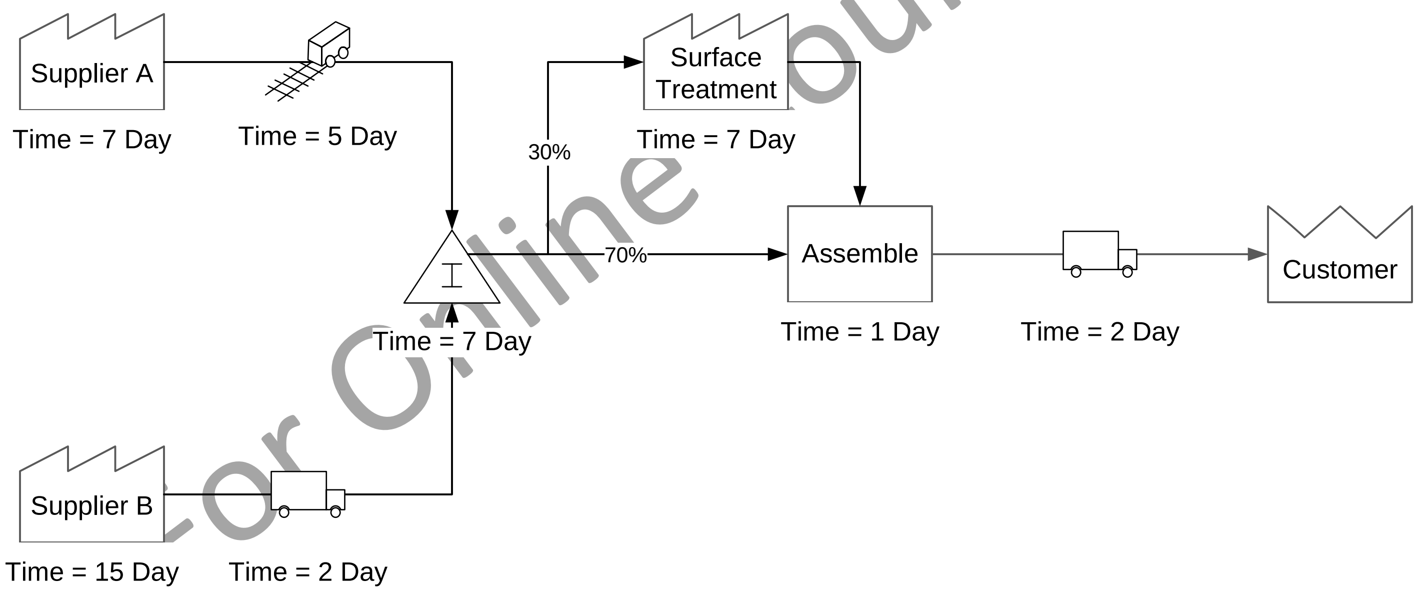


	Lead Time
Path 1	64 Min
Path 2	75 Min
Path 3	52 Min

Critical Path

Q. What is the lead time for the critical path?

- ◎ 29 Days
- ◎ 38 Days
- ◎ 31 Days
- ◎ 34 Days



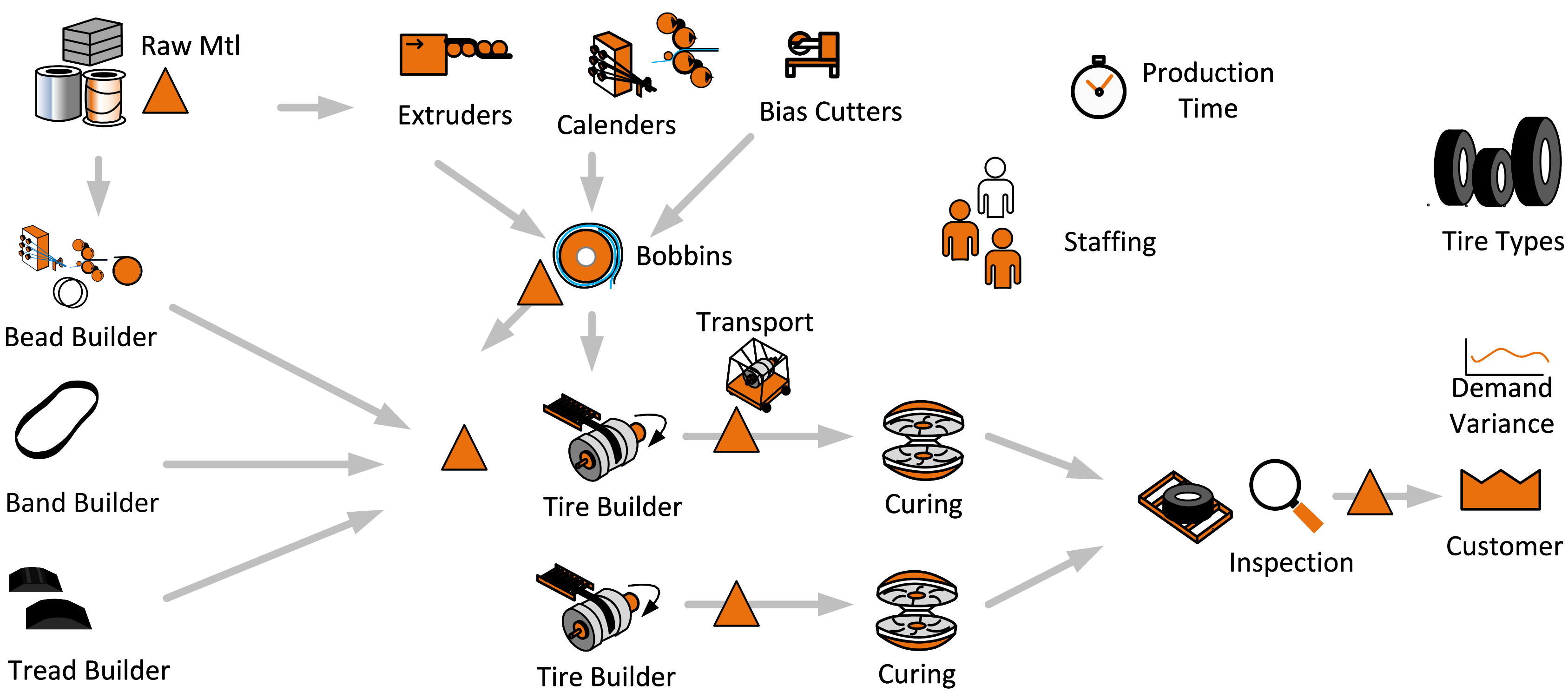
Activity Capacity

Where items are being made at a station, the following input data is needed for capacity calculation

- Qty Per Cycle: For each cycle on this station how many items are made
- Cycle Time: The time for one cycle
- Stations: Number of parallel stations represented by this activity
- OEE Input Percent: Percent of time assumed available for production after OEE losses (quality, availability, performance)

Calculation

Capacity = (Production Time * OEE Input Percent* Stations * Qty Per Cycle) / Cycle Time



Capacity

A clear understanding of capacity allows you to see if you can meet the customer demand, identify excess resources, and select the best way to address any bottlenecks. The capacity of an activity is the maximum number of "good" products it can produce in a given time period.

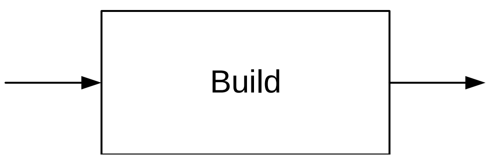
The simple calculation for capacity is:

$$\text{Capacity} = \frac{\text{Available Time}}{\text{Cycle Time}}$$

Typical value streams have losses such downtime, setup time, changeovers, process inefficiencies, rework, scrap, etc. So, a more general calculation is:

$$\text{Capacity} = \frac{\text{Available Time}}{\text{Cycle Time}} - \text{Losses}$$

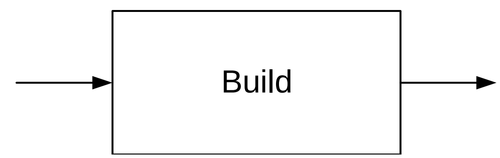
Example 1 – No Capacity Losses



Cycle Time = 10 Min/Item
Available Time = 8 Hr/Day

$$\text{Capacity} = \frac{8 * 60}{10} = 48 \text{ items/day}$$

Example 2 – Capacity with 80% OEE



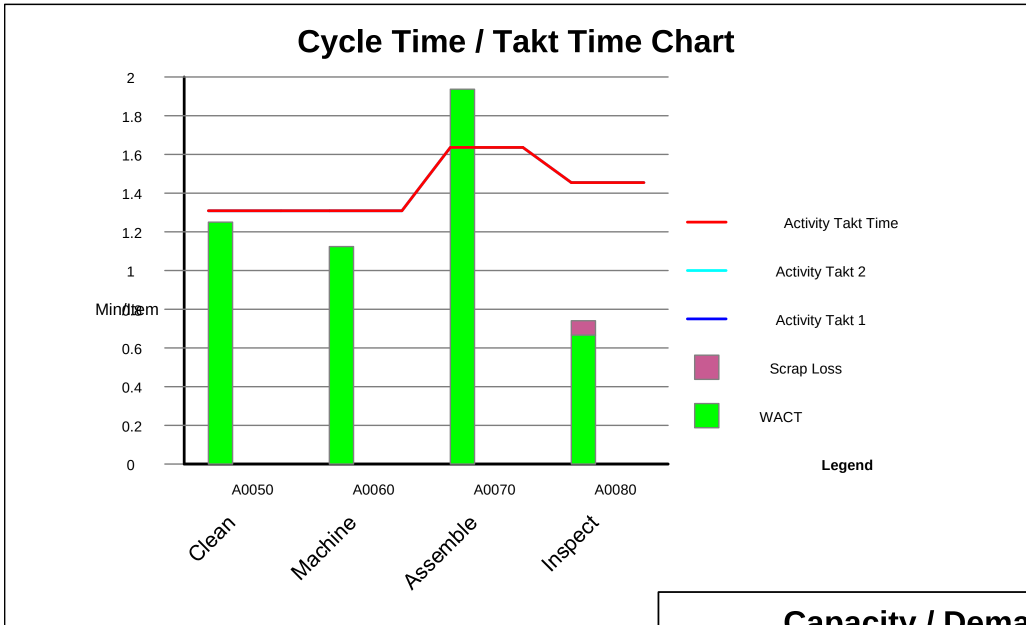
Cycle Time = 10 Min/Item
Available Time = 8 Hr/Day
OEE = 80%

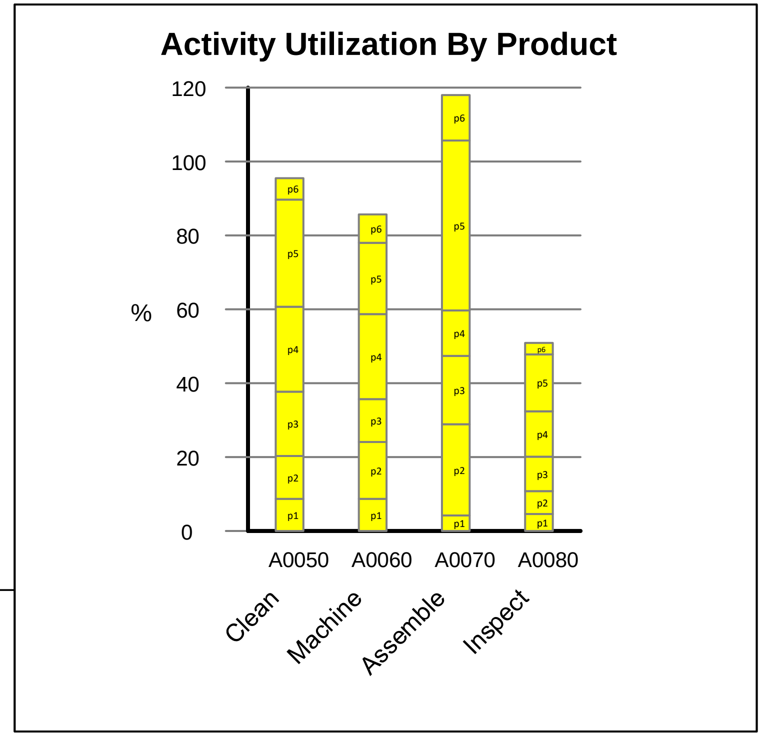
$$\text{Capacity} = \frac{8 * 60 * 0.8}{10} = 43 \text{ items/day}$$

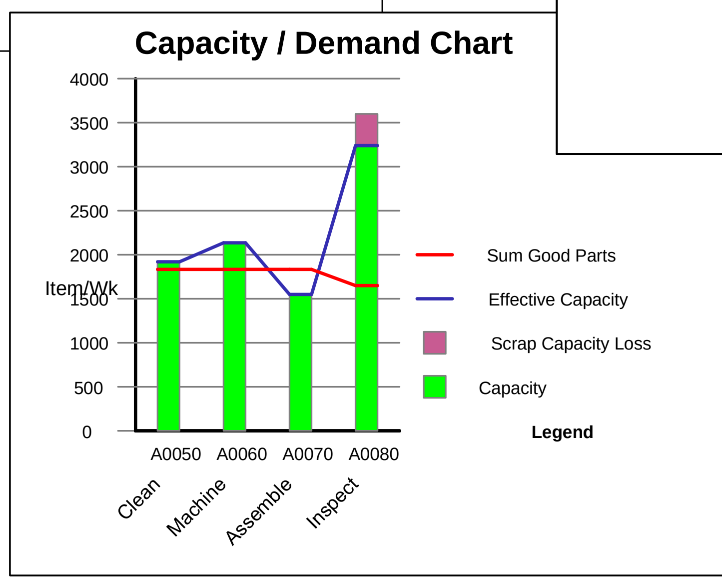
Capacity Analysis Charts

Charts are often used to visualize capacity across all activities in the value stream. These charts can help see where the bottle-neck process is, which activities have insufficient capacity, how well the line is balanced, and where there may be excess capacity. Here are some examples.

All three charts show that the Assemble activity is the bottle-neck and that the Inspect activity has excess capacity.







Q. What is the Daily Capacity of this Process?

- ◎ ~46 Item/Day
- ◎ ~28 Items/Day
- ◎ ~43 Items/Day
- ◎ ~48 Items/Day

Process

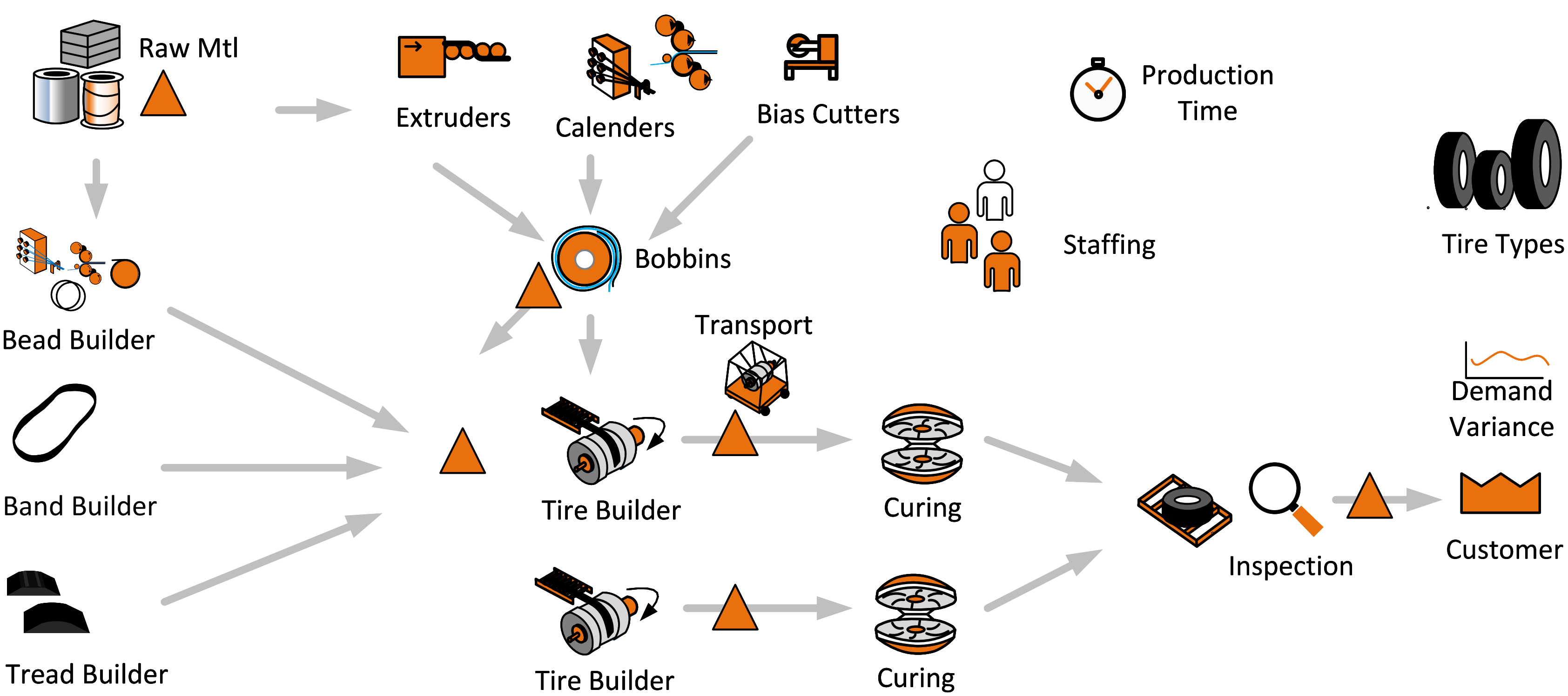
Cycle Time	10	Min
Activity Time	8	Hr
Down Time	10	%

Customer Surge, Station Surge and Inventory Surge

Customer surge is a percent value you can declare against each tire to see the impact on station utilizations and design inventory values. It reflects possible change in customer demand.

Inventory surge is a percent value you can declare when calculating downstream design inventory values. It has no impact on station capacity or station utilization. Think of it as a percent of cycle stock to act as safety stock

Station surge is a percent value you can declare for any item at a station to reflect that the station may need to (perhaps temporarily) run faster based on its history (ahead of a long preventive maintenance period, for example). You will see the impact of this on station utilization and downstream design inventory



Inventory Quantity Needed Downstream

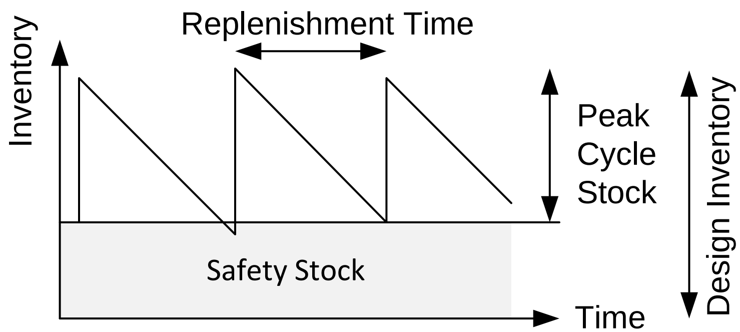
We can calculate the design quantity for products in inventory produced by the station and immediately downstream of it.

Input Values:

- Min Demand Period: What's the minimum demand period for which the inventory should be sufficient to meet downstream demand without replenishment
- Added Replenishment Interval: If there is significant delay for the station in getting a signal on what to make next or in delivering the goods downstream than this added interval can be declared and will cause additional stock to be held to cover this time.
- Inventory Surge % : This allows us to take the cycle stock calculated from the above input values and add an additional safety factor to the inventory levels.

Example

Let us say our demand from two station(s) building the same items is 100 items per day. We want at least one day of inventory downstream and can make this at our 2 stations building that inventory in 3 hours (this calculation should allow for OEE at the station). The signal on what's needed downstream and our time to deliver the inventory take an additional 5 hours. We also want an inventory safety buffer of 10% of the cycle stock. Our facility is running 24 hours per day, 7 days per week



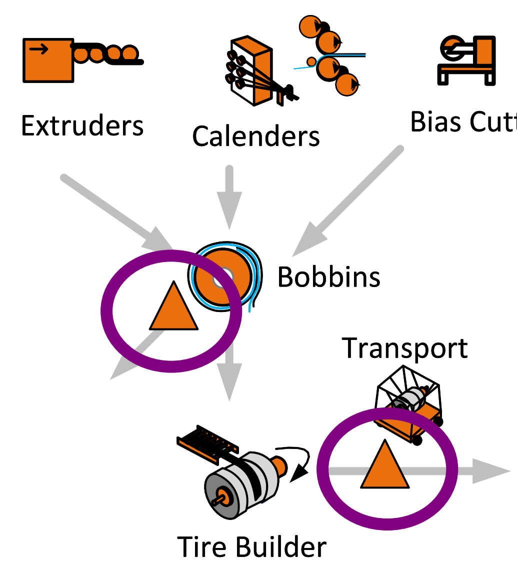
So:

Minimum Demand Period = 1 Day

Production Time to make demand for 1 day after allowing for OEE = 3 Hr

Inventory Surge % = 10 %

Replenishment Time = 3 Hr + 5 Hr = 8 Hr



Then:

Peak Cycle Stock in Inventory = 100 * 8/24

Safety Stock = 10% * Peak Cycle Stock

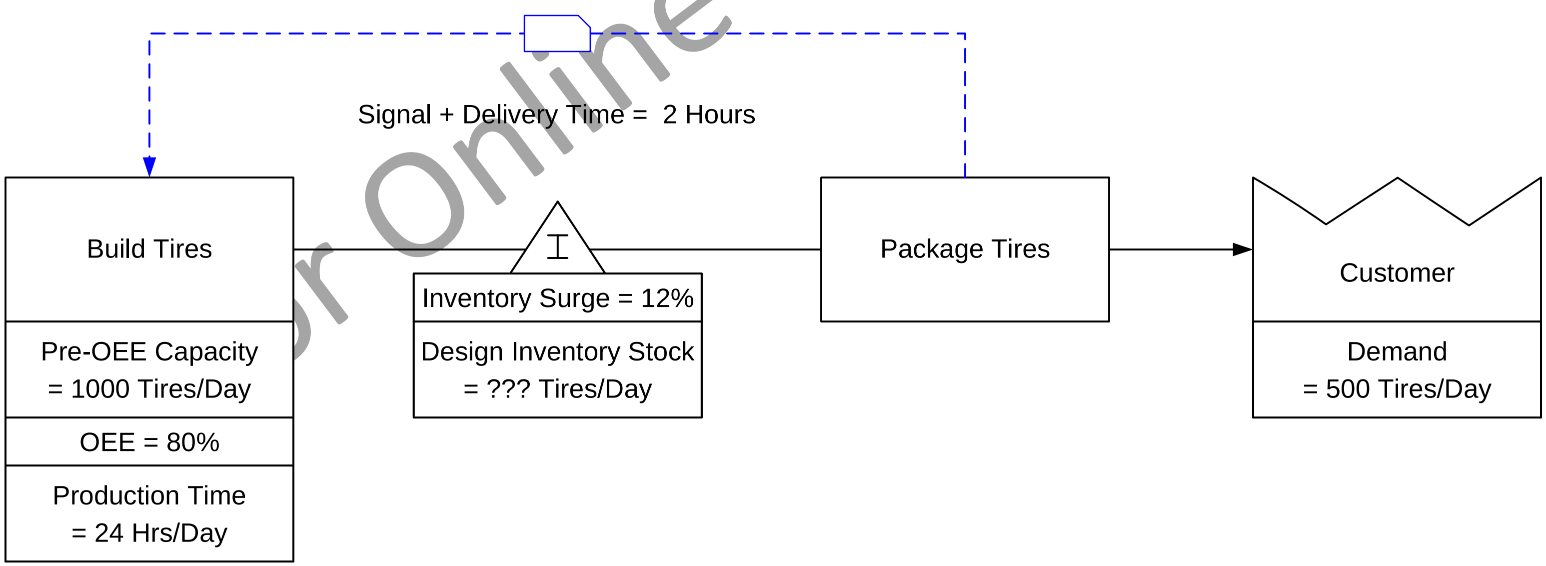
Design Inventory Stock = Peak Cycle Stock + Safety Stock = 37 Items

We round up the design inventory stock values calculated (whole bobbins or whole tires)

Design Inventory Stock

Given the data below, what is the Design Inventory Stock?

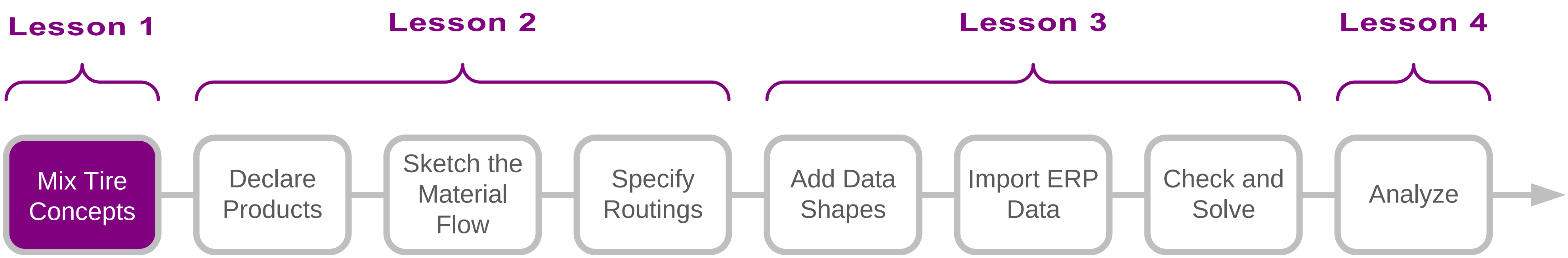
- 423 Tires/Day
- 397 Tires/Day
- 415 Tires/Day
- 406 Tires/Day



In this lesson we covered:

- An overview of the tire production process
- The key concepts tire productions value stream modelling is based on
- Calculations to estimate key metrics

Quick Mix Tire Learning Path



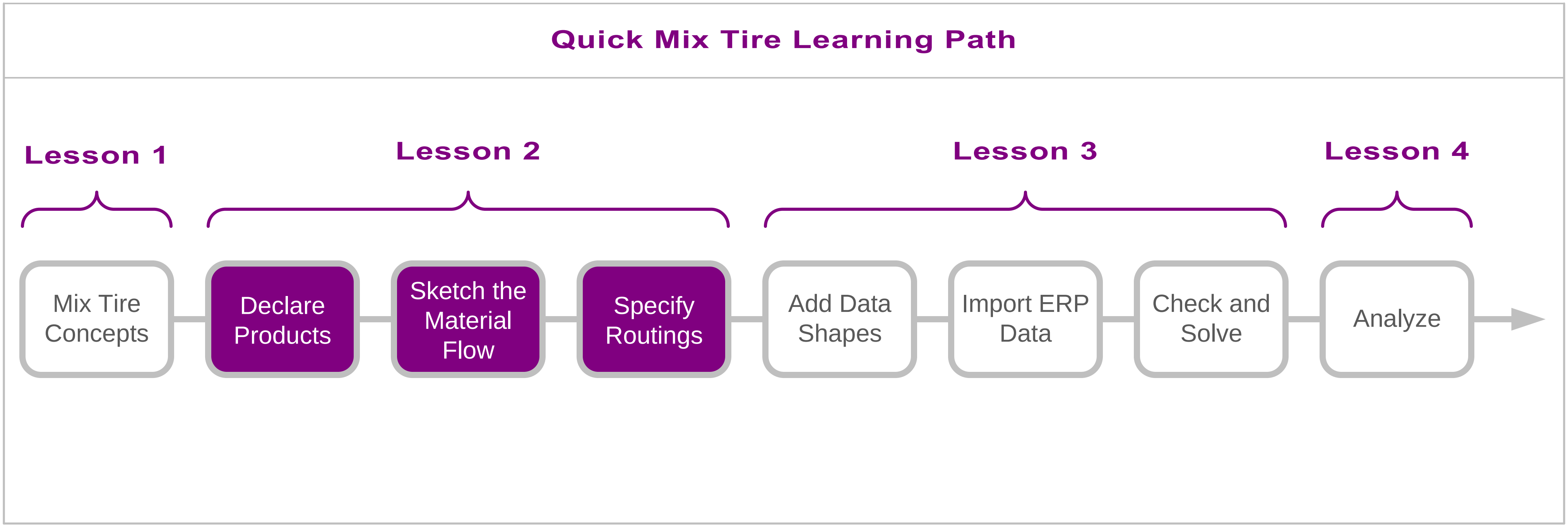
What's next:

- Learn how to initiate a Mix Tire value stream map
- Learn how to declare tires and bobbins, and group them into common routing sets
- Learn how to sketch the material flow and specify routings

Specify the Products, Flow, and Routings

In this lesson, we will introduce a map example which we will start to build here by declaring the products (bobbins and tires), sketch the material flow, and specify the routings for the different products.

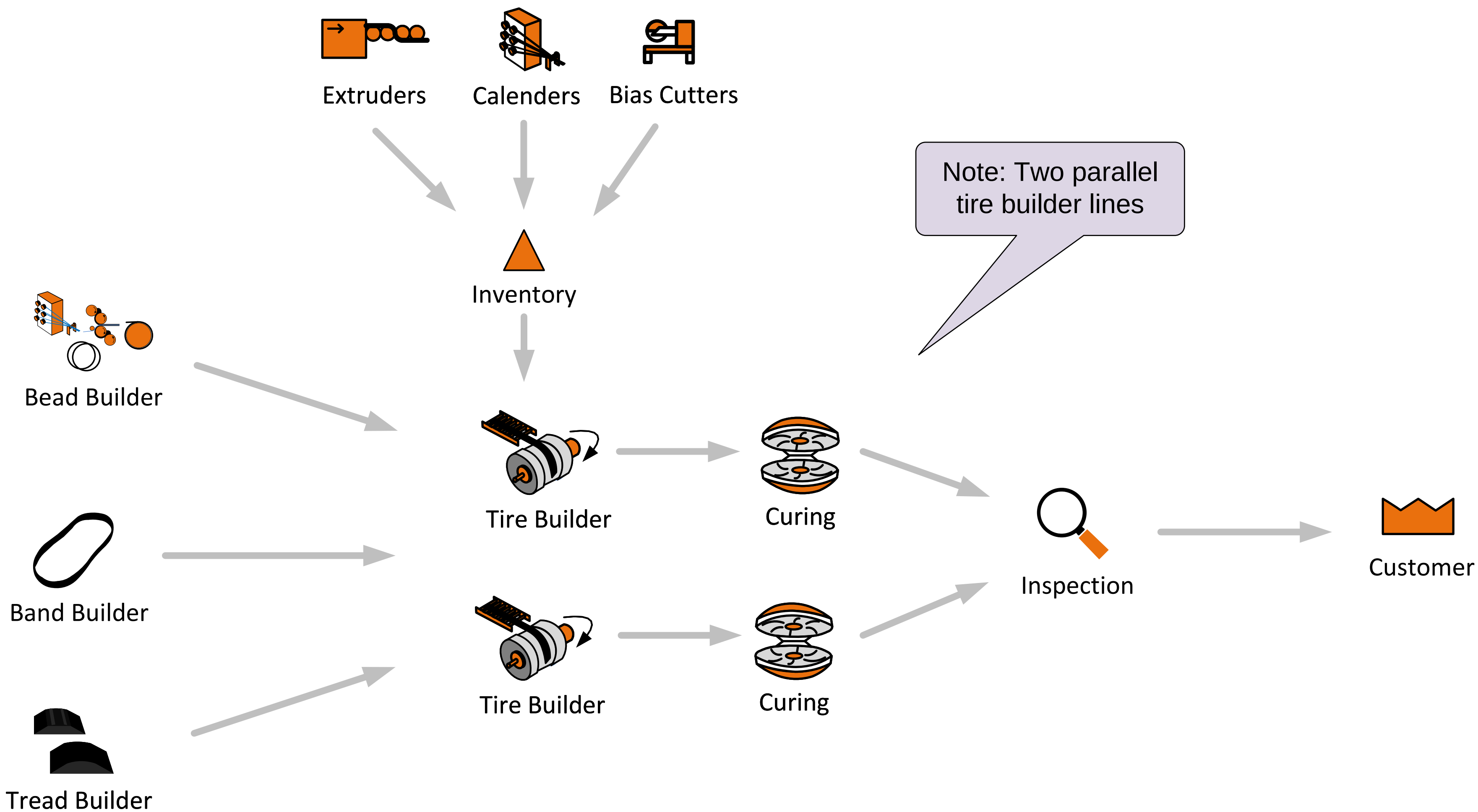
This will give us the basis model to which we can add data values and perform analysis on.



Specify the Products, Flow, and Routings

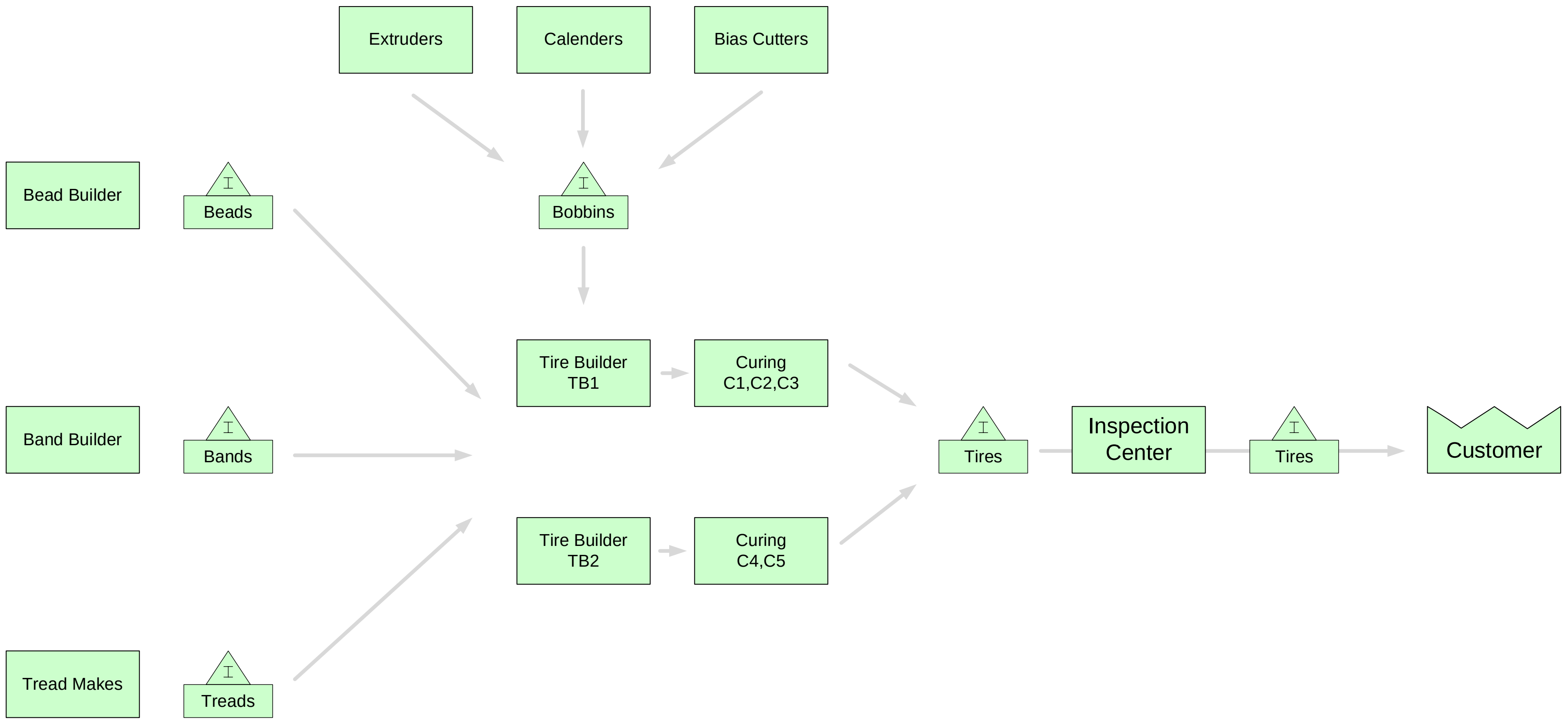
Tire Production Value Stream

This is the simple tire production value stream we will model in this course. The model will include a range of tires with tire-specific demands, inventories, and operational parameters.



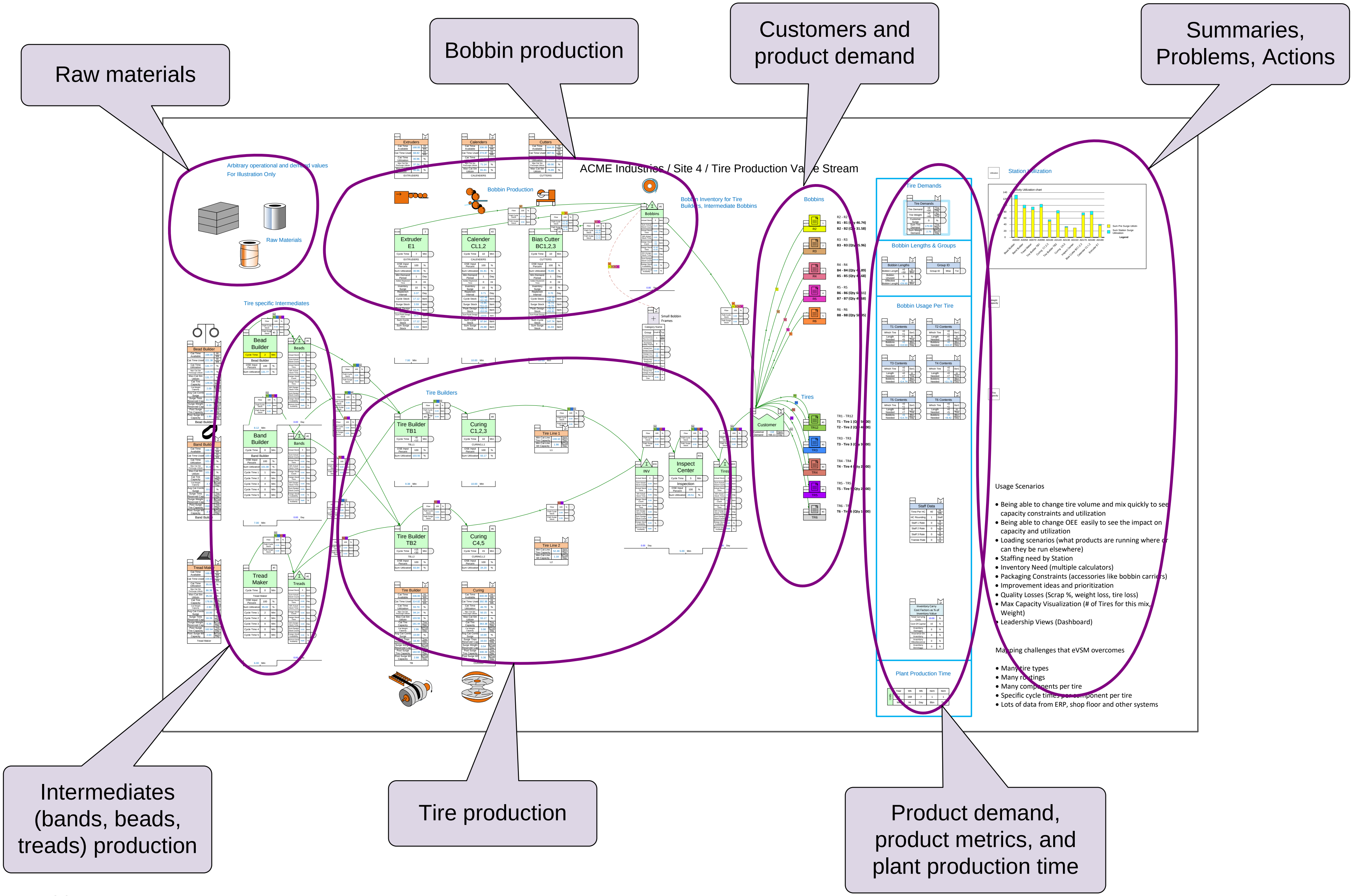
Tire Production Flowchart

Here is an initial sketch of the value stream with Mix Tire centers. Key inventory areas are included.



Tire Production Value Stream Layout

This example map shows the recommended layout for a tire production value stream.



Mix Tire Map Layout Quiz

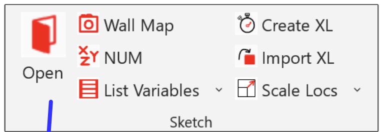
Drop each box on the appropriate part of the map layout below.

- Customers, Tires, Bobbins
- Map summaries, Key Metrics, Actions
- Accessory Production
- Bobbin Factory
- Tire Factory
- Tire/Bobbin Demand and Plant Operating Hours

Map Layout

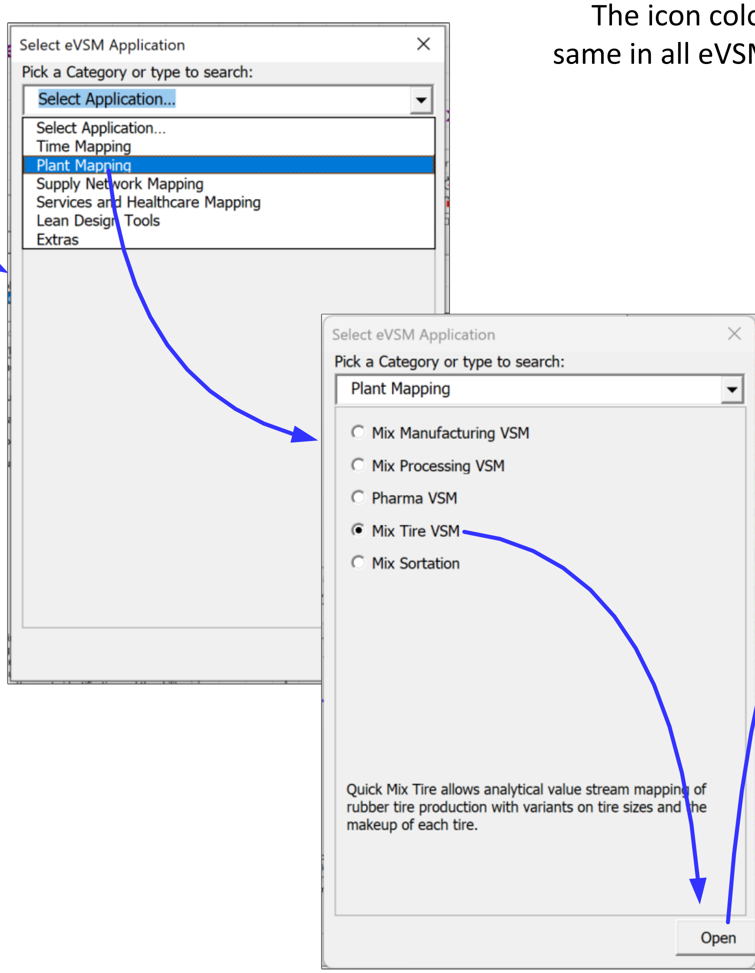
Mix Tire VSM Stencils

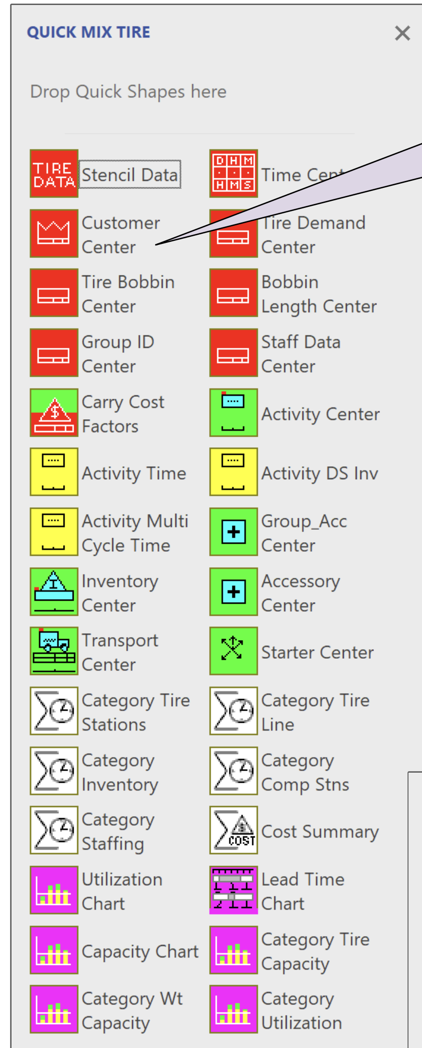
Here is how to open the Mix Tire stencils and a description of the stencils.



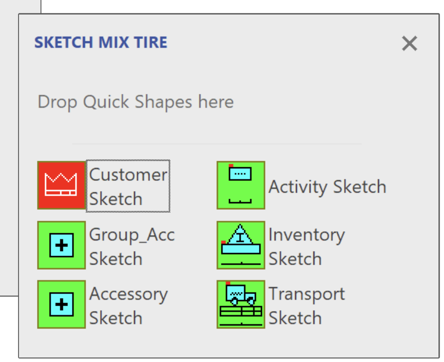
This is the main stencil and contains all icons for the Mix Tire application.

The icon color coding is the same in all eVSM applications, like Mix Time.





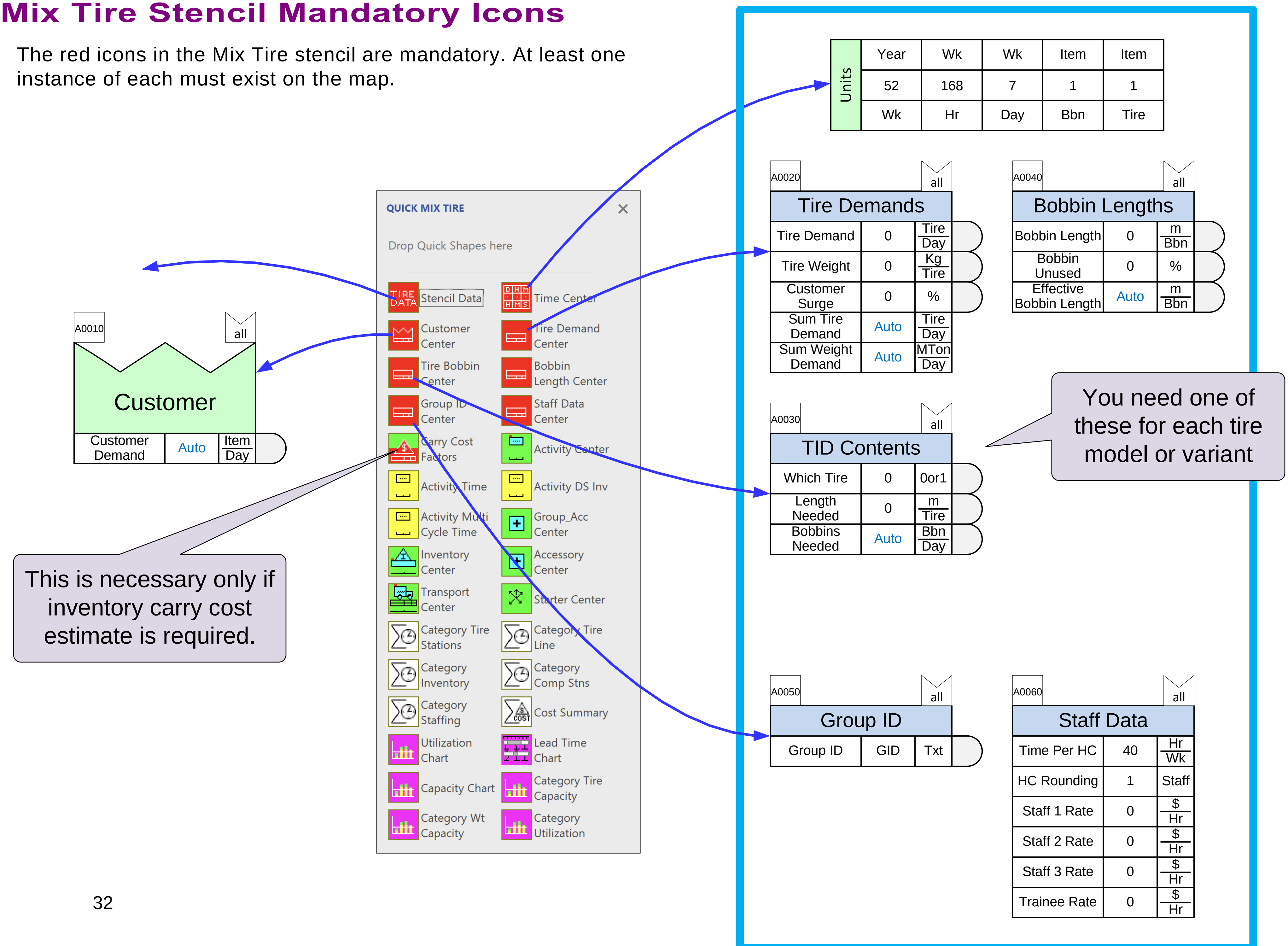
To initiate a new map, drop the customer center on the page.



The sketch stencil contains only the shapes required to create a flowchart (no data fields) of the value stream. Used to capture the flow. Once the flow is established, a single click will add all the default data shapes to the page.

Mix Tire Stencil Mandatory Icons

The red icons in the Mix Tire stencil are mandatory. At least one instance of each must exist on the map.

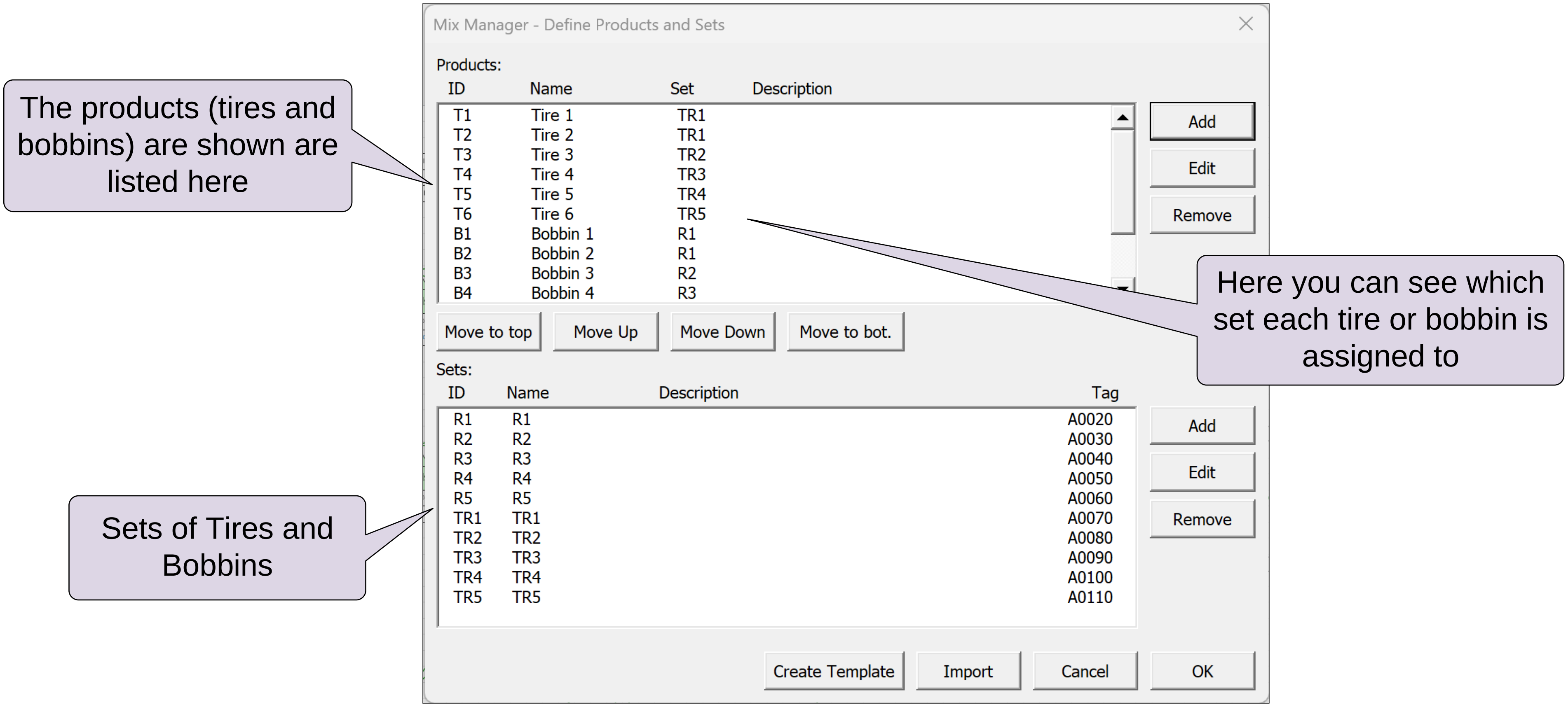


Start a Mix Tire Map

Initiate this page for a Mix Tire map

Declare Products and Sets for the Value Stream

Tires and Bobbins are considered the products. These are grouped into “Sets”. Each set contains, all the products that follow exactly the same route through the production process. The purpose of the sets groupings is to make it easier to manage and visualize long lists of products on the map.

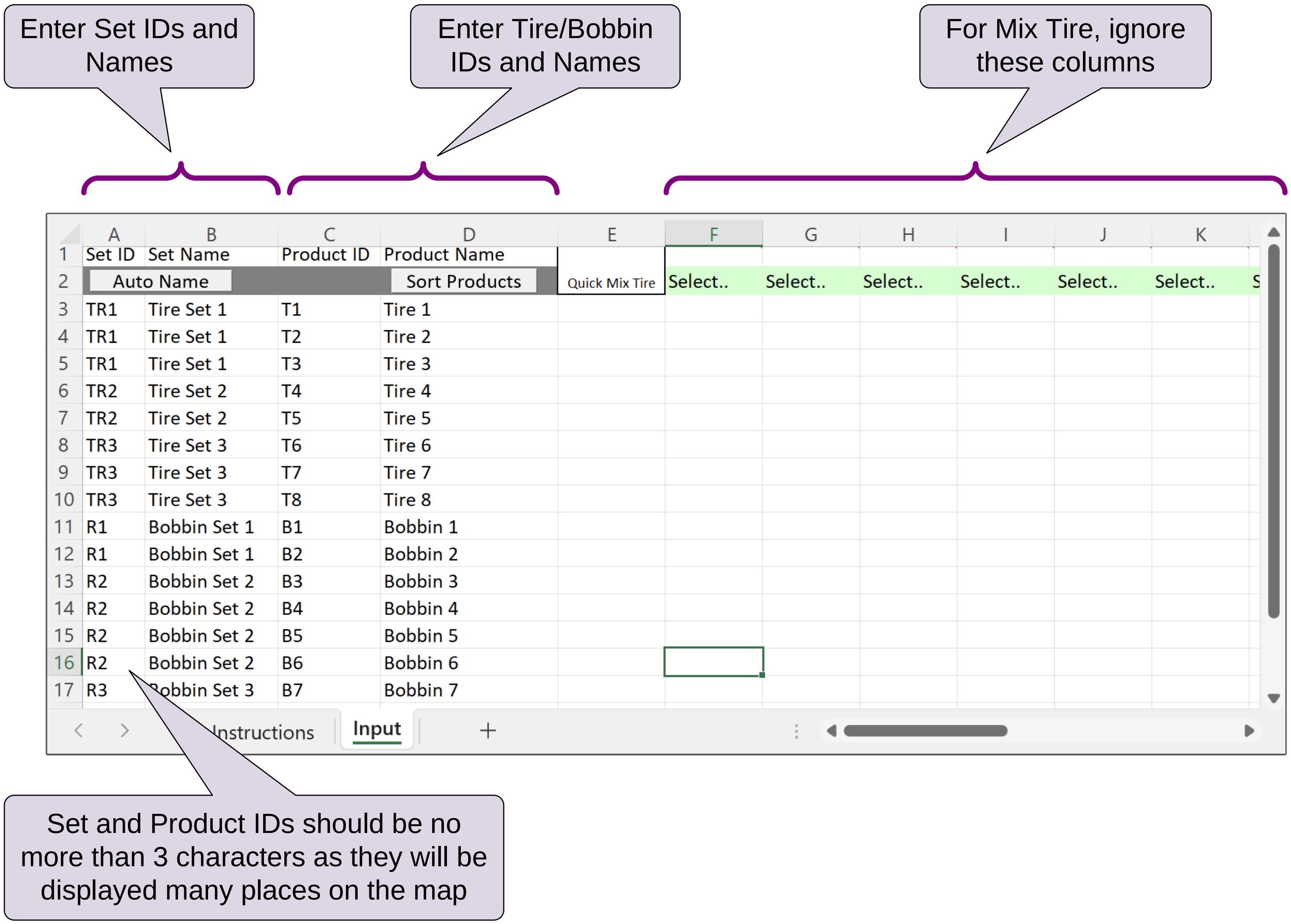


Tires, bobbins, and sets may be declared directly in this dialog as covered in the Mix Time course.

They can also be imported from Excel, as shown on the following pages.

Excel Template for Entering Tires and Bobbins IDs and Names

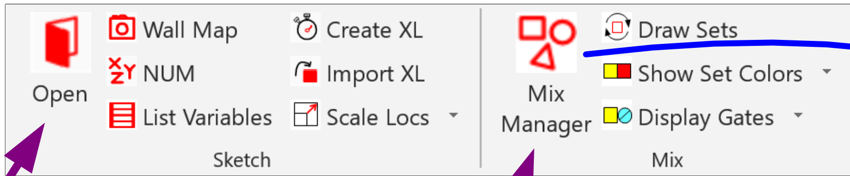
Click the “Create Template” button in the Mix Manager dialog box. This will open an Excel file similar to the one shown here. After entering the tire/bobbin names/IDs in the template file, click the “Import” button in Mix Manager dialog to import the data.

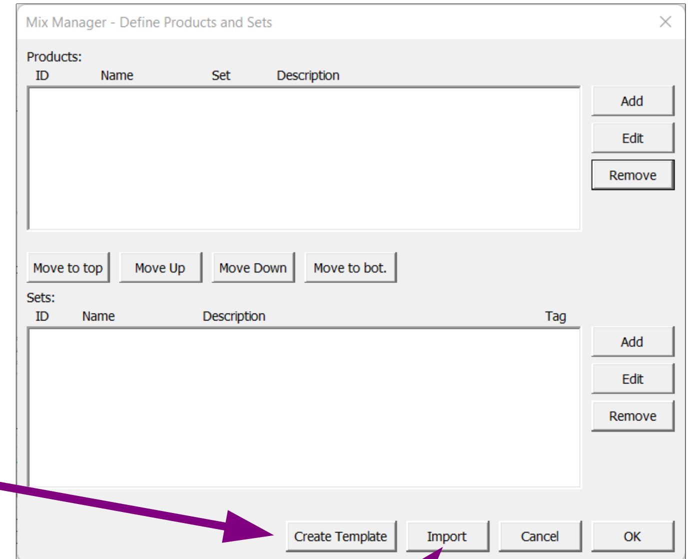


	A	B	C	D	E	F	G	H	I	J	K
1	Set ID	Set Name	Product ID	Product Name							
2	Auto Name			Sort Products	Quick Mix Tire	Select..	Select..	Select..	Select..	Select..	Select..
3	TR1	Tire Set 1	T1	Tire 1							
4	TR1	Tire Set 1	T2	Tire 2							
5	TR1	Tire Set 1	T3	Tire 3							
6	TR2	Tire Set 2	T4	Tire 4							
7	TR2	Tire Set 2	T5	Tire 5							
8	TR3	Tire Set 3	T6	Tire 6							
9	TR3	Tire Set 3	T7	Tire 7							
10	TR3	Tire Set 3	T8	Tire 8							
11	R1	Bobbin Set 1	B1	Bobbin 1							
12	R1	Bobbin Set 1	B2	Bobbin 2							
13	R2	Bobbin Set 2	B3	Bobbin 3							
14	R2	Bobbin Set 2	B4	Bobbin 4							
15	R2	Bobbin Set 2	B5	Bobbin 5							
16	R2	Bobbin Set 2	B6	Bobbin 6							
17	R3	Bobbin Set 3	B7	Bobbin 7							

Note: The purpose of this template is only to import the tires and bobbins to the map. The template is intended for one-time use only. Once imported, you can add/remove/edit directly in the Mix Manager dialog.

Declare Tires, Bobbins, and Sets with Excel





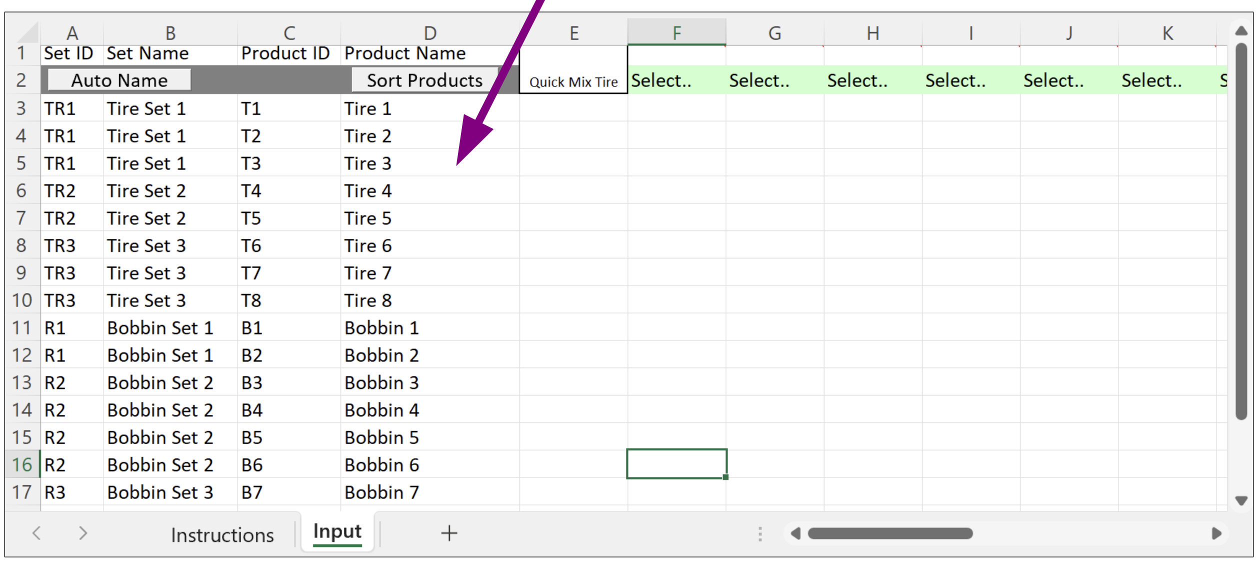
1 Initiate the page for a Mix Tire map as you did in Exercise 2.

2 Open the Mix Manager form

3 Create Template. This will open an Excel template file

4 Populate the Excel template

5 Click “Import” and close the Mix Manager form.



	A	B	C	D	E	F	G	H	I	J	K
1	Set ID	Set Name	Product ID	Product Name							
2	Auto Name			Sort Products	Quick Mix Tire	Select..	Select..	Select..	Select..	Select..	Select..
3	TR1	Tire Set 1	T1	Tire 1							
4	TR1	Tire Set 1	T2	Tire 2							
5	TR1	Tire Set 1	T3	Tire 3							
6	TR2	Tire Set 2	T4	Tire 4							
7	TR2	Tire Set 2	T5	Tire 5							
8	TR3	Tire Set 3	T6	Tire 6							
9	TR3	Tire Set 3	T7	Tire 7							
10	TR3	Tire Set 3	T8	Tire 8							
11	R1	Bobbin Set 1	B1	Bobbin 1							
12	R1	Bobbin Set 1	B2	Bobbin 2							
13	R2	Bobbin Set 2	B3	Bobbin 3							
14	R2	Bobbin Set 2	B4	Bobbin 4							
15	R2	Bobbin Set 2	B5	Bobbin 5							
16	R2	Bobbin Set 2	B6	Bobbin 6							
17	R3	Bobbin Set 3	B7	Bobbin 7							

Sets

Tires & Bobbins

Declare Tires, Bobbins, and Sets with Excel

1. Create Excel template.
2. Fill out the Excel template like this:

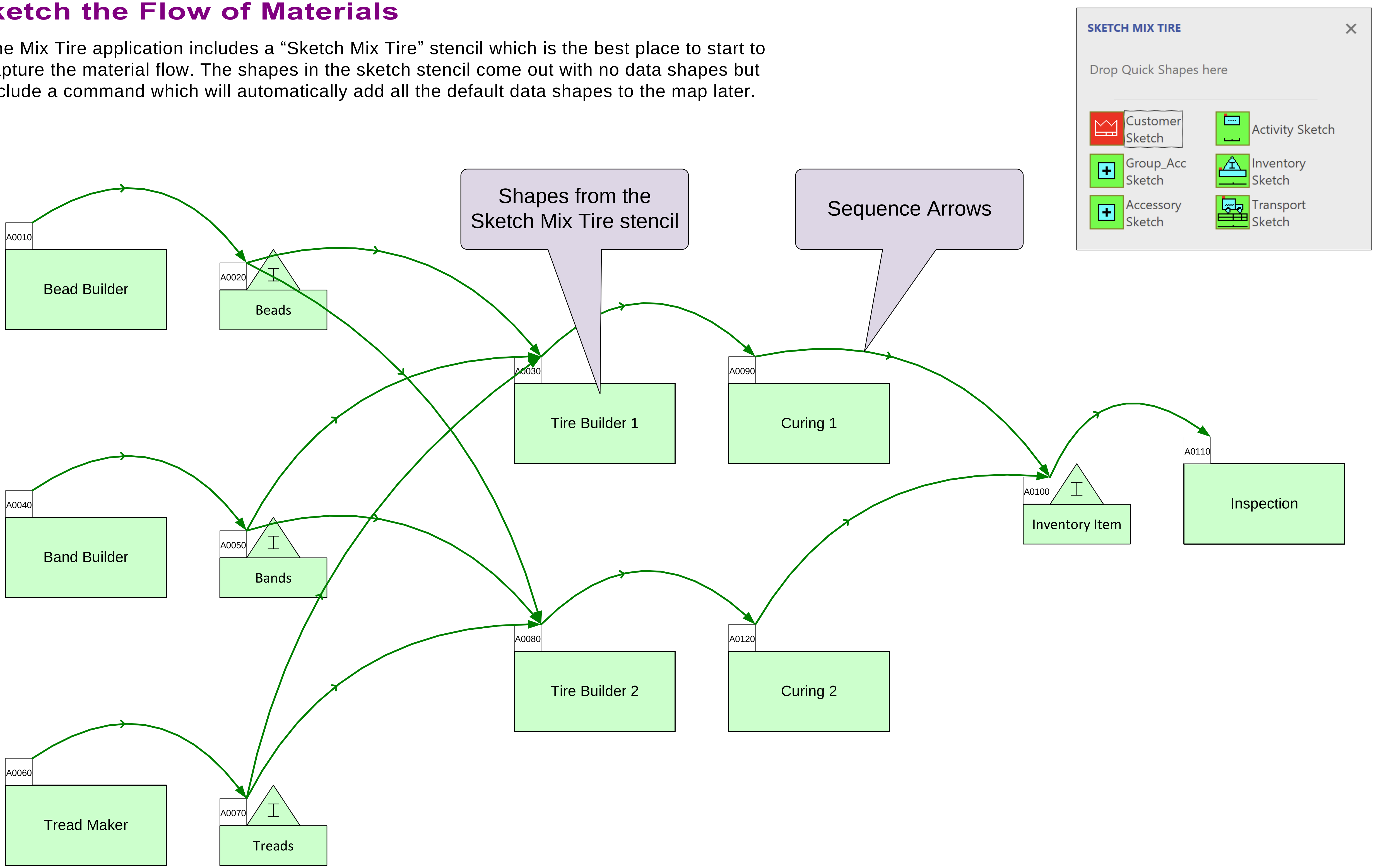
	A	B	C	D	E
1	Set ID	Set Name	Product ID	Product Name	
2	Auto Name			Sort Products	Quick Mix Tire
3	TS1	Tire Set 1	T1	Tire Model 1	
4	TS1	Tire Set 1	T2	Tire Model 2	
5	TS1	Tire Set 1	T3	Tire Model 3	
6	TS2	Tire Set 2	T4	Tire Model 4	
7	TS2	Tire Set 2	T5	Tire Model 5	
8	BS1	Bobbin Set 1	B1	Bobbin 1	
9	BS2	Bobbin Set 2	B2	Bobbin 2	
10	BS2	Bobbin Set 2	B3	Bobbin 3	

Instructions | Input

3. Import the Excel data. This will automatically draw the 4 sets beyond the right edge of the page.
4. Grade It.

Sketch the Flow of Materials

The Mix Tire application includes a “Sketch Mix Tire” stencil which is the best place to start to capture the material flow. The shapes in the sketch stencil come out with no data shapes but include a command which will automatically add all the default data shapes to the map later.



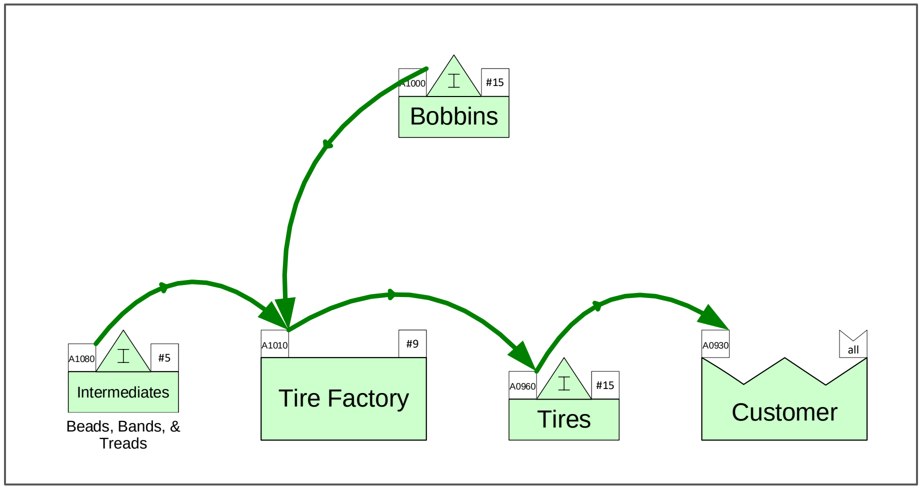
Sketch Exercise

This page has already been initiated for a Mix Tire map. Drag out shapes from the Sketch Mix Tire stencil that match the ghost shapes below. Overlay the new shapes on the ghost shapes and name them as shown.

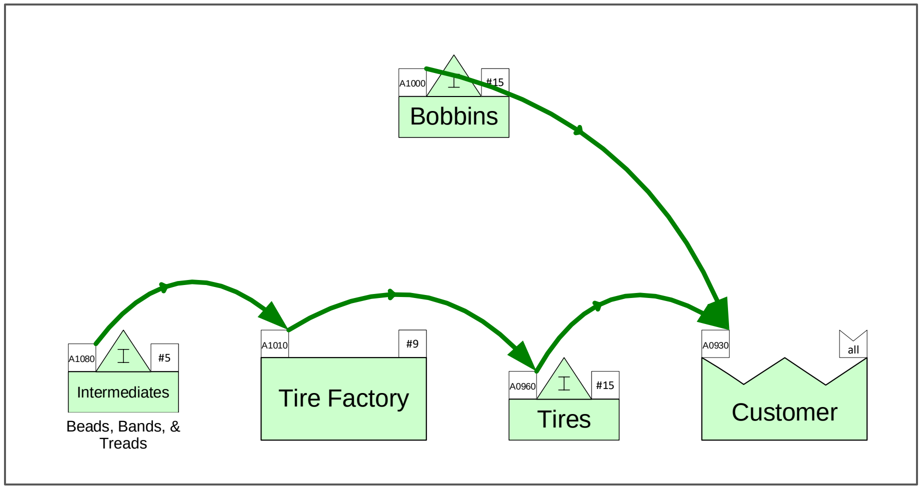
Special Note for Bobbin Factory Sequencing

The material flow through the value stream is specified with sequence arrows. The tire plant receives supplies from the bobbins factory. As a result, it would seem natural to link the bobbin inventory to the tire factory. We must, however, make an exception in this case and connect the bobbins inventory directly to the customer center. This is required for bobbin demand calculations to function properly.

Incorrect

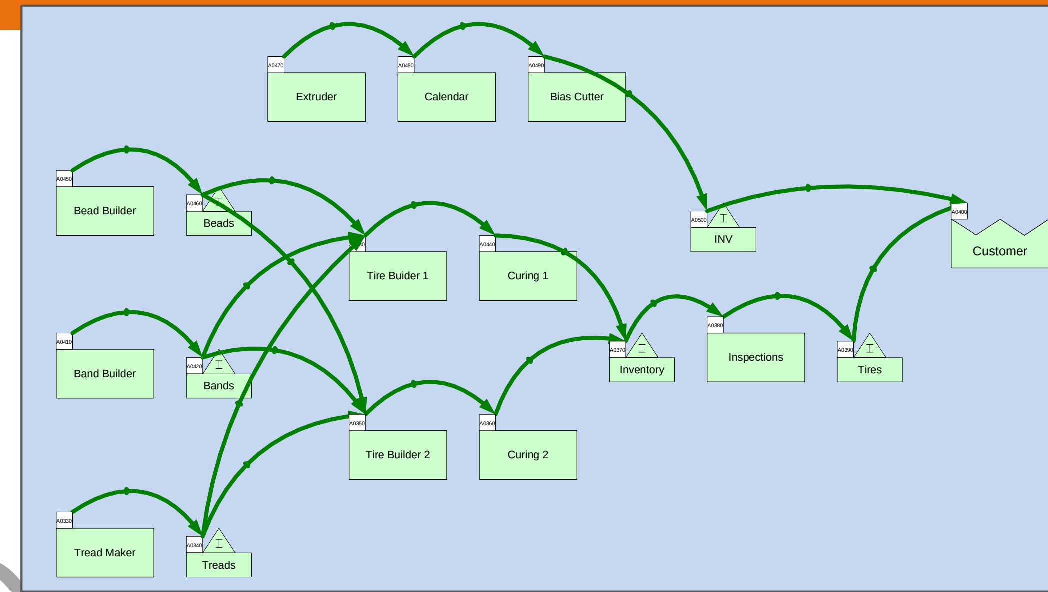


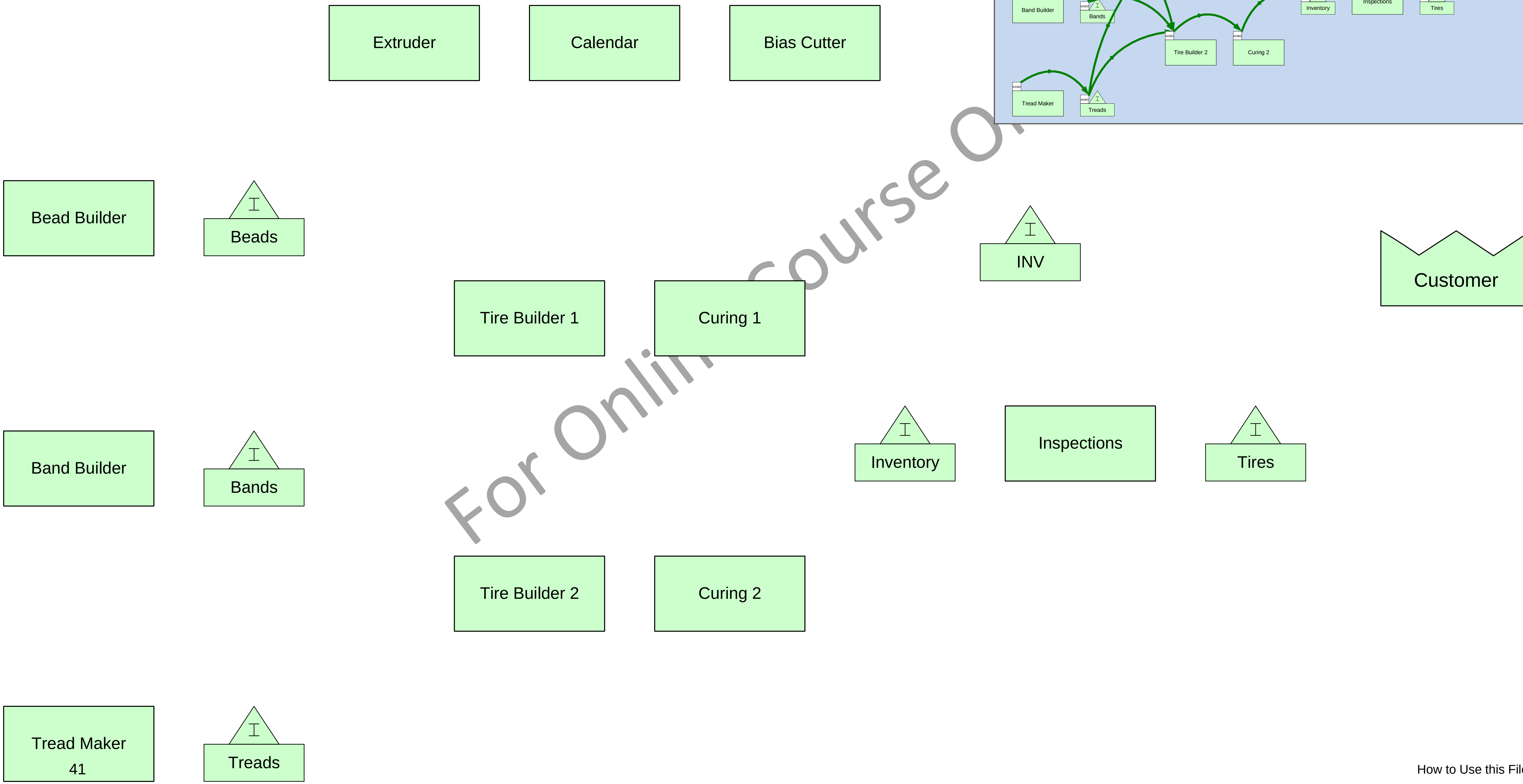
Correct



Add Sequence Arrows

Add sequence arrows to the map as shown in the thumbnail image.



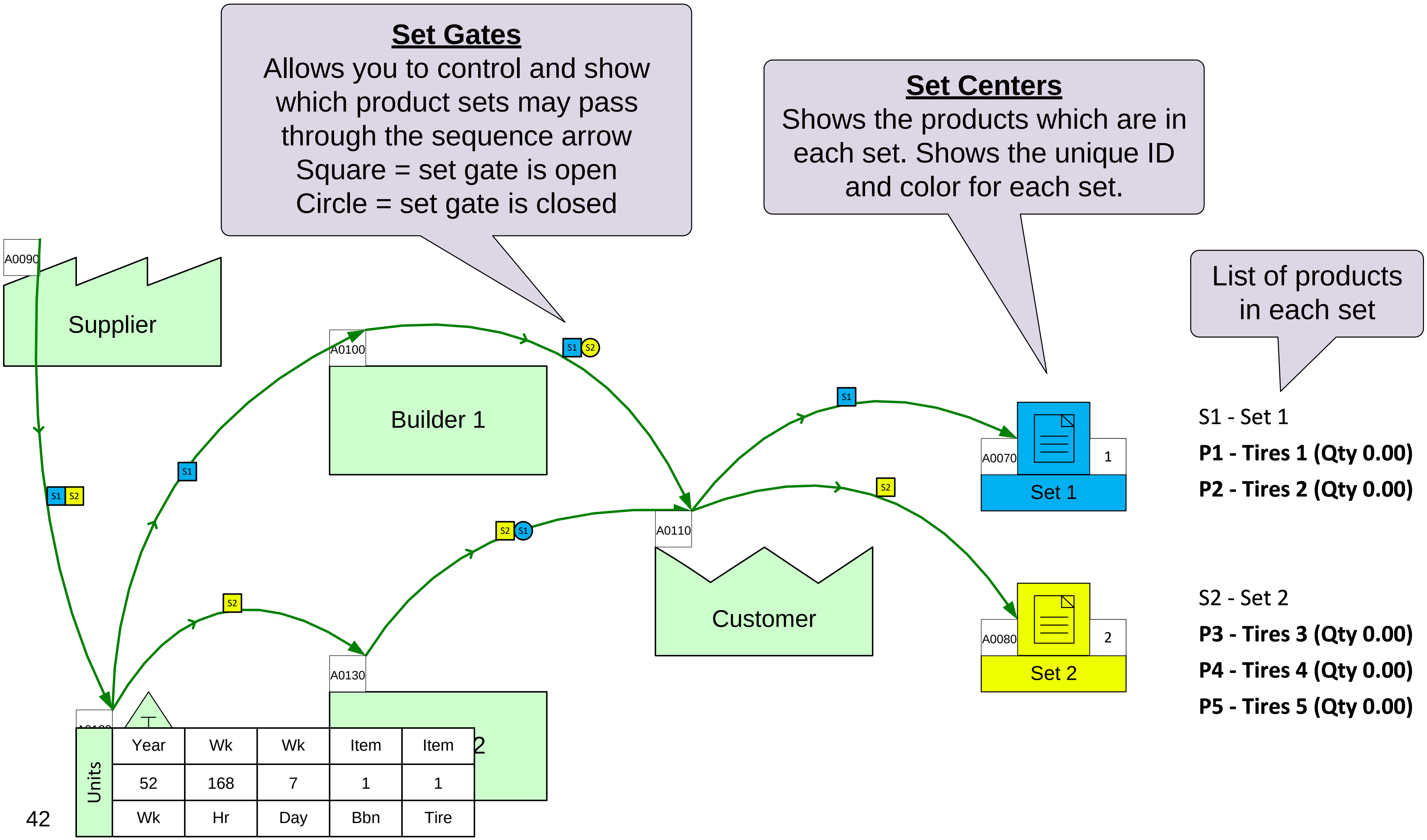


Refresher: Products, Sets, and Routings

Remember: A set is a group of products that go through exactly the same routing. Setting up routings was covered in lesson 2 in the Mix Time course. To review, download a fresh copy of the course notes from: https://evsm.com/el/v12/file=eL_MixTime.12Notes.pdf.

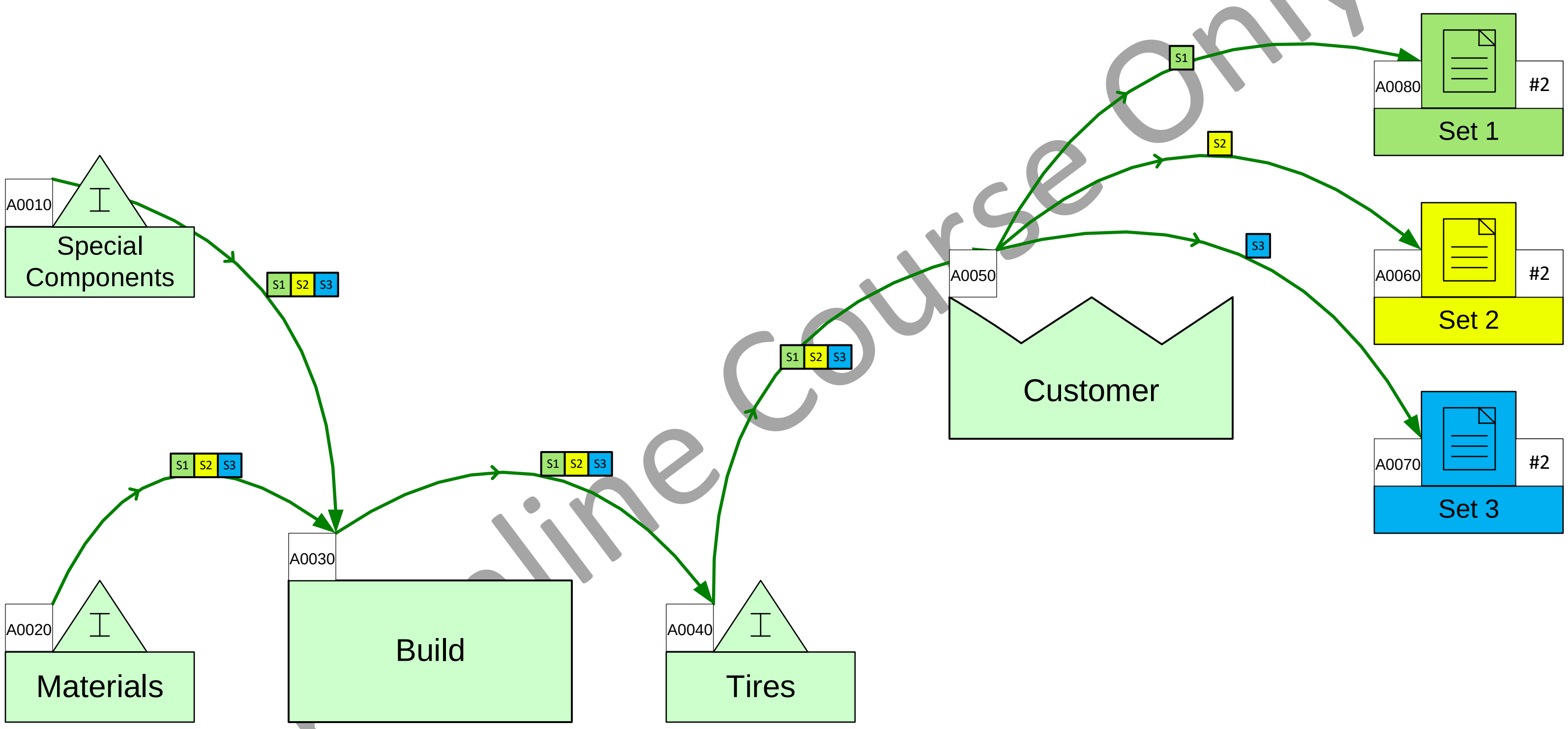
Example

In this example, the blue products (S1) go through Builder 1 while the yellow products (S2) go through Builder 2.



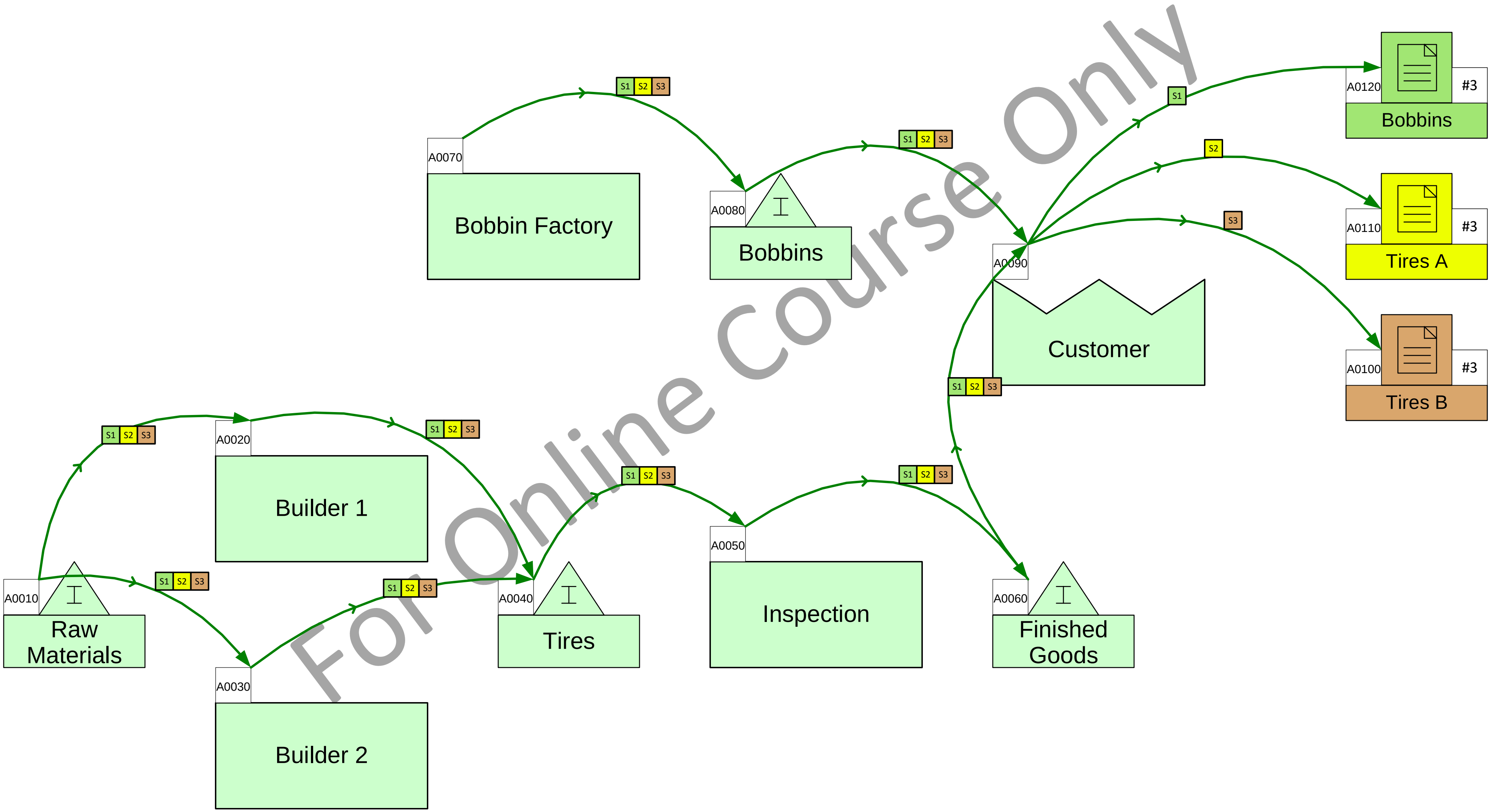
Set Routings Exercise 1

In this model, only Set 3 requires the "Special Components". Adjust the set gates to specify this in the model and then click Grade It.



Set Routings Exercise 2

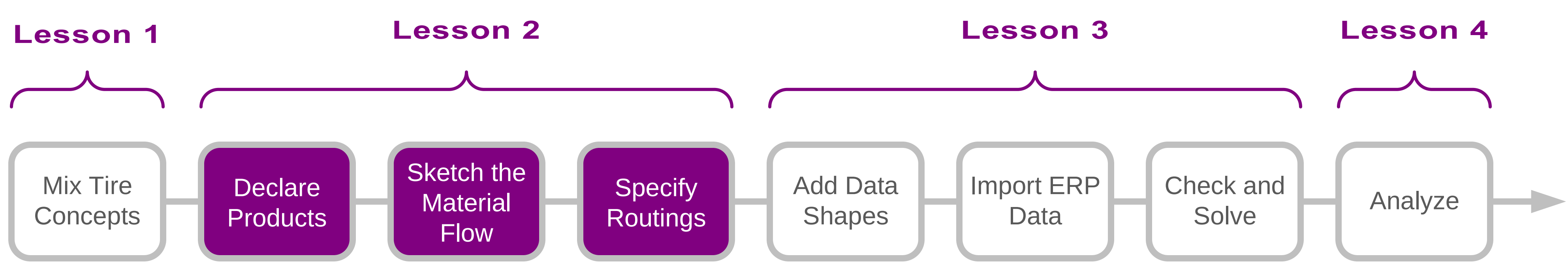
In the model we need to make S1 follow the bobbins factory route, S2 through Builder 1, and S3 through Builder 2. To make the set gates visible, click Display Gates and then tweak them as needed.



You learned that:

- How initiate a Mix Tire value stream map
- How to declare the tires and bobbins made in the value stream
- How to use the Sketch Mix Tire stencil to capture the flow of materials
- How to specify the routings for the different products through the value stream

Quick Mix Tire Learning Path



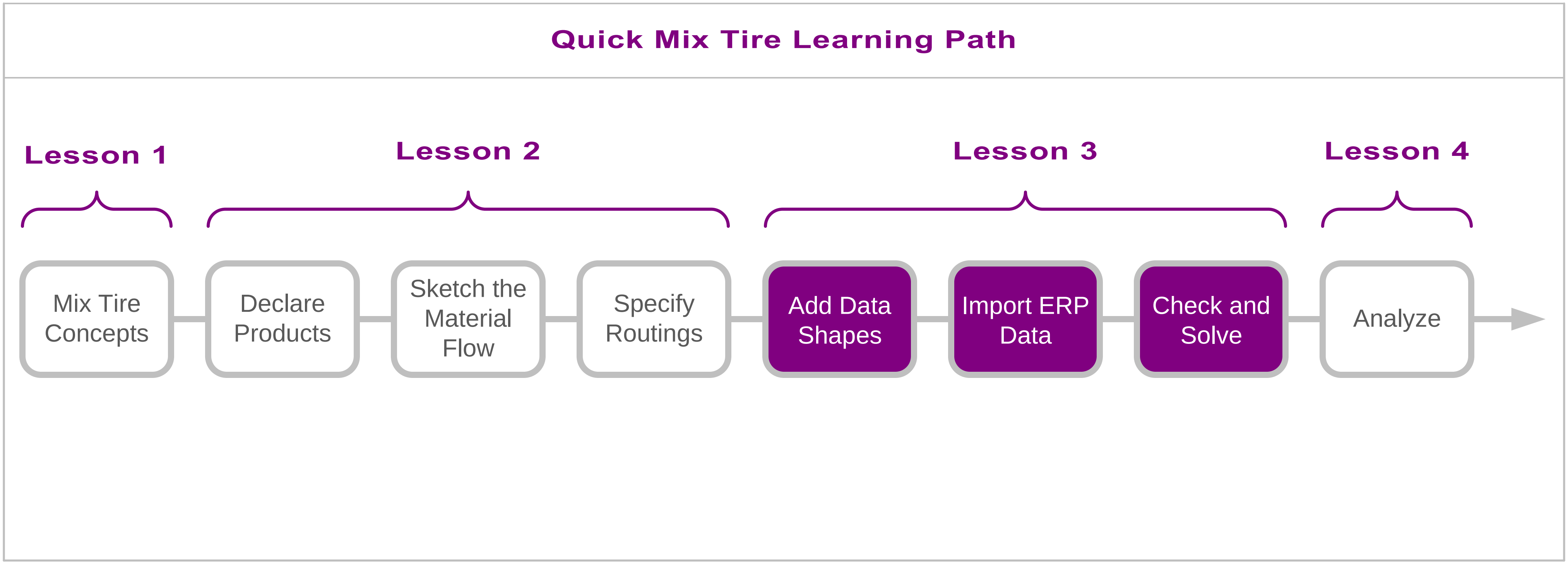
What's next:

- Learn to add data shapes to the sketch of the map
- How to import ERP data to the map
- How to check and solve the value stream model

Add Data and Solve the Model

In the last lesson you learnt to build a tire production value stream sketch and then to specify material flow and routings.

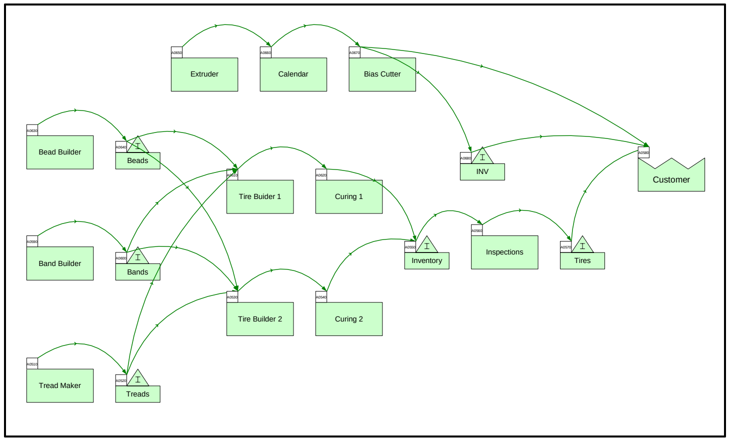
In this lesson, we will introduce a map example which we will start to build here by declaring the products (bobbins and tires), sketch the material flow, and specify the routings for the different products.

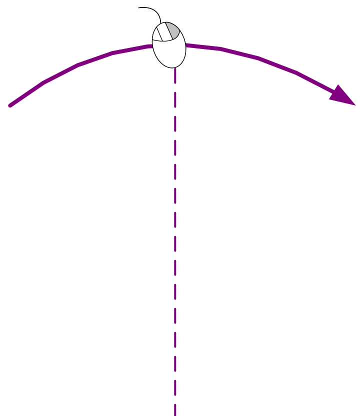


Add Data Shapes to the Sketch

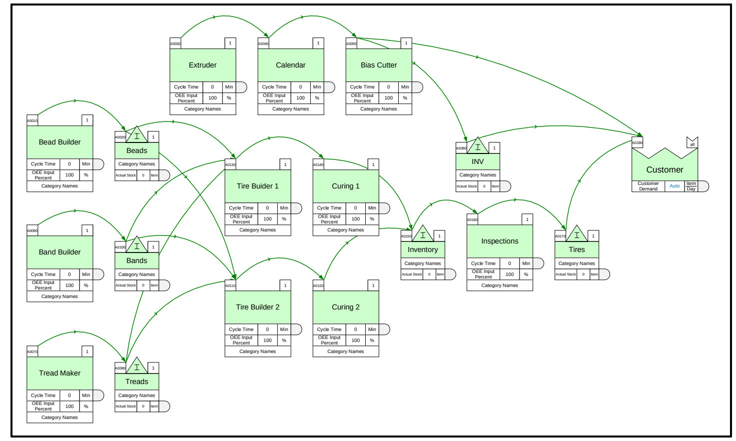
If the flow has been drawn with the Sketch Mix Tire stencil, then you can automatically turn the flowchart into a data based value stream model. Just right-mouse click on any Sketch Mix Transactional shape on the page and use the Add All Data commands.

Sketch Map





After "Add all data"



Right-mouse menu, available on all center parent shapes (green shapes)

.
.
.
.

Add all data (shape)
Add all data (page)

Remove all data (shape)
Remove all data (page)

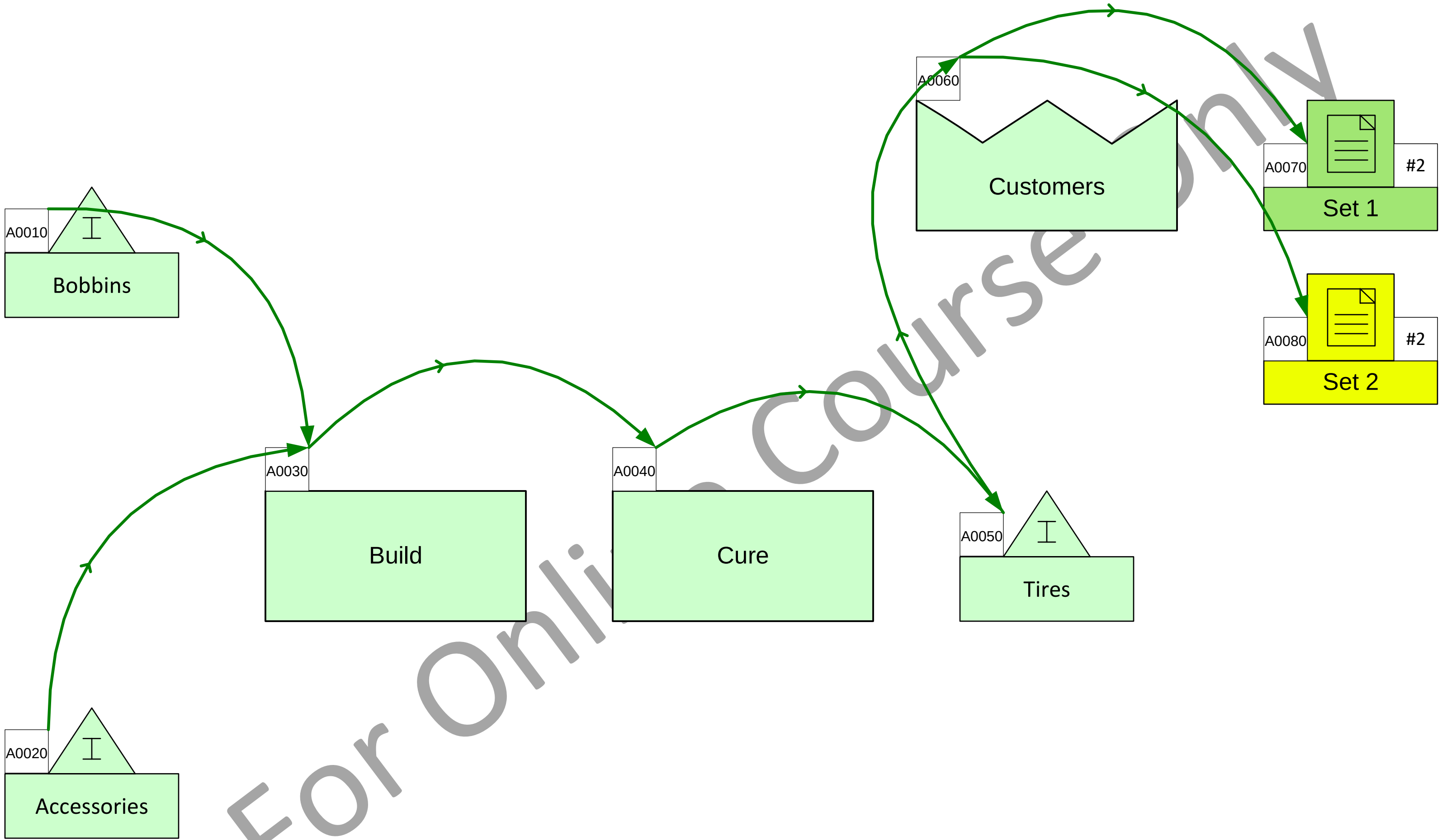
Click this command to automatically add data shapes to all centers on the page.

Data shapes can be removed too. Note that this will delete any entered and calculated values.

The above and the general topic of working with data shapes was covered in lesson 4 of the Mix Time Mapping course. If you need to review, download a fresh copy of the course notes here https://evsm.com/el/v12/file=eL_MixTime.12Notes.pdf

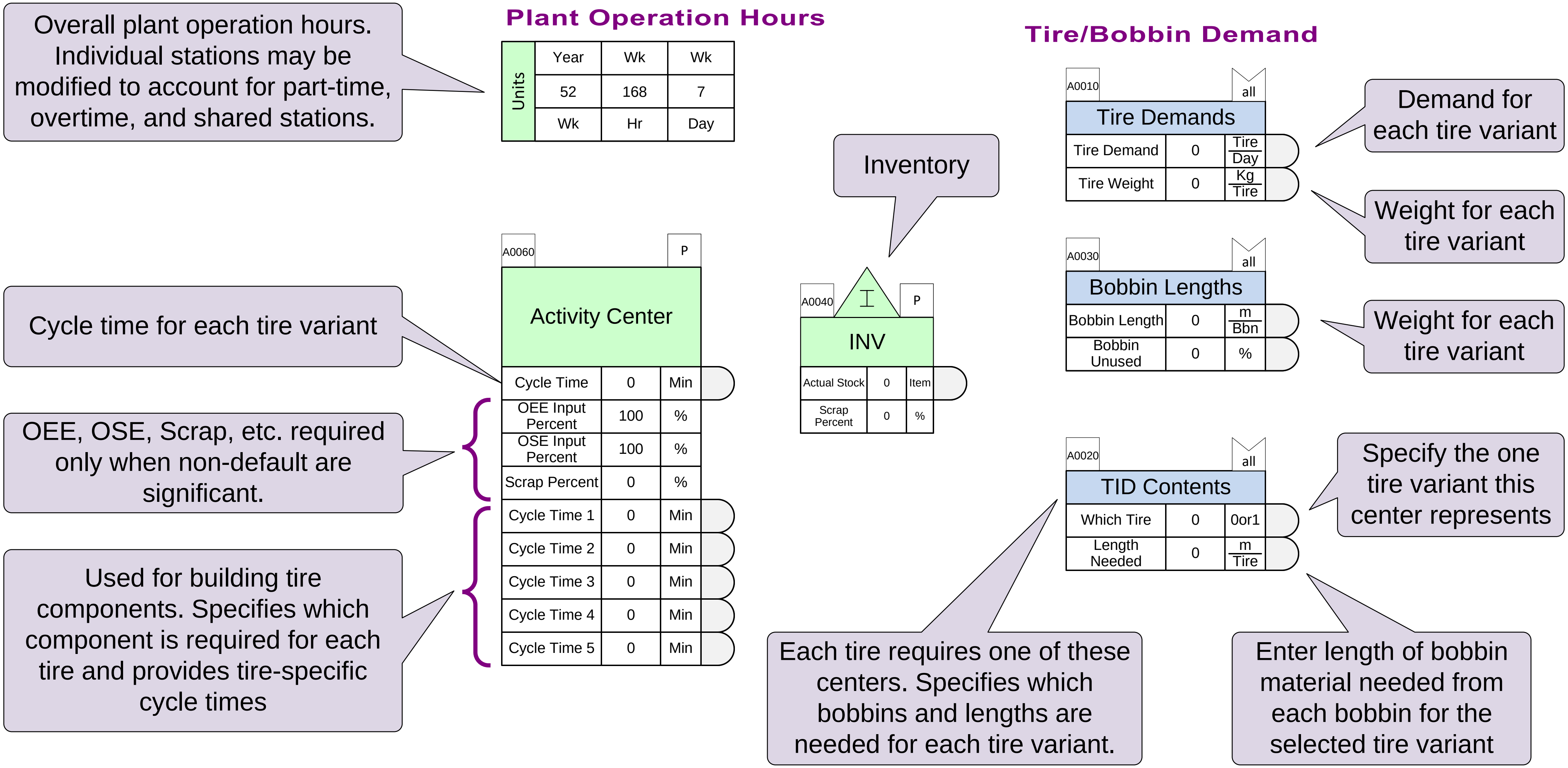
Add Data Shapes to this Sketch

This sketch was drawn with the Sketch Mix Tire stencil. Use the “Add all data (page)” command to automatically add the default data shapes.



What Data is Needed to Create a Tire Map?

Along with the list of tires and bobbins, activity flow sequence, and routings, the following data is necessary to build a value stream model which can then be used to explore what-if scenarios.



Optional calculations, such as staffing, costs, transport, requires additional data.

Map Data Entry Methods

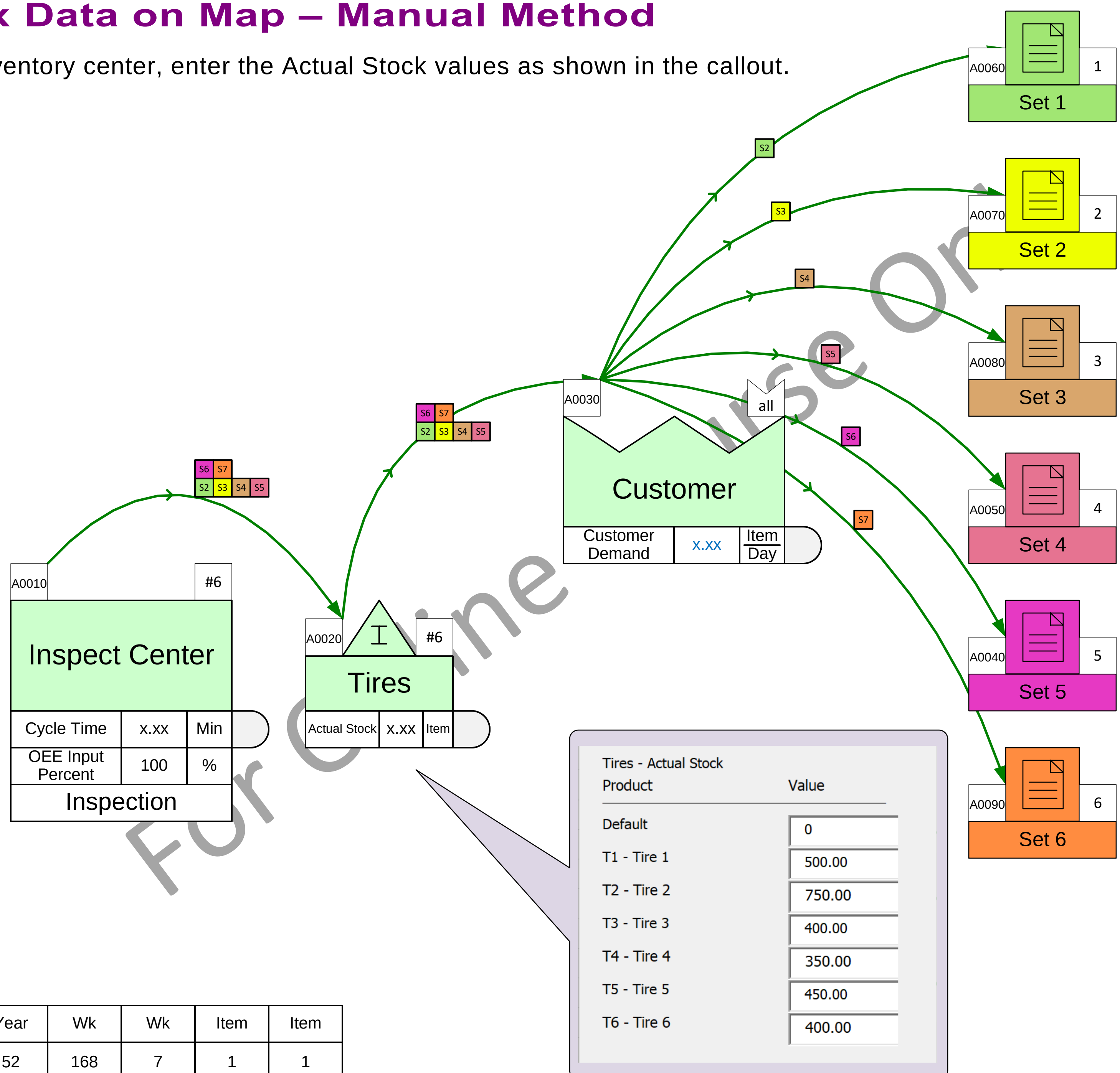
There are several ways to input data on the map.

Method	Pros	Cons
1. Direct entry on the map Simply click on data shapes and type the data values.	No learning involved	Gets very tedious for large amounts of input data.
2. CreateXL/ImportXL When the map is ready, click Create XL to generate an Excel file which represents all the input data for the map. Enter data values in Excel and then click Import XL to populate the map.	Allows some automation and checking of data before importing to the map.	Data has to be in an eVSM-specific format. All data must be in a single worksheet.
3. ERP Import See what this can do and how it works in the ERP Import course.	Data can be dumped from the ERP system into Excel in any logical format. Data may be organized across multiple worksheets and files. User-defined rules convert into eVSM format.	Need to go through a 2 hour ERP Data Import course to learn.

Methods 1 and 2 were coved in the Mix Time course. Current course notes at https://evsm.com/el/v12/file=eL_MixTime.12Notes.pdf

Enter Mix Data on Map – Manual Method

In the Tires inventory center, enter the Actual Stock values as shown in the callout.



S2 - Set 1

T1 - Tire 1 (Qty 0.00)

S3 - Set 2

T2 - Tire 2 (Qty 0)

S4 - Set 3

T3 - Tire 3 (Qty 0)

S5 - Set 4

T4 - Tire 4 (Qty 0)

S6 - Set 5

T5 - Tire 5 (Qty 0)

S7 - Set 6

T6 - Tire 6 (Qty 0)

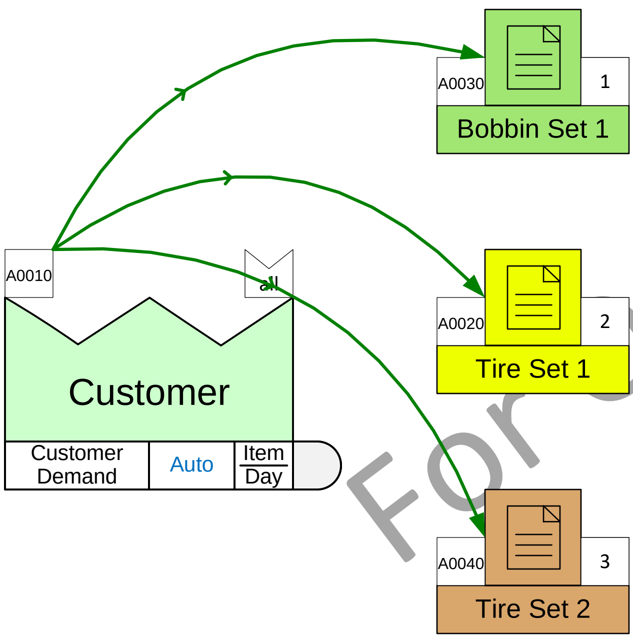
Units	Year	Wk	Wk	Item	Item
	52	168	7	1	1
	Wk	Hr	Day	Bbn	Tire

Centers for Tire Demand and Bobbin Data

This page already has Tires and Bobbins declared. You need to add centers in the pane on the right to enter the following data.

- Tire Demand & Weight
- Bobbin Length & Unused %
- Bobbin Length needed for each tire

Actual data values will not be entered but you must ensure the correct number of centers.



BS1 - Bobbin Set 1

B1 - Bobbin 1 (Qty 0.00)

B2 - Bobbin 2 (Qty 0.00)

B3 - Bobbin 3 (Qty 0.00)

B4 - Bobbin 4 (Qty 0.00)

TS1 - Tire Set 1

T1 - Tire 1 (Qty 0.00)

T2 - Tire 2 (Qty 0.00)

TS2 - Tire Set 2

T3 - Tire 3 (Qty 0.00)

Units	Year	Wk	Wk	Item	Item
	52	168	7	1	1
	Wk	Hr	Day	Bbn	Tire

Demand Data and Related Input

Here is an example of the data required to start a tire value stream model.

Bobbin sizes. Length of each bobbin.

List of bobbins

Percentage of bobbin not used.

List of tire SKUs

			B1	B2	B3	B4	B5	B6	B7	B8
Bobbin Length (m/Bbn)			22	28	30	32	25	22	25	28
Bobbin Unused (%)			5	5	5	5	5	5	5	5
	Demand (Tire/Day)	Weight Kg/Tire								
T1	40	20	3.2	5.3		5.1		4.2	3.8	3.7
T2	150	25	4.2	2.3	4.1	2.2		3.8	3.8	3.7
T3	75	22	2.8		3.6	2.8	3.2		3.8	3.3
T4	90	30		3.4	3.6	4.2		3.2	3.8	2.8
T5	120	28	2.4	3.4		3.2	4.1	2.7	3.8	2.8

Customer demand for each SKU

Weight of each SKU

Which bobbin is required for each SKU

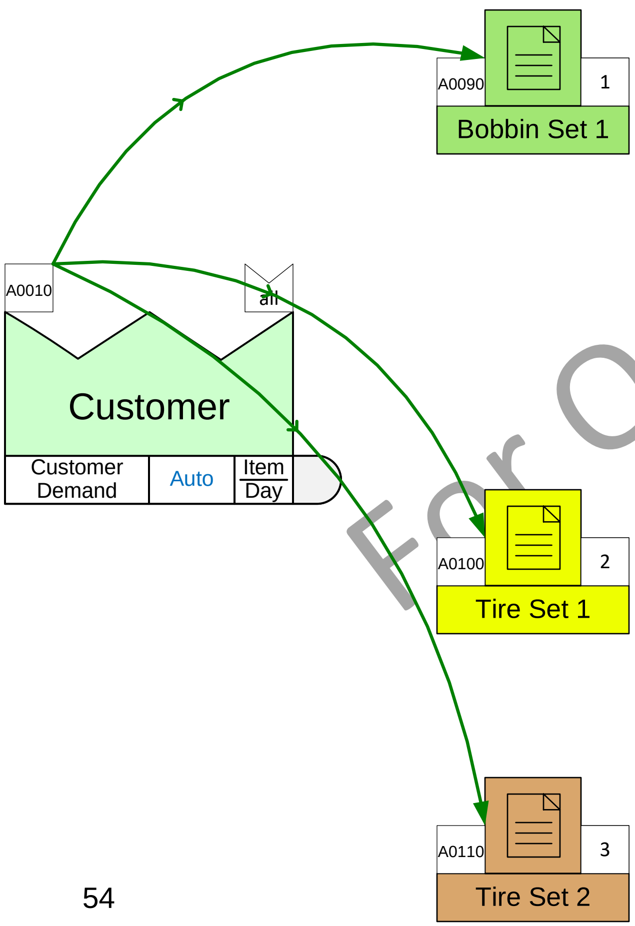
Bobbin material length required for each SKU

Enter Demand and Bobbin Data

In this exercise, the list of tires and bobbins has already been entered in the Mix Manager. You need to enter these data values in the centers on the right for:

- Tire Demand
- Tire Weight
- Bobbin Length
- Bobbin Unused (5% for all bobbins)
- Length Needed (bobbin material length needs for each tire SKU)

All data values are in the grid on the previous page. A copy is also pasted above the top of the current page.



BS1 - Bobbin Set 1

B1 - Bobbin 1 (Qty 0.00)

B2 - Bobbin 2 (Qty 0.00)

B3 - Bobbin 3 (Qty 0.00)

B4 - Bobbin 4 (Qty 0.00)

B5 - Bobbin 5 (Qty 0.00)

B6 - Bobbin 6 (Qty 0.00)

B7 - Bobbin 7 (Qty 0.00)

B8 - Bobbin 8 (Qty 0.00)

TS1 - Tire Set 1

T1 - Tire 1 (Qty 0.00)

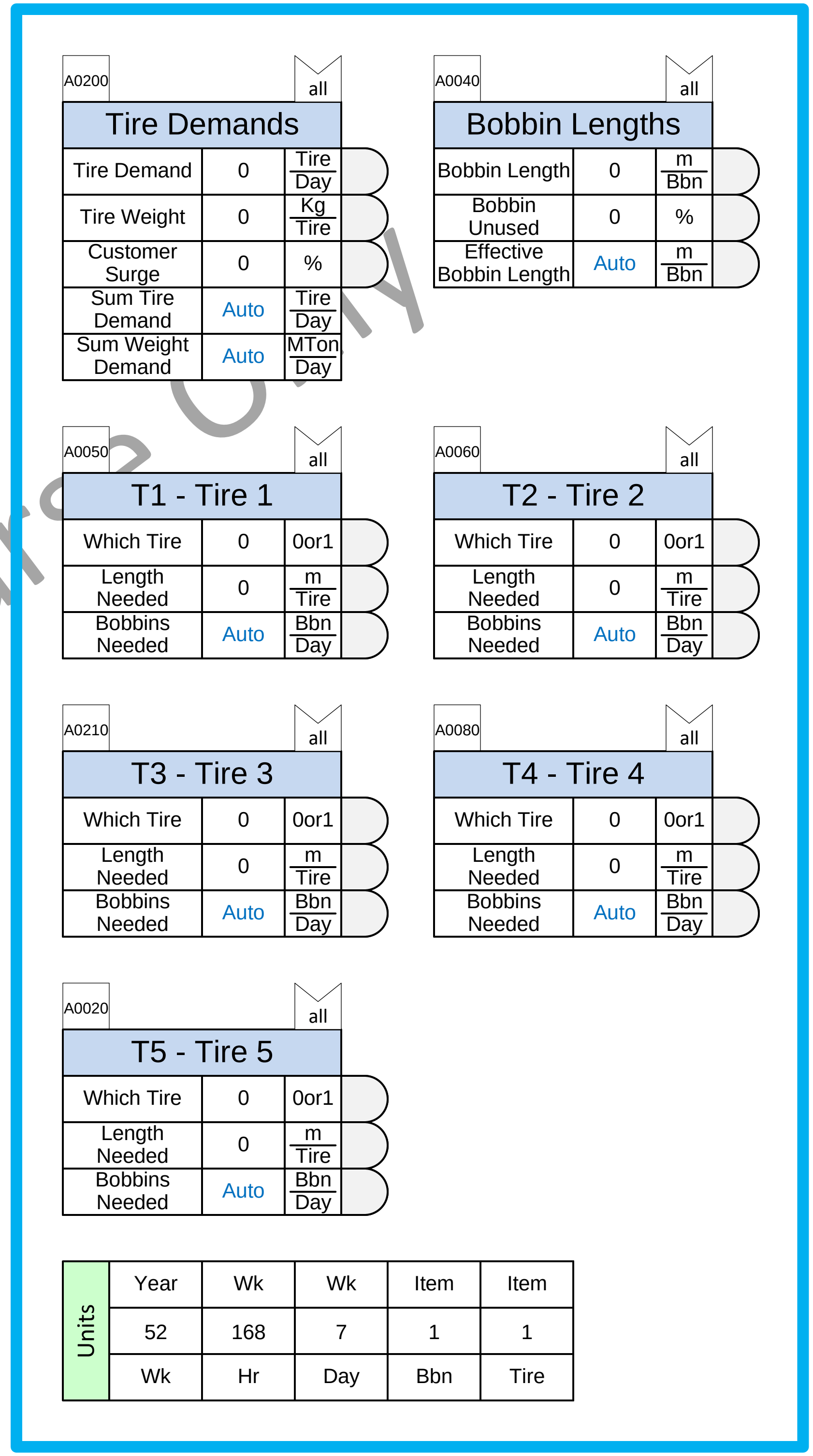
T2 - Tire 2 (Qty 0.00)

T3 - Tire 3 (Qty 0.00)

TS2 - Tire Set 2

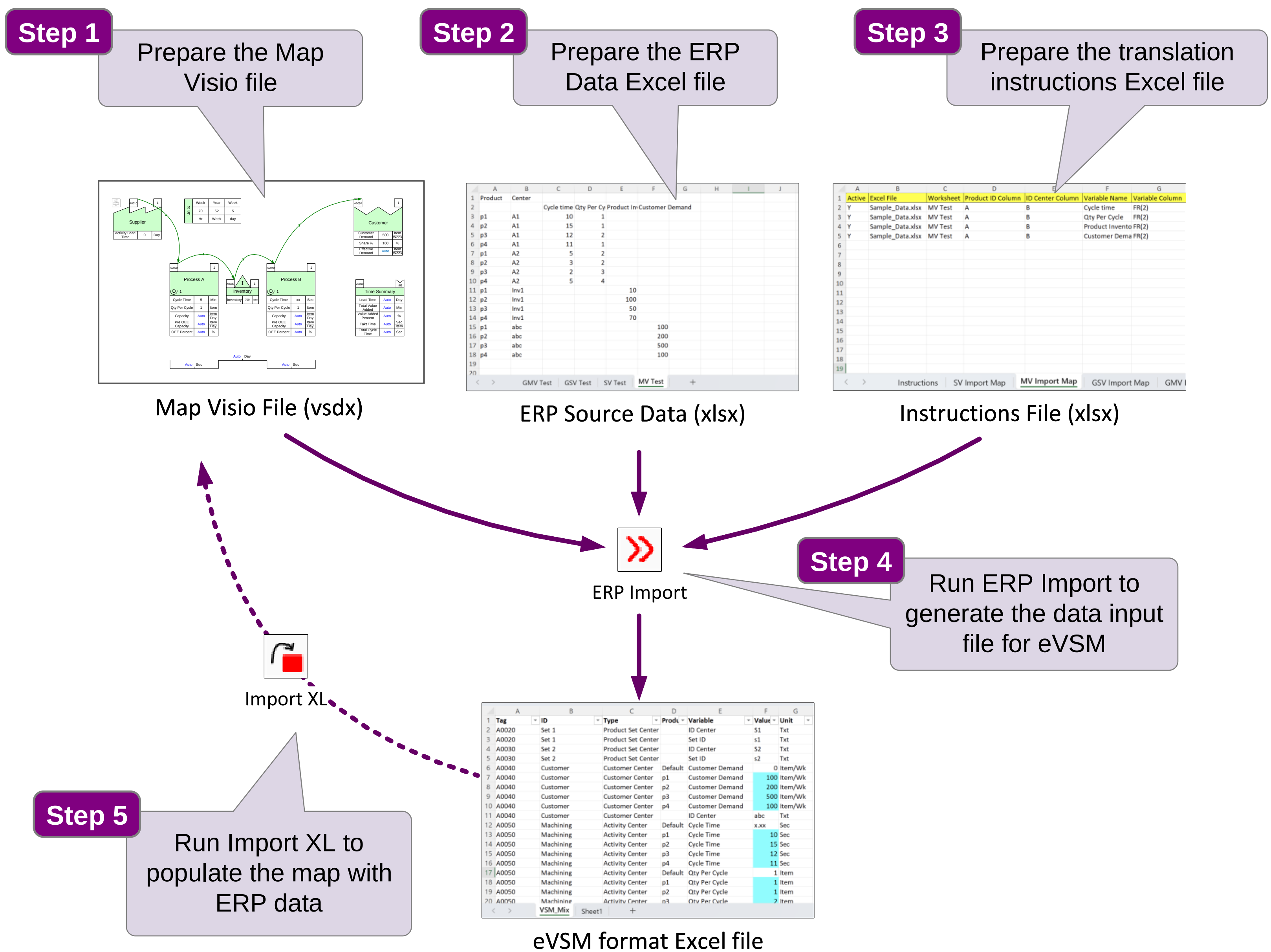
T4 - Tire 4 (Qty 0.00)

T5 - Tire 5 (Qty 0.00)



ERP Data Import Refresher

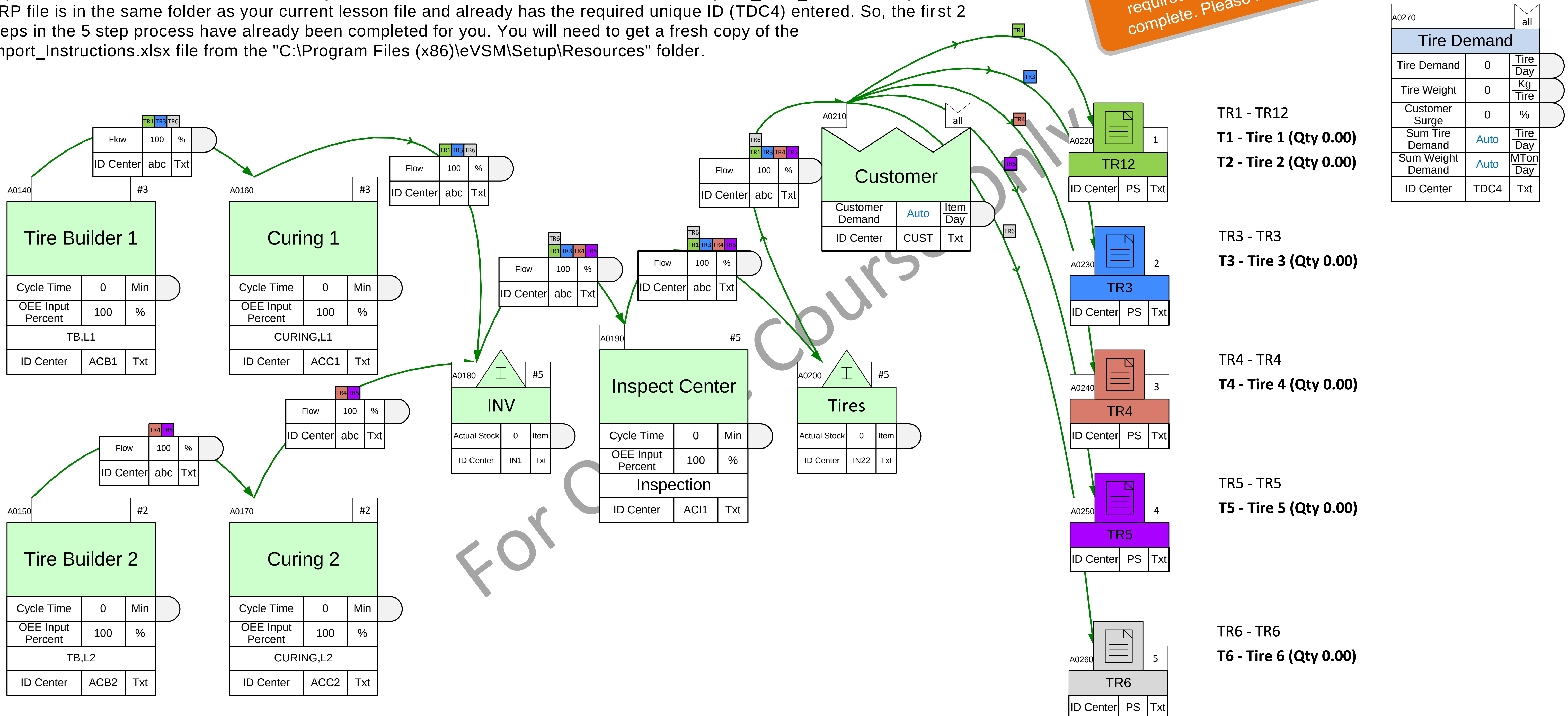
Here are the 5 steps to import ERP (Enterprise Resource Planning) data to the value stream map. If you are unclear, download the ERP Data Import course notes from: https://evsm.com/el/v12/file=eL_ERPImport.12Notes.pdf



Import ERP Demand Data

Import the “Tire Demand” and the “Tire Weight” data from the “Mix” worksheet in the ERP file (ERP_Data_Lesson3.xlsx). The ERP file is in the same folder as your current lesson file and already has the required unique ID (TDC4) entered. So, the first 2 steps in the 5 step process have already been completed for you. You will need to get a fresh copy of the Import_Instructions.xlsx file from the "C:\Program Files (x86)\eVSM\Setup\Resources" folder.

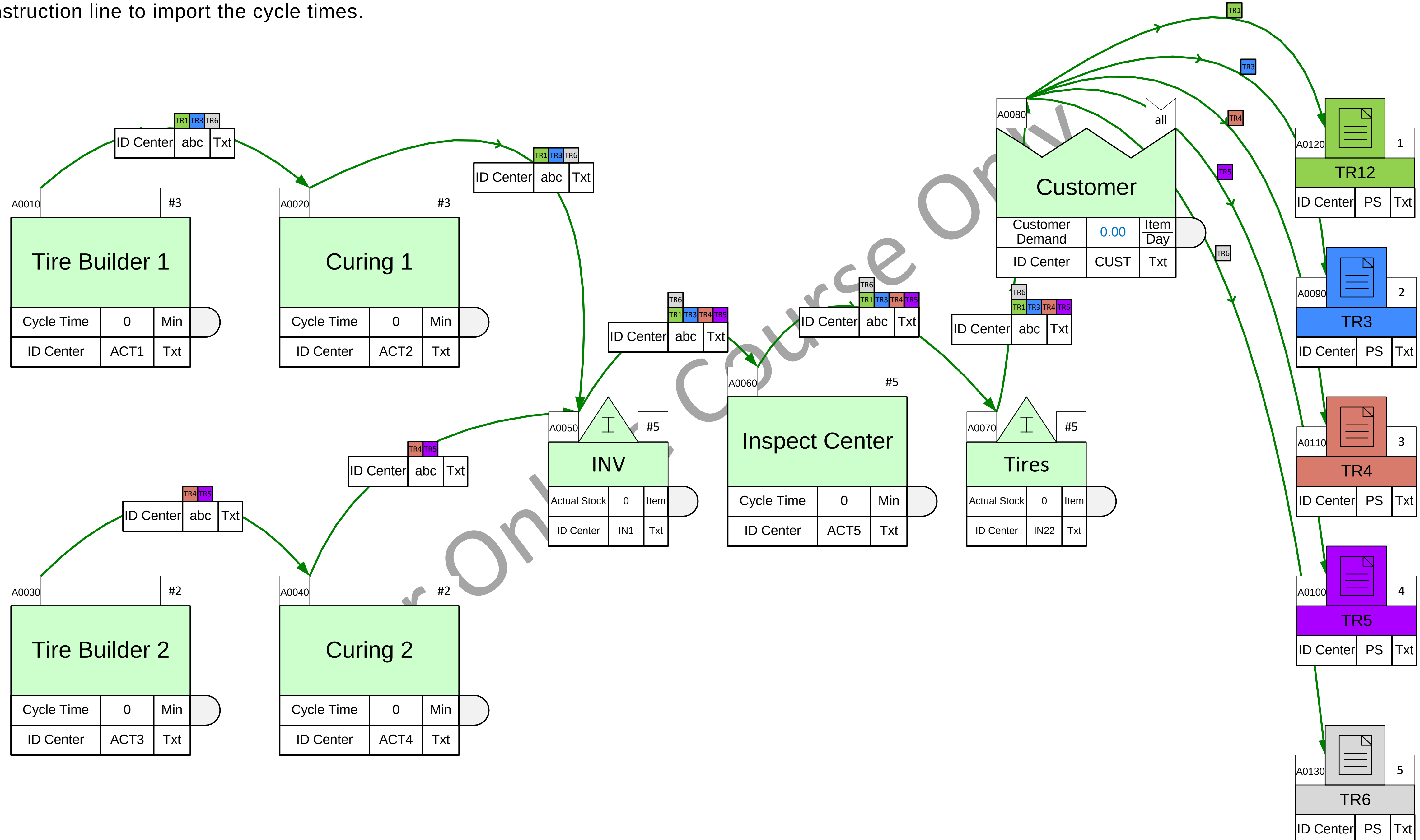
Note: Import XL is slow and requires several minutes to complete. Please be patient.



Units	Year	Wk	Wk	Item	Item
	52	168	7	1	1
	Wk	Hr	Day	Bbn	Tire

Enter Mix Data on Map – Import Cycle Times

Import the “Cycle Time” data from the “Cycle Times” worksheet in the ERP file (ERP_Data_Lesson3.xlsx). The ERP file already has the required unique IDs entered. You can use the same instructions file you used in the last exercise. Just ad a new instruction line to import the cycle times.



Units	Year	Wk	Wk	Item	Item
	52	168	7	1	1
	Wk	Hr	Day	Bbn	Tire

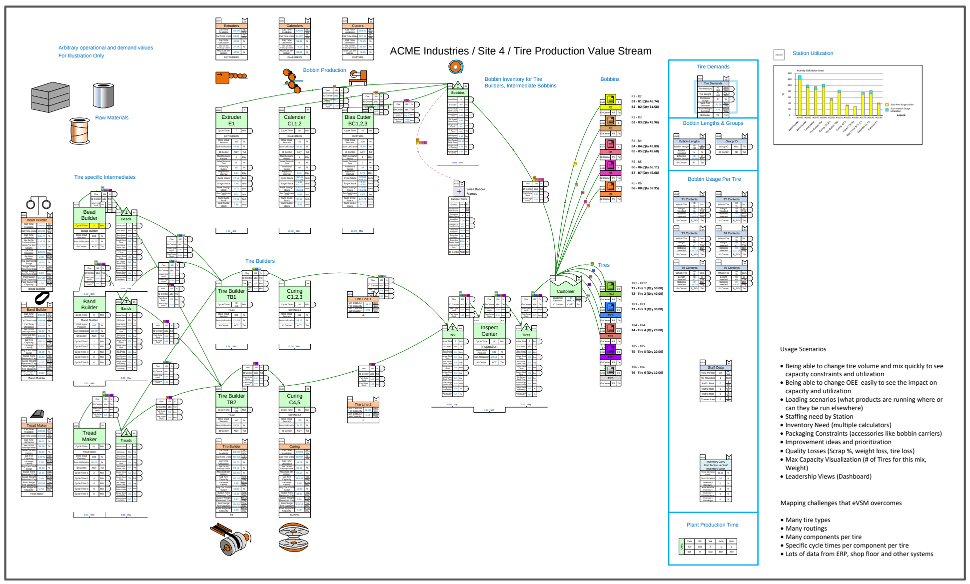
Solving the Map

Solve

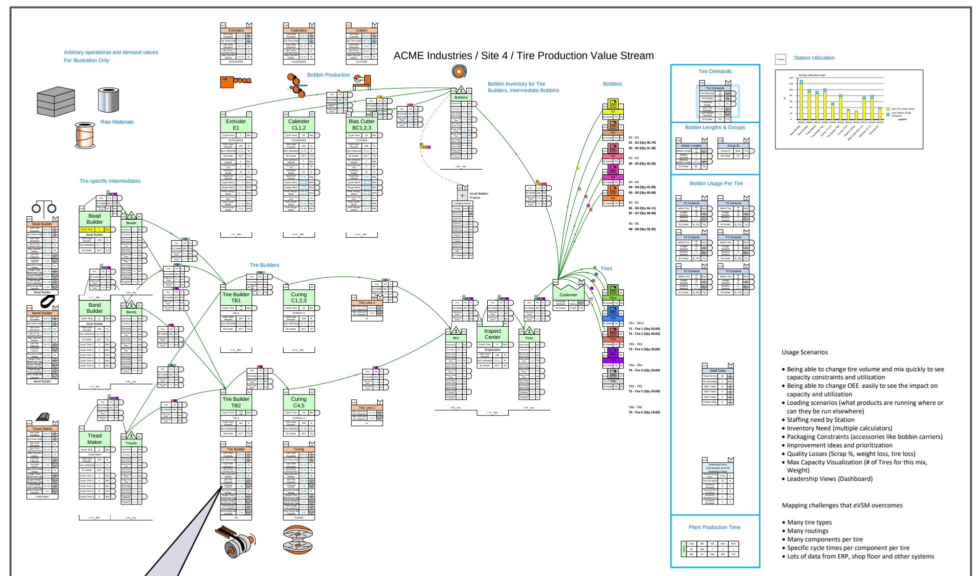
The Solve button checks the map for completeness, automatically fixes some common mistakes, and reports and problems the user needs to fix. If there are no problems, it proceeds to solve. The final result has all the calculations complete, and charts updated.

Solve

Before Solve



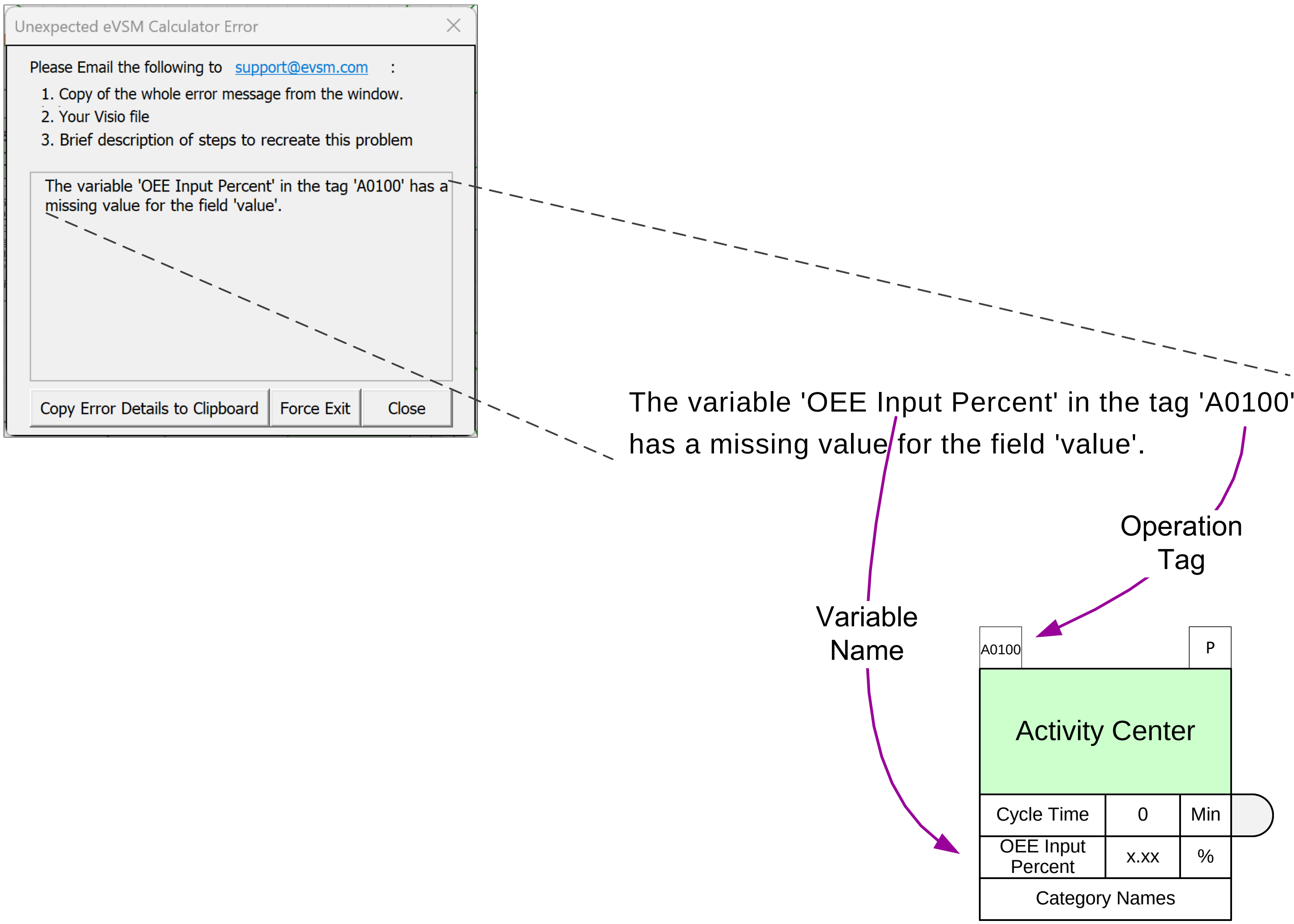
After Solve



The blue values represent auto calculation results

Solving the Map – Error Messages

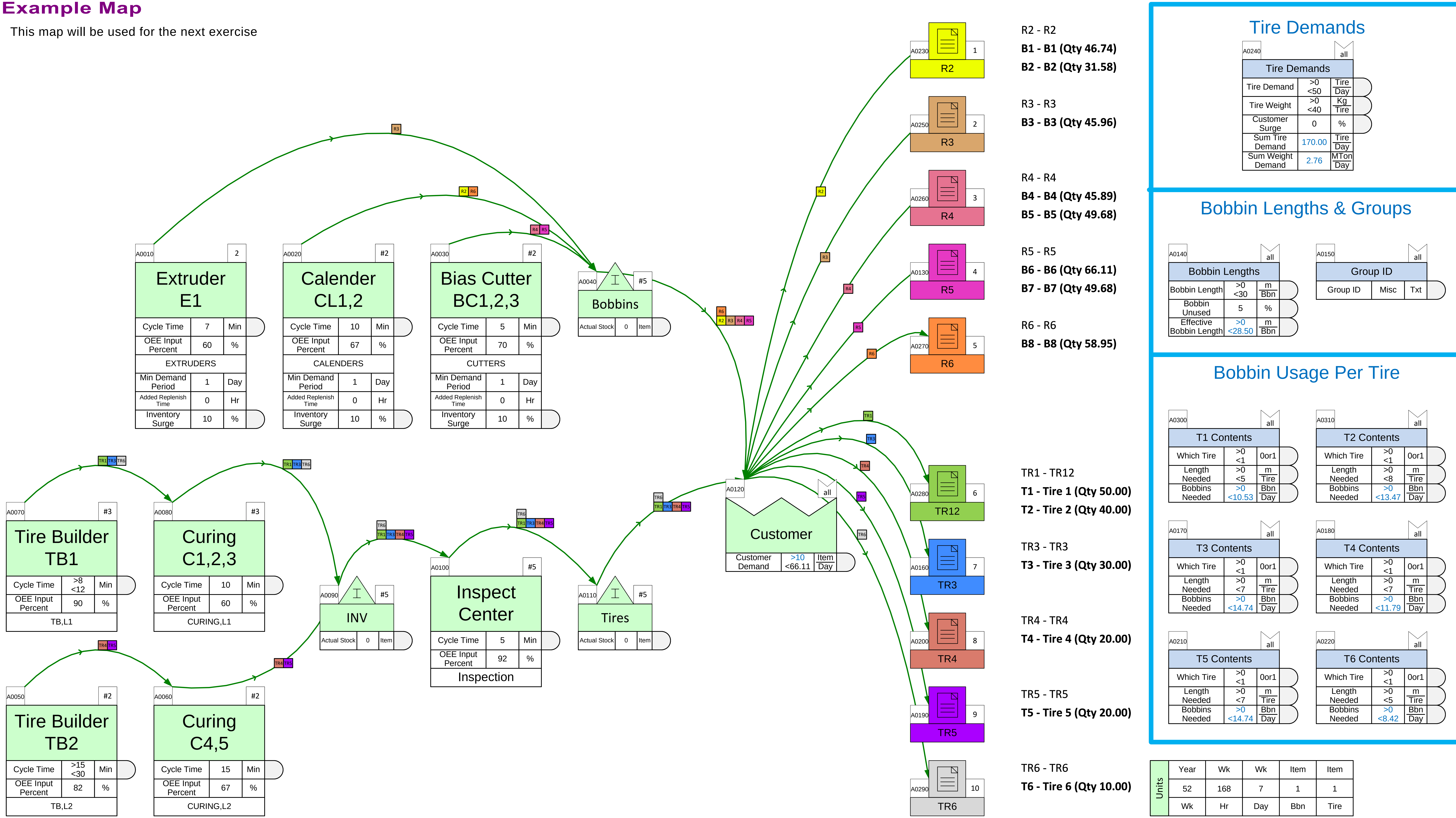
The Solve function first checks the map for completeness and some common errors. If there are problems, the Solver aborts with an error message. Some of these error messages are in simple programming language. Here is an example.



Use the Visio Find command to find the operation tag on the map.

Example Map

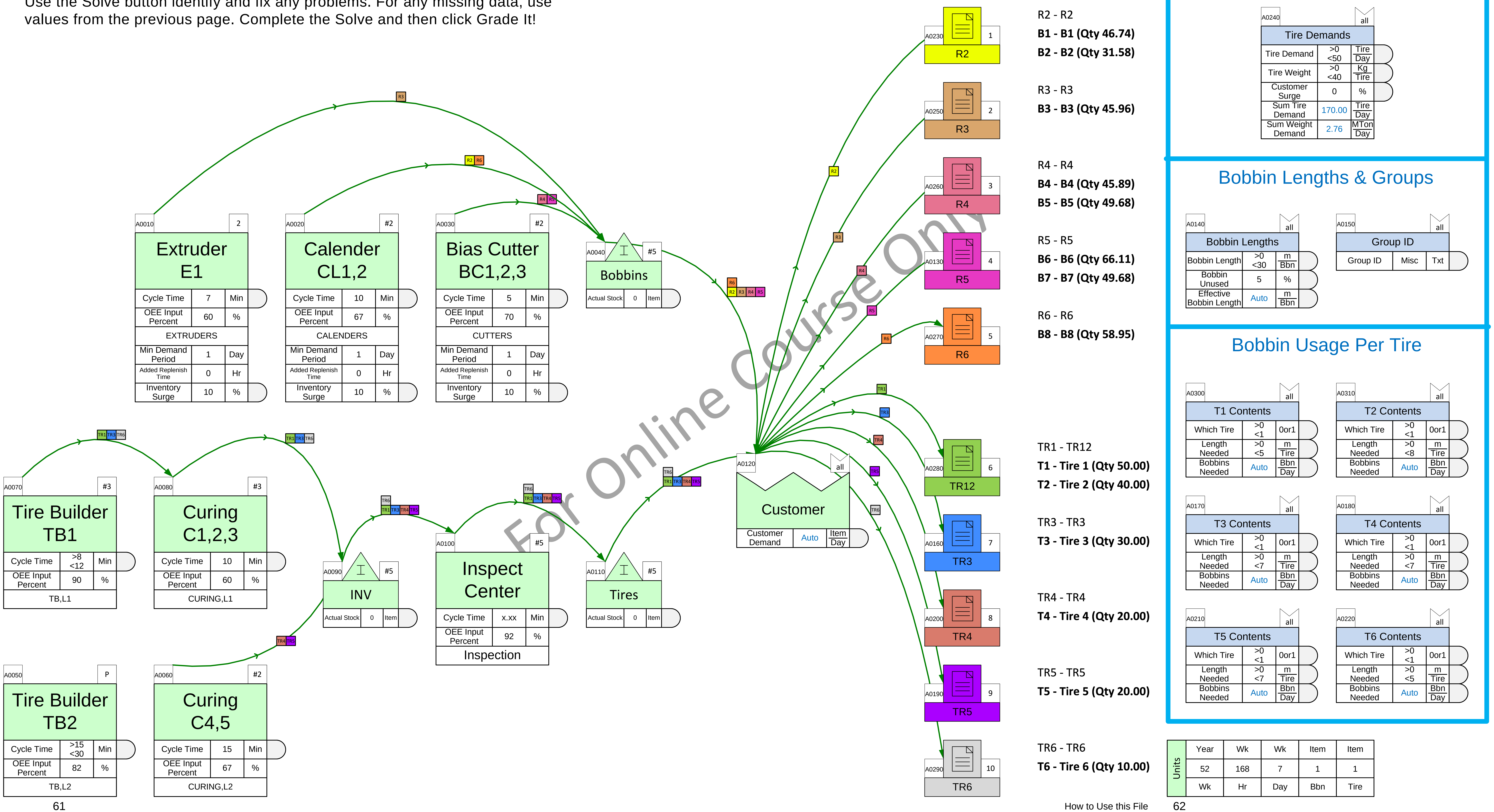
This map will be used for the next exercise



How to Use this File

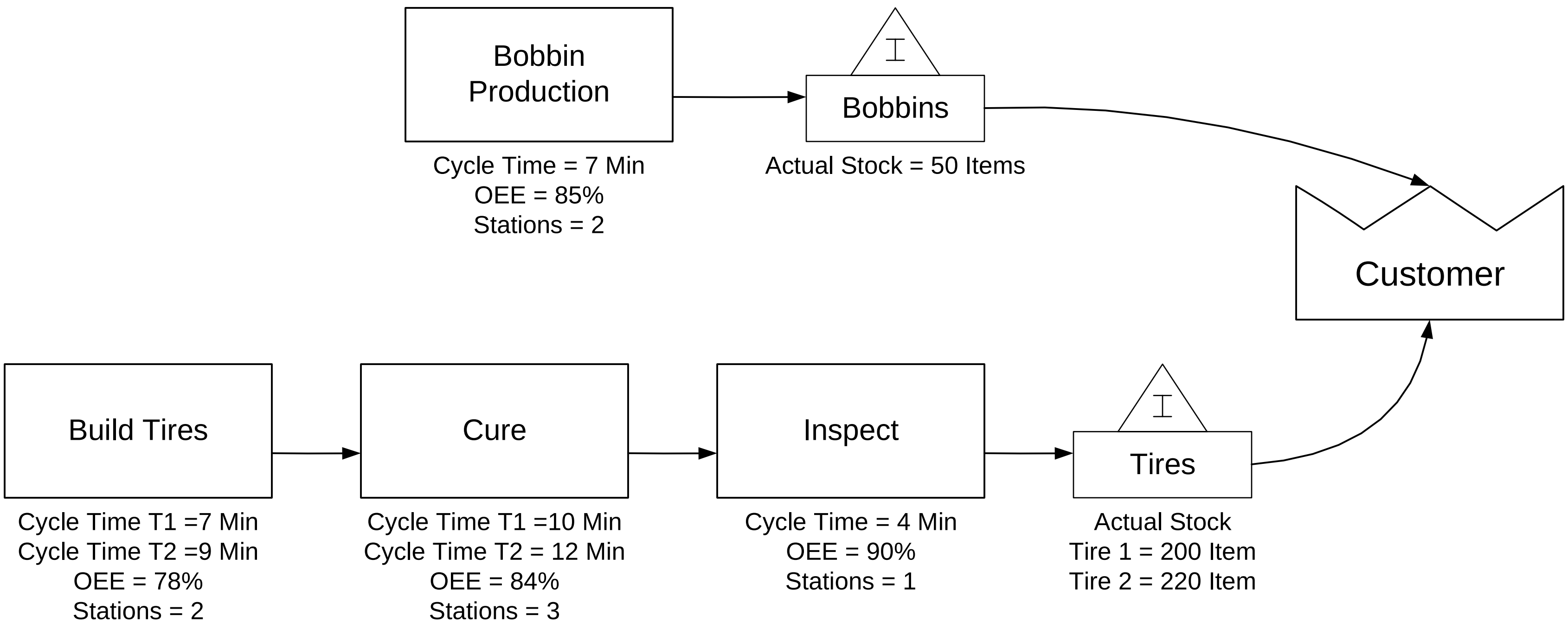
Solve this Map

Use the Solve button identify and fix any problems. For any missing data, use values from the previous page. Complete the Solve and then click Grade It!



Tire Mapping Project

In the next exercise, you need to build the model shown here with the data below.



Model has just 1 bobbin

Tire and Bobbins Data

			B1
Bobbin Length (m/Bbn)			28
Bobbin Unused (%)			5
	Demand (Tire/Day)	Weight (Kg/Tire)	Bobbin Length Needed per Tire (m)
T1	100	30	6
T2	150	25	5

2 Tires

Plant Hours
168 hours per week
7 days per week
52 weeks per year

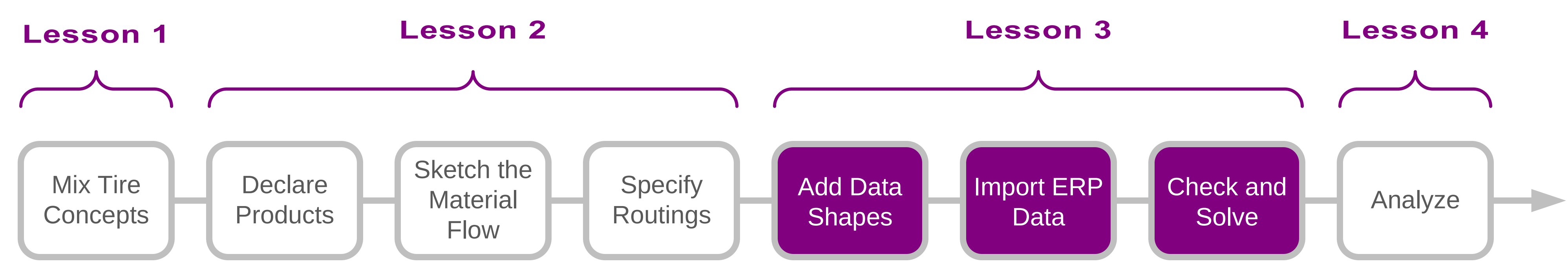
Build the Tire Model Shown on the Previous Page

Follow all the steps, starting with initializing the page for a Mix Tire map, through to final Solve. If you get stuck, click the Video button in the eLeanor control panel and watch the silent movie.

You learned that:

- How add data shapes to a Mix Tire flowchart
- How to import tire demand and production data from the ERP system
- How to check the model for completeness and errors, and then Solve.

Quick Mix Tire Learning Path

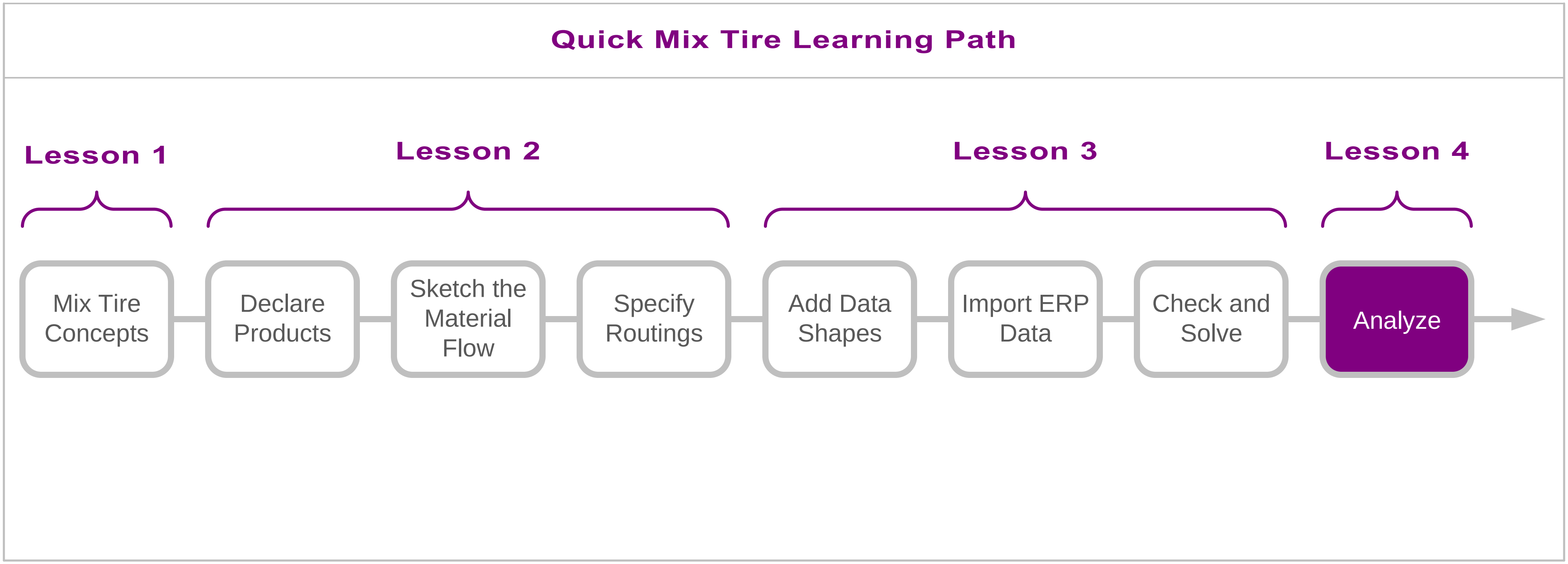


What's next:

- Learn how to analyze the model for key metrics such us demand, capacity, lead time, and cost.
- Learn how to extend the model to explore "What-If" scenarios.

Mix Tire Model Analyses

You have now learnt the Mix Tire mapping concepts and how to build a base Mix Tire model. In this lesson, you will learn how to analyze the model and how to use it to explore what-if future state scenarios.



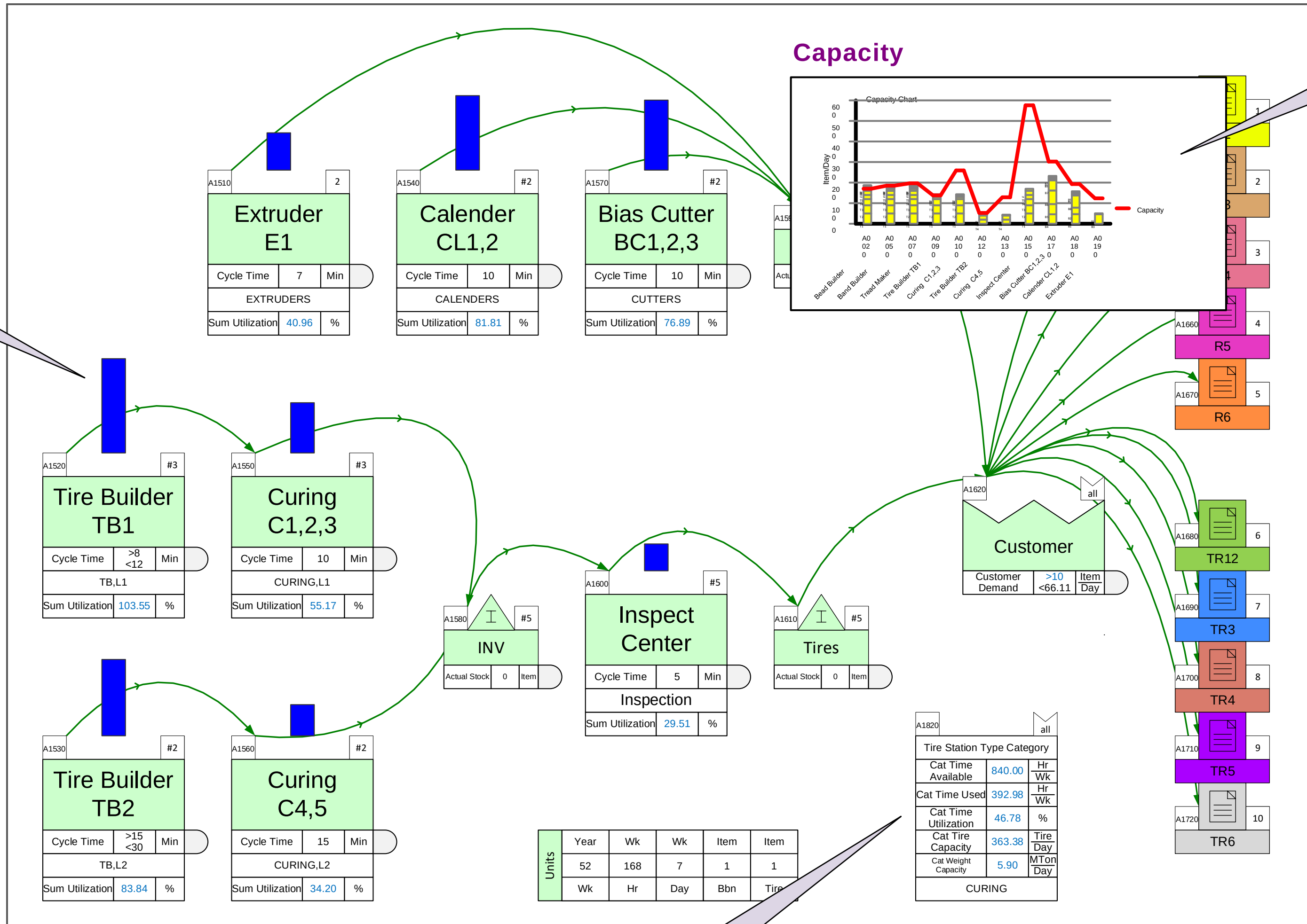
Mix Tire Model Analyses

Map Analysis

Once the model has successfully solved, the results can be summarized using summary tables, charts, and gadgets. The model can be expanded to perform additional analyses.

Gadgets

See the Mix Time course for instructions



Charts

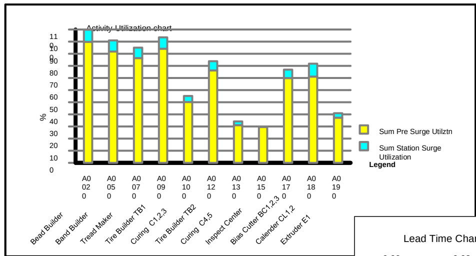
Summary Tables

Summary tables might be for the entire map, limited by path, or restricted to a specific area of the map. The latter are called as Category Summaries.

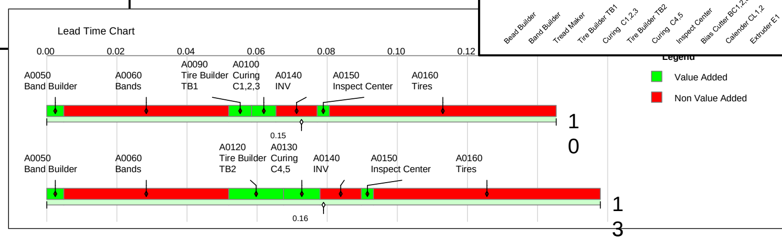
Mix Tire Charts

Mix Tire includes several built in charts.

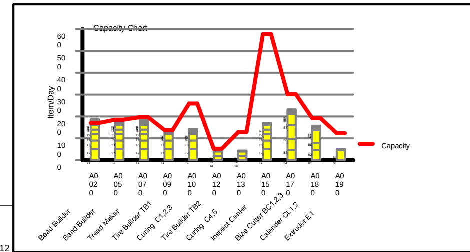
Utilization



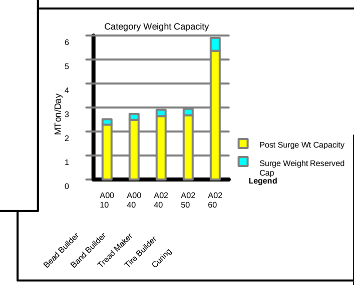
Lead Time



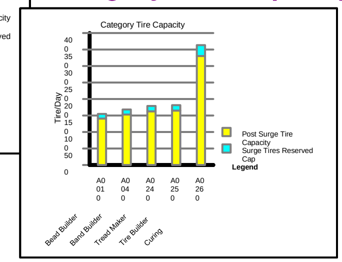
Capacity



Category Weight Capacity



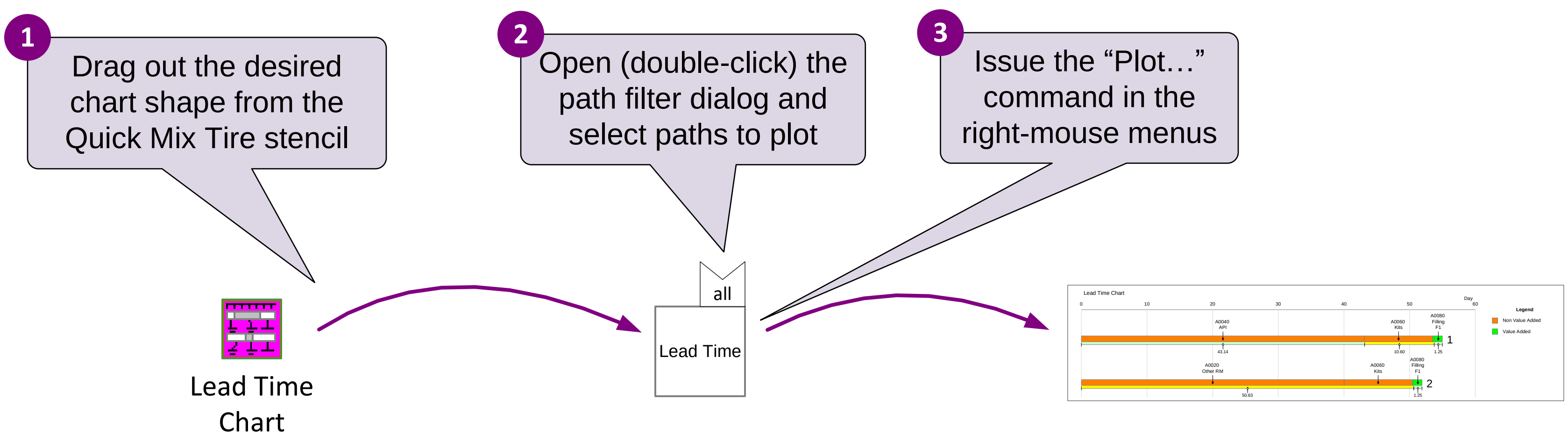
Category Tire Capacity



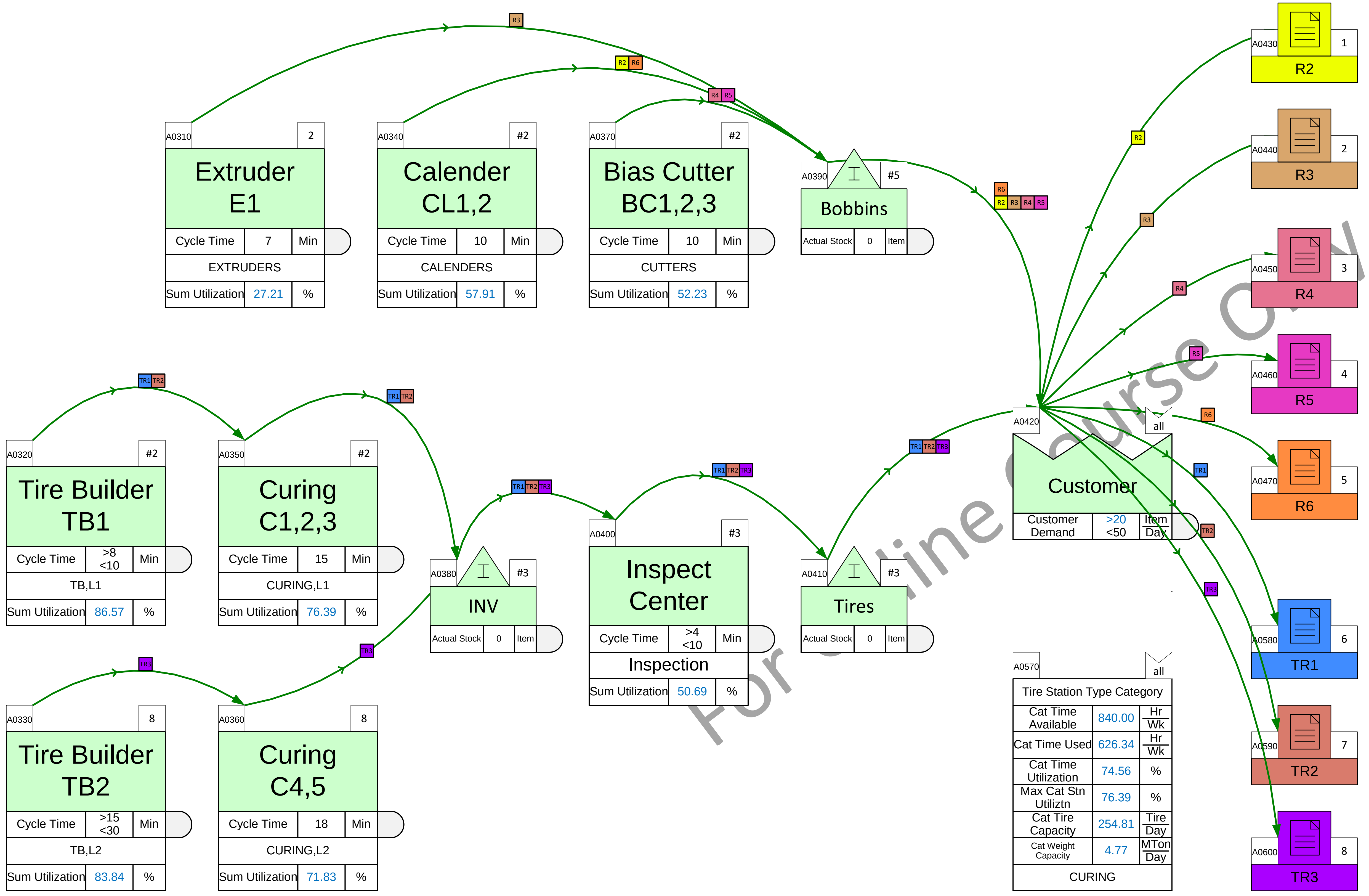
Plotting Charts…

These charts can only be plotted if the required data is available on the map and after the model has solved successfully.

To plot a chart, simply drop the chart icon on the drawing page, and use the plot command in the right mouse menus of the chart shape. Some charts (e.g. Lead time chart) include a filter to remove unwanted paths from the chart.



Plot the Utilization Chart for this Value Stream



R2 - R2

B1 - B1 (Qty 32.42)

B2 - B2 (Qty 22.22)

R3 - R3

B3 - B3 (Qty 30.53)

R4 - R4

B4 - B4 (Qty 31.16)

B5 - B5 (Qty 34.53)

R5 - R5

B6 - B6 (Qty 41.68)

B7 - B7 (Qty 36.21)

R6 - R6

B8 - B8 (Qty 42.53)

TR1 - TR1

T1 - Tire 1 (Qty 50.00)

T2 - Tire 2 (Qty 40.00)

TR2 - TR2

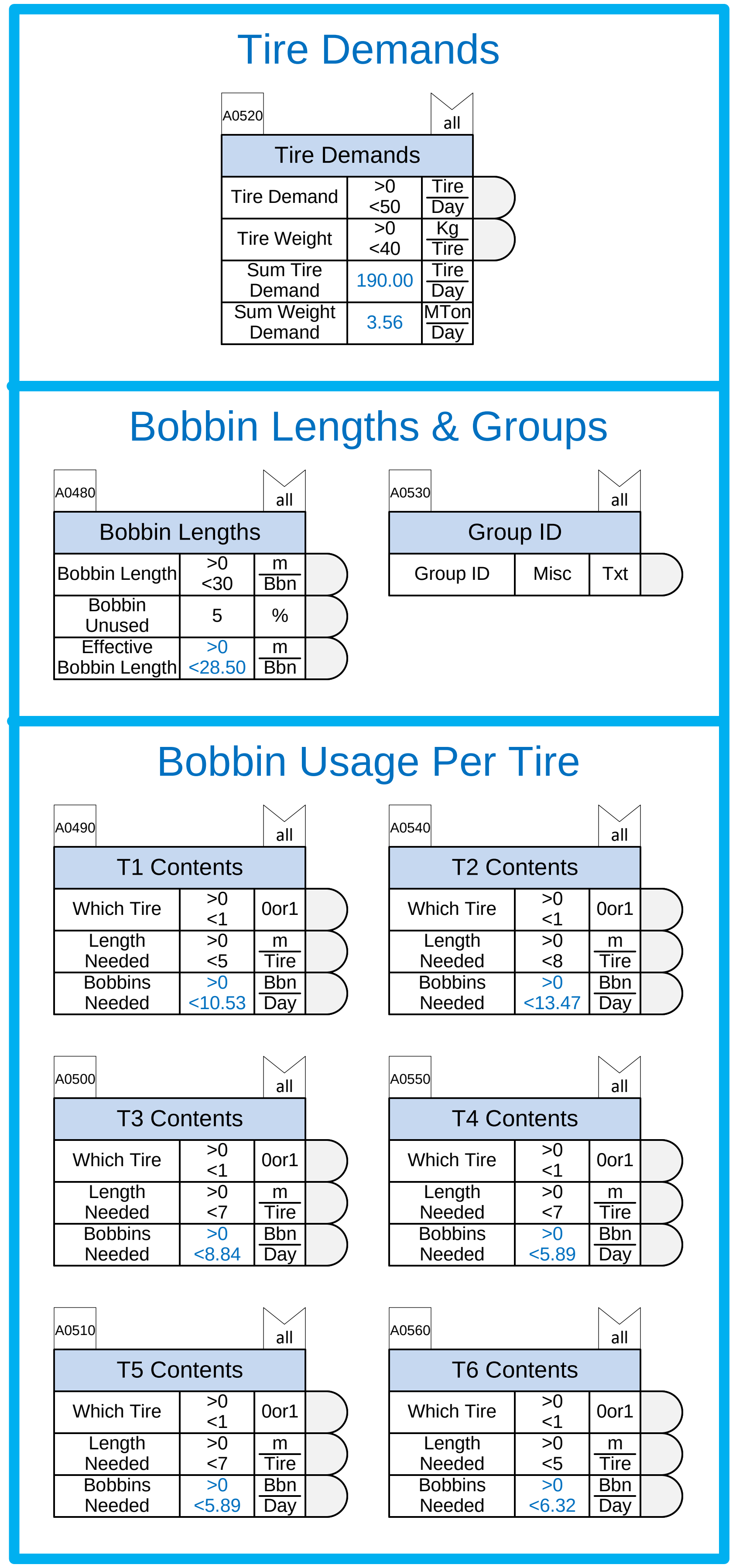
T3 - Tire 3 (Qty 30.00)

TR3 - TR3

T4 - Tire 4 (Qty 20.00)

T5 - Tire 5 (Qty 20.00)

T6 - Tire 6 (Qty 30.00)



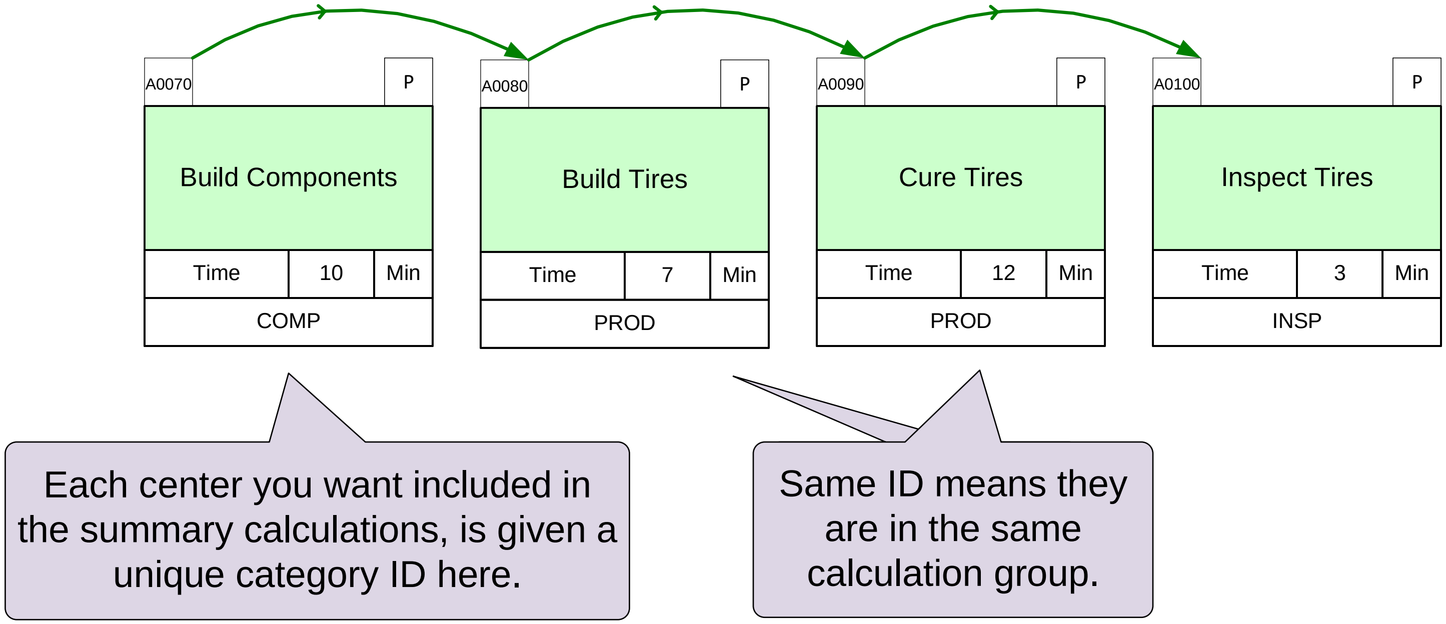
Units					
	Year	Wk	Wk	Item	Item
	52	168	7	1	1
	Wk	Hr	Day	Bbn	Tire

How to Use this File

Category Summary

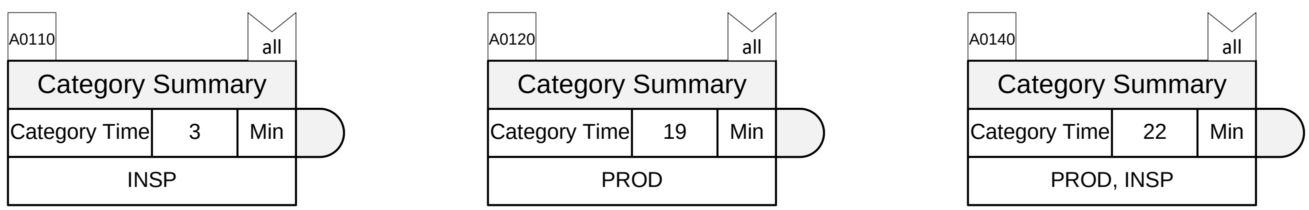
The Category Summary function allows you to create summaries of a part of the value stream. Here is an example.

Example



Summary Calculations

Summaries can be for single or combined ID groups.



Total for Inspection process

Total for all "PROD" processes

Total for all "PROD" processes plus "INSP"

For a step-by-step guide to set up category summaries, see the Mix Time course notes at https://evsm.com/MixTimeV12

Category Summary Centers in Mix Tire

In the Mix Tire application, category summaries are pre-set for some common calculations.

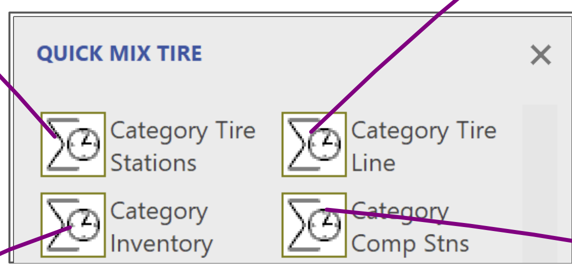
Here are the Category summary centers that are currently available in Mix Tire. All require only the Category ID as input; no further information is necessary.

A0010		all
Tire Station Type Category		
Cat Time Available	Auto	Hr/Wk
Cat Time Used	Auto	Hr/Wk
Cat Time Utilization	Auto	%
Max Cat Stn PreSurge Utiliztn	Auto	%
Max Cat Stn Utiliztn	Auto	%
Cat Tire Capacity	Auto	Tire/Day
Cat Weight Capacity	Auto	MTon/Day
Avg Cat Comb Surge	Auto	%
Surge Tires Reserved Cap	Auto	Tire/Day
Surge Weight Reserved Cap	Auto	MTon/Day
Post Surge Tire Capacity	Auto	Tire/Day
Post Surge Wt Capacity	Auto	MTon/Day
Station Type Category ID		

Used for main production multi-line stations, e.g. Curing activity across all lines

Partial line capacity summary

A0020		all
Tire Line ID Category		
Min Cat Line Tire Capacity	Auto	Item/Day
Min Cat Line Wt Capacity	Auto	MTon/Day
Line Category ID		



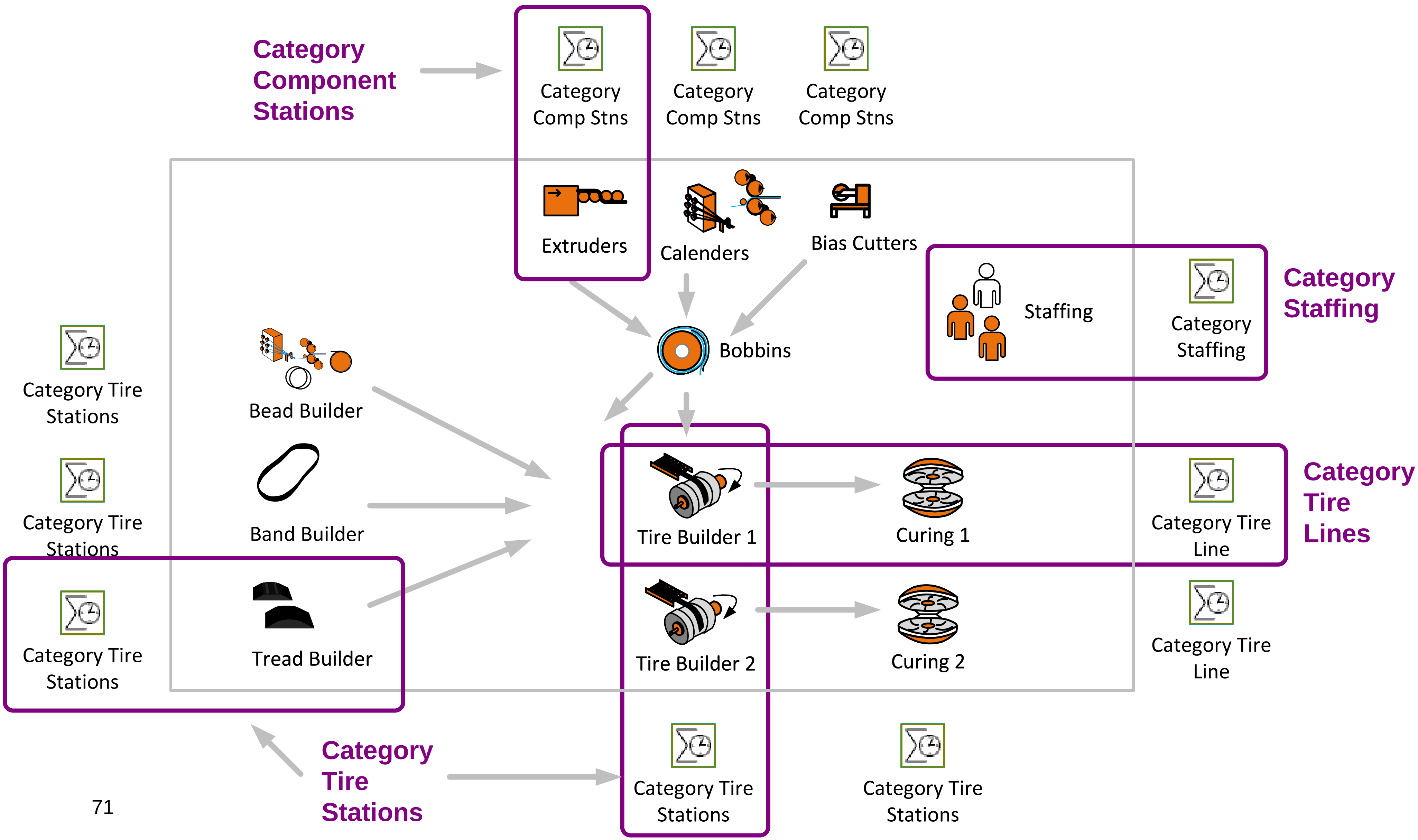
A0040		all
Component Stn Category		
Cat Time Available	Auto	Hr/Wk
Cat Time Used	Auto	Hr/Wk
Cat Time Utilization	Auto	%
Max Cat Stn PreSurge Utiliztn	Auto	%
Max Cat Stn Utiliztn	Auto	%
Station Type Category ID		

Used for component stations, e.g. Bias Cutter

A0030		all
Inventory Category		
Cat Actual Stock	Auto	Item
Cat Design Stock	Auto	Item
Cat Actual Stock Time	Auto	Day
Cat Design Stock Time	Auto	Day
Inventory Category ID		

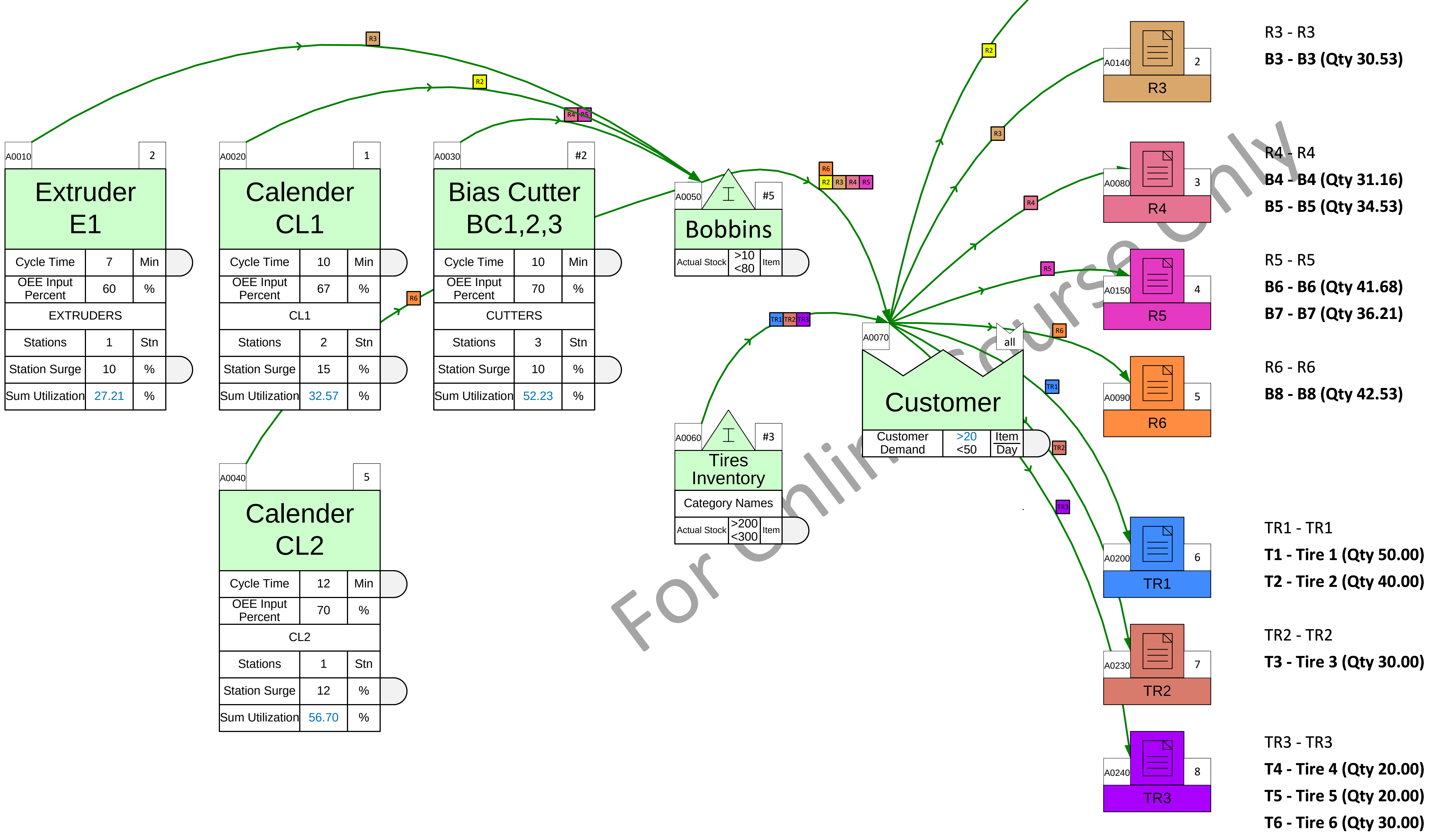
Category Based Capacity Analysis

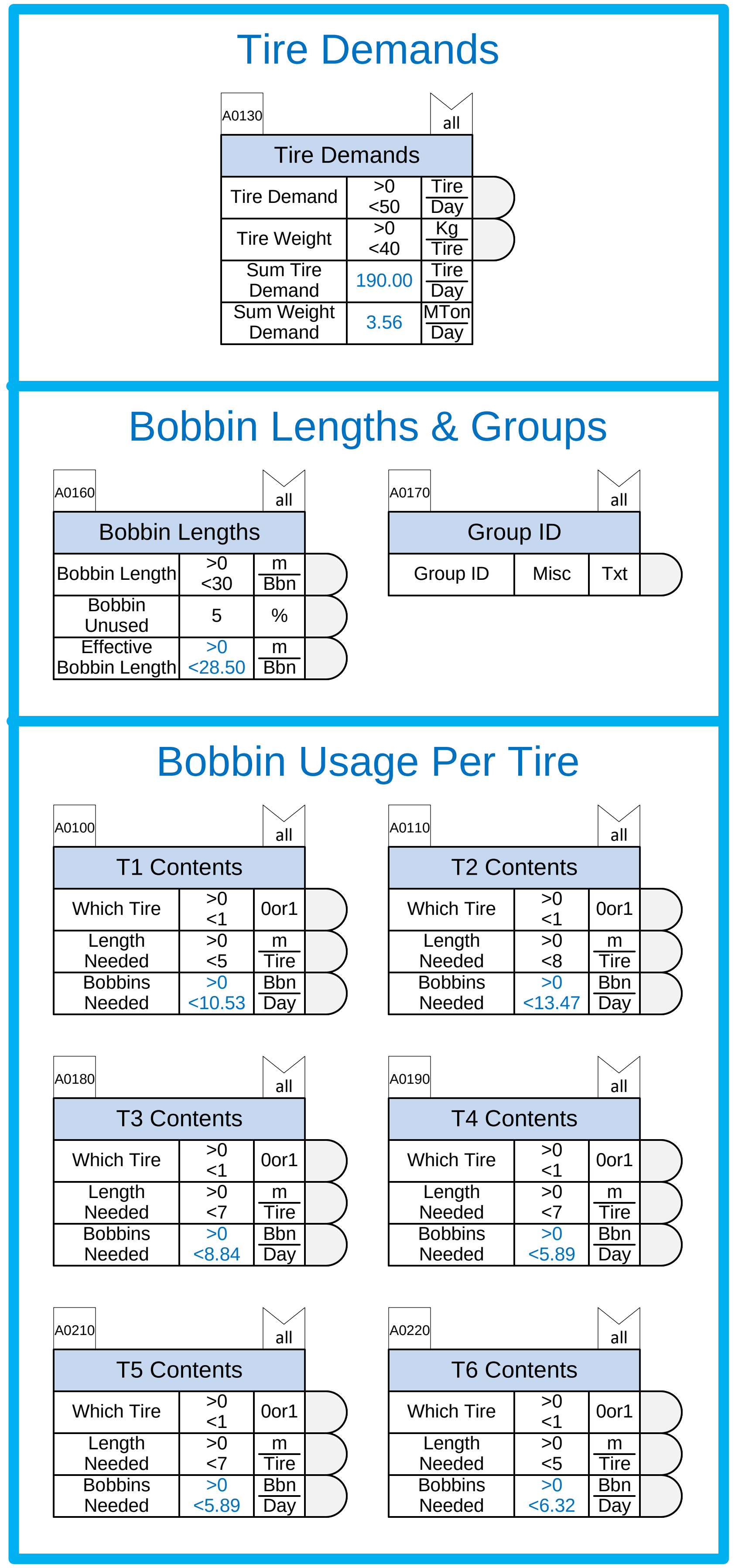
The utilization values of individual stations can be analyzed, but due to the large number of stations in a tire plant, it is often more practical to categorize these stations (using text category IDs) and then review the summaries for each category. The category masters displayed below are accessible in the Quick Mix Tire application, and we typically position them on the periphery of the main map.



What is the total time available and the max utilization for the Calenders?

1. Drag out a “Category Comp Stns” (Category Component Stations) summary
2. Set it’s Category ID to “CL1,CL2” to match the Calender canter's IDs
3. Solve and examine the results
4. Grade It!



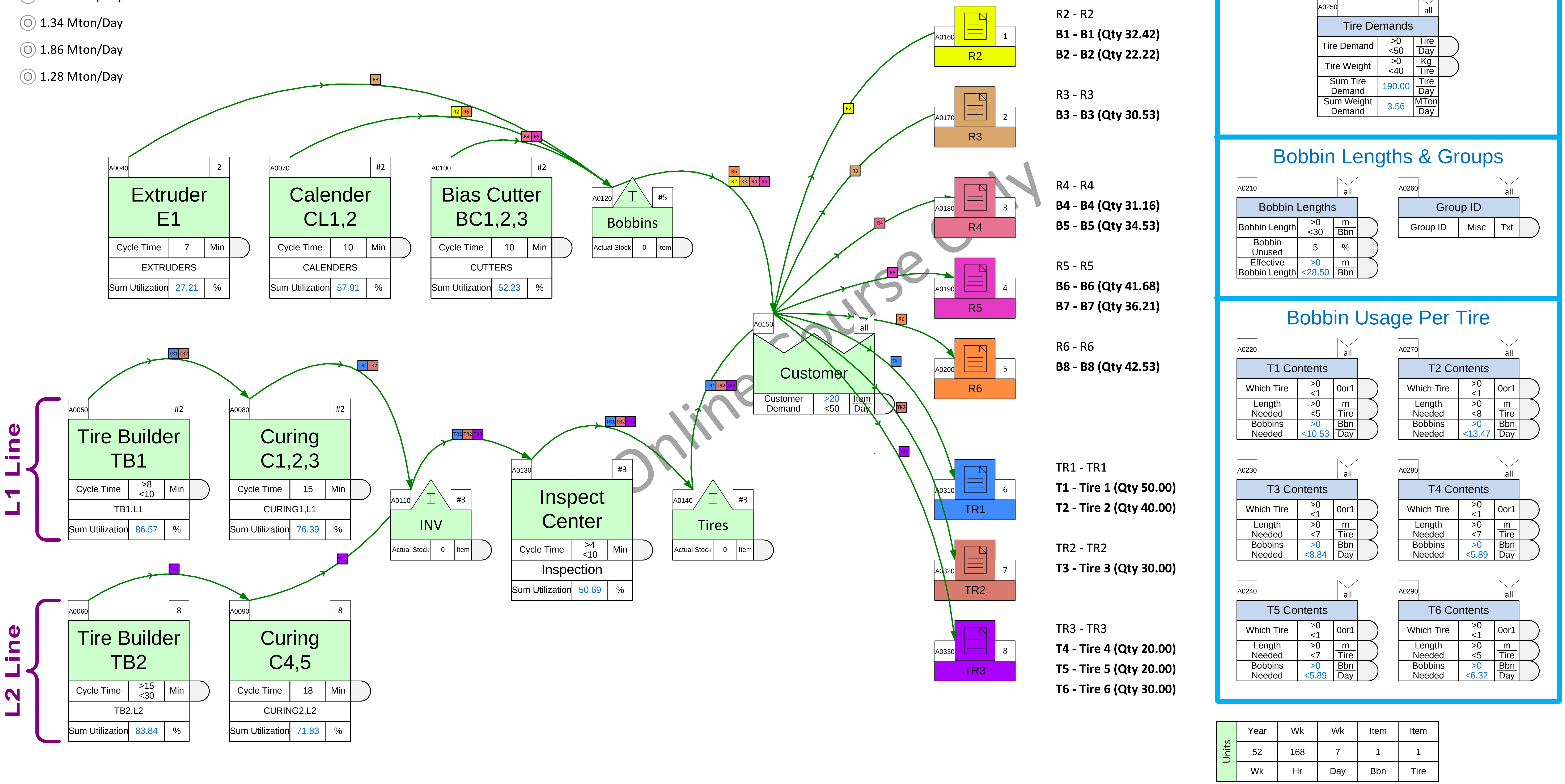


Units	Year	Wk	Wk	Item	Item
	52	168	7	1	1
	Wk	Hr	Day	Bbn	Tire

How to Use this File

What is the Daily Weight Capacity of the L1 Tire Production Line?

- 1.02 Mton/Day
- 1.34 Mton/Day
- 1.86 Mton/Day
- 1.28 Mton/Day



The demand forecast for next month is shown in the blue table. Which of the following statements is true (select ALL 3 that apply)?

☐ We can meet the forecast demand with no changes on the floor

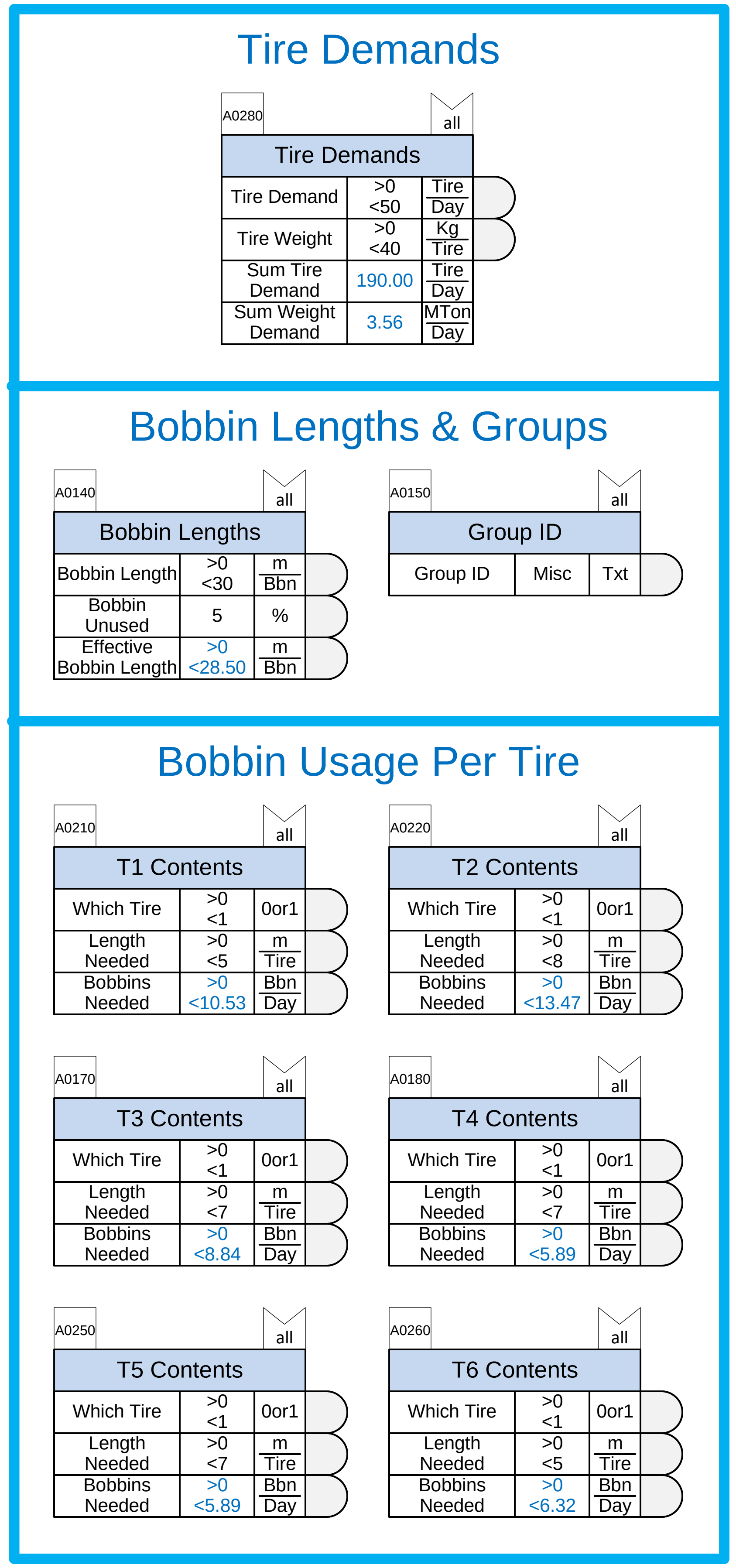
☐ Bias Cutter stations cannot meet the extra demand

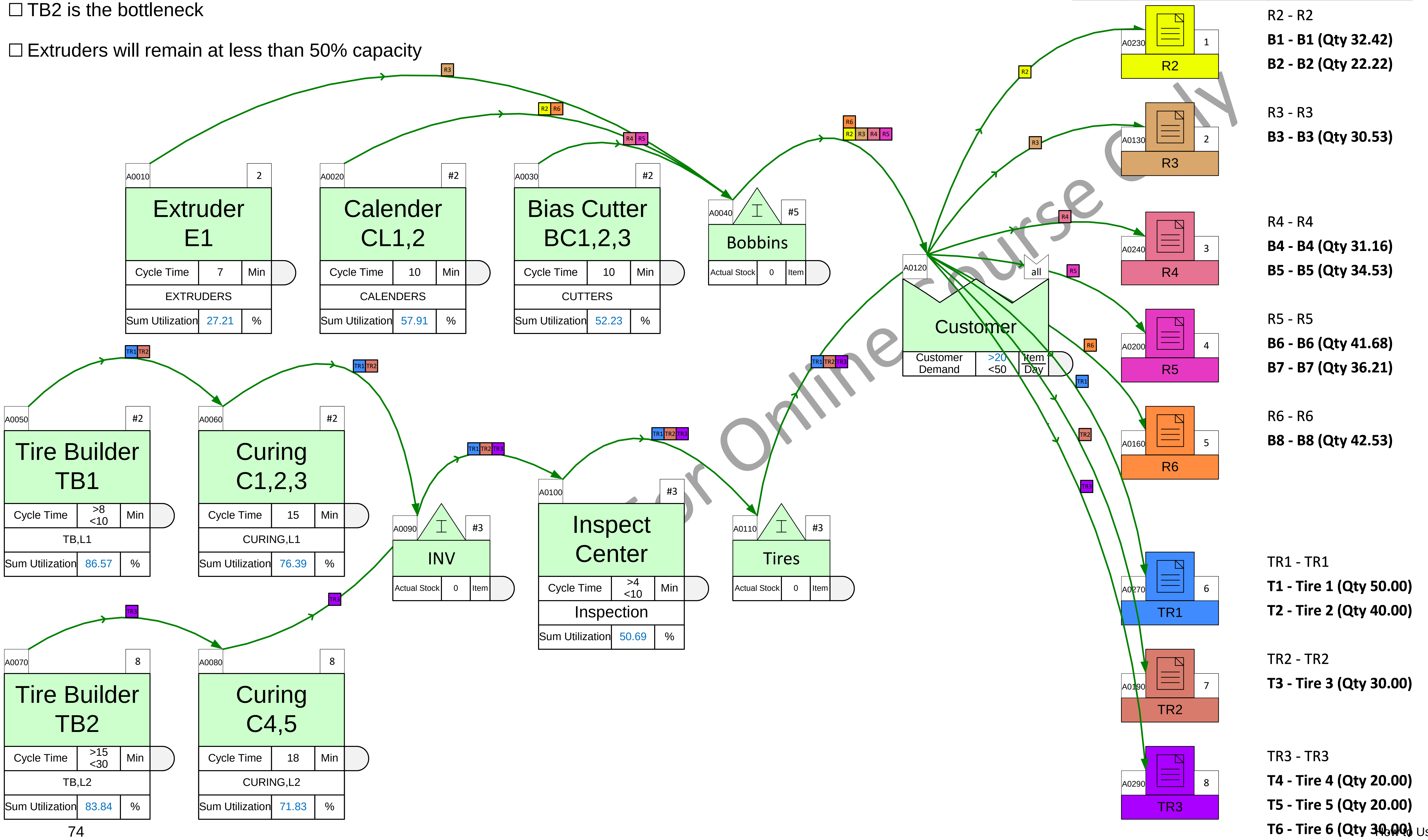
☐ Inspection Center has sufficient capacity to cover the forecast

☐ TB2 is the bottleneck

☐ Extruders will remain at less than 50% capacity

	Current Demand (Tire/Day)	Demand Forecast (Tire/Day)
T1	50	50
T2	40	60
T3	30	30
T4	20	80
T5	20	40
T6	30	30



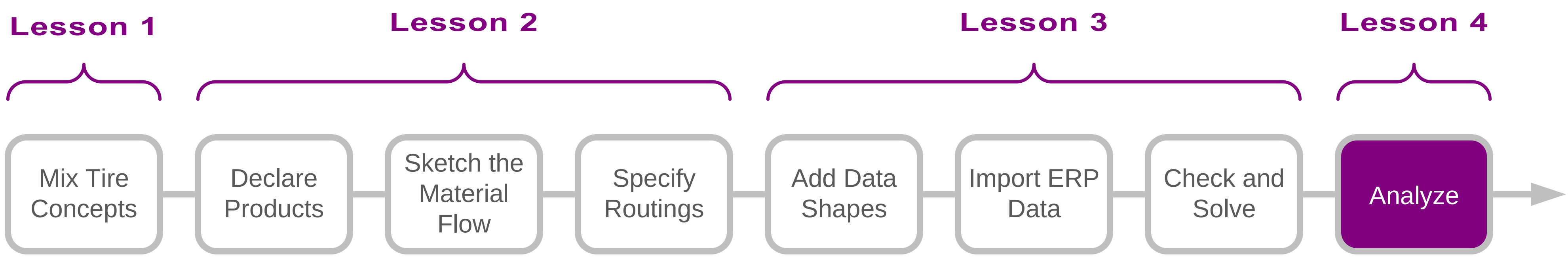


Lesson 4: Summary

In this course you learned:

- Mix Tire modeling concepts
- How to build a tire production value stream model
- How to use the model to understand the current state and how to explore future state possibilities

Quick Mix Tire Learning Path



What's next:

1. Capture your value stream in eVSM Mix Tire
2. Email any questions/problems to support@evsm.com
3. Go through the eVSM Improvement Framework course to see how tools in eVSM can help design the future state
4. Purchase the coaching package at https://evsm.com/shop for assistance with capturing your value stream

Useful Links

eVSM Toolbar Guide	evsm.com/toolbarguide
eVSM Productivity Guide	evsm.com/productivity
eVSM Blogs	evsm.com/blog
eVSM Support FAQ	evsm.com/support
Download the Latest Version	evsm.com/install